

GEOTECHNICAL ENGINEERING REPORT



I-45 FRONTAGE ROAD EXTENSION

WILLIS, TEXAS

TXDOT HOUSTON DISTRICT

CSJ NO. 0675-08-114

GEOTECHNICAL ENGINEERING REPORT

I-45 Frontage Road Extension

Willis, Texas

TXDOT Houston District

CSJ NO. 0675-08-114

Prepared by:



Gorrondona Engineering Services, Inc.

Prepared for:

L Squared Engineering
21123 Eva Street, Suite 200
Montgomery, Texas 77356

Attention: Mr. Jonathan White

November 1, 2021

GES Project No. 21-0320



Gorrondona
Engineering Services, Inc.

Geotechnical Engineering • Construction Materials Testing & Inspection

November 1, 2021

Mr. Jonathan White
L Squared Engineering
21123 Eva Street, Suite 200
Montgomery, Texas 77356

**Re: GEOTECHNICAL ENGINEERING REPORT
I-45 Frontage Road Extension
Willis, Texas
TXDOT Houston District
CSJ NO. 0675-08-114
GES Project No. 21-0320**

Dear Mr. White:

Gorrondona Engineering Services, Inc. (GES) is pleased to submit this Geotechnical Engineering Report for the referenced project. We appreciate the opportunity of working with you. Please contact us if you have any questions or require additional services.

Respectfully submitted,

Anahita Goudarzi, Ph.D., P.E.
Austin Operations Manager

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1.0 INTRODUCTION

Project Location. The project is located on the east side of I-45 between West Montgomery Street and FM 830 in Willis, Texas (CSJ: 0675-08-114). The general location and orientation of the site are provided in Appendix A - Project Location Diagrams.

Project Description. The project consists of TxDOT public utilities, roadway embankments, and culverts, totaling approximately 6,500 LF.

Project Authorization. This geotechnical investigation was authorized by Mr. Jonathan White with L Squared Engineering and performed in accordance with GES Proposal P19-0623 – Rev. 1 dated June 20, 2021.

Purpose and Methodology. The principal purposes of this investigation were to evaluate the general soil conditions at the proposed site and to develop geotechnical engineering design recommendations. To accomplish its intended purposes, the study was conducted in the following phases:

1. Drill sample borings to evaluate the soil conditions at the boring locations and to obtain soil samples;
2. Conduct laboratory tests on selected samples recovered from the borings to establish the pertinent engineering characteristics of the soils; and
3. Perform engineering analyses, using field and laboratory data, to develop design criteria.

Cautionary Statement Regarding Use of this Report. As with any geotechnical engineering report, this report presents technical information and provides detailed technical recommendations for civil and structural engineering design and construction purposes. GES, by necessity, has assumed the user of this document possesses the technical acumen to understand and properly utilize information and recommendations provided herein. GES strives to be clear in its presentation and, like the user, does not want potentially detrimental misinterpretation or misunderstanding of this report. Therefore, we encourage any user of this report with questions regarding its content to contact GES for clarification. Clarification will be provided verbally and/or issued by GES in the form of a report addendum, as appropriate.

Report Specificity. This report was prepared to meet the specific needs of the client for the specific project identified. Recommendations contained herein should not be applied to any other project at this site by the client or anyone else without the explicit approval of GES.

2.0 FIELD INVESTIGATION

Subsurface Investigation. The subsurface investigation for this project is summarized in the following table. Boring locations are provided in Appendix B - Boring Location Diagram.

Boring Nos.	Depth, feet bgs ¹	Station	Offset	Surface Elevation Above (msl) ² , (feet)	Date Drilled	Location ³	
						Northing	Easting
B-01	20	21+69.59	27.27R	340.10	8/24/2021	10142504.81	3824278.98.
B-02	20	27+03.03	1.41L	345.98	8/24/2021	10143035.54	3824217.89
B-03	20	31+41.80	27.97R	331.92	8/24/2021	10143475.71	3824201.49
B-04	20	34+44.04	41.52R	329.38	8/24/2021	10143777.83	3824168.19
B-05	20	38+30.20	20.78R	340.46	8/24/2021	10144153.61	3824070.39
B-06	20	44+73.47	13.49R	338.78	8/24/2021	10144772.66	3823890.68.
B-07	20	49+63.91	2.36L	335.26	8/25/2021	10145232.90	3823720.21
B-08	20	51+67.90	16.84L	333.94	8/25/2021	10145421.75	3823641.75
B-09	20	55+83.00	3.17R	337.71	8/25/2021	10145821.75	3823529.00
B-10	20	60+60.9	7.34R	354.12	8/25/2021	10146276.40	3823382.02.
B-11	20	64+98.09	15.38R	359.51	8/25/2021	10146696.05	3823259.32
B-12	20	69+72.11	26.41R	365.40	8/25/2021	10147151.82	3823128.60
B-13	20	74+80.63	8.08L	368.29	8/26/2021	10147626.96	3822944.14
B-14	20	79+94.96	4.56R	368.50	8/26/2021	10148121.89	3822803.86
B-15	20	84+82.60	11.28R	369.95	8/26/2021	10148592.94	3822677.78.
<u>Notes:</u> 1. bgs = below ground surface 2. msl = mean sea level 3. NAD83 Texas State Plane surface coordinates							

Boring Logs. Subsurface conditions were defined using the sample borings. Boring logs generated during this study are included in Appendix C - Boring Logs and Laboratory Results. Borings were advanced between sample intervals using continuous flight auger drilling procedures.

Cohesive Soil Sampling. Cohesive soil samples were generally obtained using Shelby tube samplers in general accordance with American Society for Testing and Materials (ASTM) D1587. The Shelby tube sampler consists of a thin-walled steel tube with a sharp cutting edge connected to a head equipped with a ball valve threaded for rod connection. The tube is pushed into the undisturbed soils by the hydraulic pulldown of the drilling rig. The soil specimens were extruded from the tube in the field, logged, tested for consistency using a hand penetrometer, sealed and packaged to maintain "in situ" moisture content.

Consistency of Cohesive Soils. The consistency of cohesive soil samples was evaluated in the field using a calibrated hand penetrometer. In this test a 0.25-inch diameter piston is pushed into the undisturbed sample at a constant rate to a depth of 0.25-inch. The results of these tests are tabulated at the respective sample depths on the boring logs. When the capacity of the penetrometer is exceeded, the value is tabulated as 4.5+.

Granular Soil Sampling. Granular soil samples were generally obtained using split-barrel sampling procedures in general accordance with ASTM D1586. In the split-barrel procedure, a disturbed sample is obtained in a standard 2-inch outside diameter (OD) split barrel sampling spoon driven 18-inches into the ground using a 170-pound (lb) hammer falling freely 24 inches. The number of blows for the last 12-inches of a standard 18-inch penetration is recorded as the Standard Penetration Test resistance (N-value). The N-values are recorded on the boring logs at the depth of sampling. Samples were sealed and returned to our laboratory for further examination and testing.

Texas Cone Penetration (TCP) Test. Texas Cone Penetration Test (TCP) was used to assess the apparent in-place strength characteristics of the rock type materials. In the TCP test procedure, a 3-inch outside diameter (OD) steel cone driven by a 170-pound hammer dropped 24 inches (340 ft-pounds of energy). The number of blows of the hammer required to provide 12 inches of penetration, or the inches of penetration of the cone due to 100 blows of the hammer (whichever occurs first) are recorded on the field logs (reference: TxDOT Geotechnical Manual, March 2018 edition).

Groundwater Observations. Groundwater observations are shown on the boring logs.

Borehole Plugging. Upon completion of the borings, the boreholes were backfilled from the top and plugged at the surface.

3.0 LABORATORY TESTING

GES performs visual classification and any of a number of laboratory tests, as appropriate, to define pertinent engineering characteristics of the soils encountered. Tests are performed in general accordance with ASTM or other standards and the results included at the respective sample depths on the boring logs or separately tabulated, as appropriate, and included in Appendix C - Boring Logs and Laboratory Results. Laboratory tests and procedures routinely utilized, as appropriate, for geotechnical investigations are tabulated in the following table.

Test Procedure	Description
Tex-103-E	Standard Test Method for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
Tex-104-E	Standard Test Methods for Determining Liquid Limit of Soils
Tex-105-E	Standard Test Methods for Determining Plastic Limit of Soils

Test Procedure	Description
Tex-106-E	Standard Test Methods for Calculating the Plasticity Index of Soils
Tex-110-E	Standard Test Method for Particle-Size Analysis of Soils (including Part - 1 & Part - 2)
Tex-111-E	Standard Test Methods for Amount of Material in Soils Finer than the No. 200 (75-µm) Sieve
Tex-142-E	Laboratory Classification of Soils for Engineering Purposes
ASTM D2166	Standard Test Method for Unconfined Compressive Strength of Cohesive Soil
ASTM D2487	Standard Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D2488	Standard Practice for Description and Identification of Soils (Visual-Manual Procedure)
ASTM D4220	Standard Practices for Preserving and Transporting Soil Samples
ASTM D4643	Standard Test Method for Determination of Water (Moisture) Content of Soil by the Microwave Oven Method
ASTM D4546	Standard Test Methods for One-Dimensional Swell or Settlement Potential of Cohesive Soils
ASTM D2435	Standard Test Methods for One-Dimensional Consolidation Properties of Soils Using Incremental Loading
ASTM D4767 / Tex-131-E	Standard Method for Consolidated Undrained Triaxial Compression Test for Cohesive Soils
ASTM D3080	Standard Test method for Direct Shear Test of Soils Under Consolidated Drained Conditions

4.0 SITE CONDITIONS

Site Photographs. Representative photographs of the site at the time of this investigation are provided in “Appendix D - Site Photographs”.

Geologic Formation. Based on available surface geology maps and our experience, it appears this site is located in the Willis Formation. A geologic atlas and USGS formation description are provided in “Appendix G - Geologic Information”. Soils within the Willis can generally be characterized as clay, silt, sand and gravel.

Geologic Faults. A geologic fault study was beyond the scope of this investigation.

4.1 Soil

Stratigraphy. Descriptions of the various strata and their approximate depths and thickness per the Unified Soil Classification System (USCS) are provided on the boring logs included in “Appendix C - Boring Logs and Laboratory Results”. Terms and symbols used in the USCS are presented in “Appendix H - Unified Soil Classification System”. A brief summary of the stratigraphy indicated by the borings is provided in the following table.

Generalized Subsurface Conditions Along the Alignment of Proposed Storm Sewer Line (Borings B-01 to B-15)			
Nominal Depth, feet bgs (Except as Noted)		General Description	Detailed Description of Soils/Materials Encountered
Top of Layer	Bottom of Layer		
0	20	PREDOMINANTLY SAND WITH SOME LEAN AND FAT CLAY	Very loose to very dense, CLAYEY SAND (SC), loose to dense SILTY SAND (SM), slightly compact to dirt SILTY, CLAYEY SAND (SC-SM), compact POORLY GRADED SAND WITH SILT, soft to very stiff SANDY LEAN CLAY (CL), stiff to very stiff SANDY FAT CLAY (CH).
Note: Boring Termination Depth = 20 feet bgs.			

Moisture Change Susceptibility of Near Surface Soils. The sandy soils encountered at and near the ground surface at this site are very susceptible to changes in moisture. The presence of surface water due to precipitation or groundwater may result in a decrease in the ability to compact and work with the soil. It is common for these soils to pump when subjected to high levels of moisture. In addition, these soils located at and near the ground surface will allow surface water to infiltrate until the water becomes perched on a less permeable layer at depth. Soils of this type are especially prone to requiring the implementation of wet weather/soft subgrade recommendations provided in this report.

Swell Potential based on Atterberg Limits. Atterberg (plastic and liquid) limits were performed on 13 shallow soil samples obtained at depths between 0 and 8-feet bgs. The plasticity index of the samples was between 7 and 31 with an average of 18 indicating that the soils have a moderate potential for shrinking and swelling with changes in soil moisture content.

Swell Tests. Swell tests were performed on selected clay soil samples. Swell test details are provided in "Appendix C - Boring Logs and Laboratory Results". The results of the tests are summarized in the following table.

Boring No.	Avg. Depth (ft.)	Moisture Content, w, %	Liquid Limit, LL	Plasticity Index, PI	Applied Overburden Stress (psi)	Swell (%)
B-01	1	13	36	22	0.8	0.56
B-01	4.5	19	46	29	3.8	0.01
B-02	3	10	16	7	2.5	0.00
B-02	7	14	29	17	5.8	0.00
B-03	1	11	Non-Plastic	Non-Plastic	0.8	0.00
B-03	5	15	27	10	4.2	0.00
B-04	3	13	Non-Plastic	Non-Plastic	2.5	0.00
B-04	7	20	46	31	5.8	0.00
B-05	1	34	43	24	0.8	0.00
B-05	4	16	-	-	3.3	0.00
B-07	1	11	21	10	0.8	0.00
B-10	7	13	30	14	5.8	0.00

Boring No.	Avg. Depth (ft.)	Moisture Content, w, %	Liquid Limit, LL	Plasticity Index, PI	Applied Overburden Stress (psi)	Swell (%)
B-11	9	16	36	20	7.5	1.74
B-12	7	13	42	24	5.8	0.00
B-13	3	10	Non-Plastic	Non-Plastic	2.5	0.00
B-13	7	10	27	14	5.8	0.00
B-14	1	8	28	15	0.8	0.00
B-14	7	13	-	-	5.8	0.00
B-15	9	14	30	14	7.5	0.00
Note: "-": Test was not performed.						

Soluble Sulfate Testing. Soluble sulfate testing (Tex-145-E) was performed for this project. The results are included in Appendix C - Boring Logs and Laboratory Results and are summarized in the table below.

Boring No.	Average Depth (feet, bgs)	Sulfate Content (ppm)
B-01	3	167
B-01	7	353
B-02	1	633
B-02	5	147
B-03	3	160
B-04	5	527
B-05	3	153
B-05	7	207
B-06	1	353
B-06	7	187
B-07	3	440
B-07	9	427
B-08	5	840
B-09	3	493
B-11	1	413
B-12	3	113
B-13	1	487
B-15	1	520
B-15	3	513
B-15	5	487

4.2 Groundwater

Groundwater Levels. The borings were advanced using auger drilling and intermittent sampling methods in order to observe groundwater seepage levels. Groundwater levels encountered in the borings during this investigation are identified in the following table.

Boring No.	Depth Groundwater Initially Encountered (feet, bgs)	Groundwater Depth after 15 Minutes (feet, bgs)
B-07	13.5	11.7
B-09	13.0	6.8
All other borings	Not Encountered	Not Encountered

Long-term Groundwater Monitoring. Piezometers PZ-01, PZ-02, and PZ-03 were installed at the location of the boring B-04, B-08 and B-12, respectively. The piezometer configuration is provided in table below.

Piezometer Location	Total Depth of Piezometer (feet, bgs)	Screen Length (feet, bgs)	Riser Length (feet, bgs)	Stickup Length (feet, above the ground surface)
PZ-01 (B-04)	20.0	10.0	10.0	2.0
PZ-02 (B-08)	20.0	10.0	10.0	2.0
PZ-03 (B-12)	20.0	10.0	10.0	2.0

Groundwater level measurements in the piezometer are presented below.

Piezometer	Date Installed	Date Measured	Groundwater Depth (feet, bgs)
PZ-01 (B-04)	8/27/2021	8/29/2021	Not Encountered
		9/3/2021	Not Encountered
		9/24/2021	Not Encountered
PZ-02 (B-08)	8/27/2021	8/29/2021	8.3
		9/3/2021	9.5
		9/24/2021	9.9
PZ-03 (B-12)	8/27/2021	8/29/2021	Not Encountered
		9/3/2021	Not Encountered
		9/24/2021	Not Encountered

Long-term monitoring can reveal groundwater levels materially different than those encountered during measurements taken while drilling the borings and in the piezometer. *The piezometer installed as part of this study should be monitored over a period of time to assess groundwater fluctuations.*

Groundwater Fluctuations. Future construction activities may alter the surface and subsurface drainage characteristics of this site. It is difficult to accurately predict the magnitude of subsurface water fluctuations that might occur based upon short-term observations. The groundwater level should be expected to fluctuate throughout the years with variations in precipitation.

5.0 ANALYSIS AND RECOMMENDATIONS

5.1 Seismic Site Classification

The seismic site classification is based on the 2018 International Building Code (IBC) and is a classification of the site based on the type of soils encountered at the site and their engineering properties. Per Table 20.3-1 of ASCE 7-10, the seismic site classification for this site is D.

5.2 Potential Vertical Soil Movements

Problem Discussion. Most clay soils, when subjected to increases in moisture content, swell. Swelling clay soils exert an outward pressure that can easily exceed 5,000 psf when subjected to moisture increases. Swell potential and swell pressures are a function of several factors including clay mineralogy and antecedent moisture condition. Generally, for a given clay soil, the drier the soil the greater its potential to swell and the higher its swell pressure. Conversely, wetter soils generally have a lower potential to swell and have lower swell pressures. The potential for a clay soil to swell is a variable and cannot be separated from its moisture condition.

The overburden pressure at a given depth above the groundwater table is calculated as the unit weight of the soil times the depth. For a soil with a unit weight of 125 pcf, the overburden pressure at 10-feet would be 1250 psf ($125 \text{ pcf} \times 10\text{-feet}$). As stated previously, clay soils can have a swell pressure that exceeds 5,000 psf. Thus, the swell pressure can exceed the overburden at depths of over 40-feet. This means soils at 40-feet exposed to changes in moisture can impact movements at the ground surface.

For a clay soil to swell or shrink, it must be subjected to increases or decreases in moisture content, respectively. The predominant way clay soils are subjected to increases or decreases in moisture content is the weather. As would be expected, extended periods of wet weather cause soil to get wetter and extended dry weather cause soil to get drier. The longer the period of wet or dry weather, the deeper the influence of the weather. Vegetation also causes variations in soil moisture content. Shallow rooted grass and bushes have a shallower impact, deep rooted trees have a deeper impact.

For a clay soil at a given depth to influence surface heave, two things must happen: (1) the soil must be subjected to an increase in moisture, and (2) the swell pressure of the soil must exceed the overburden pressure. Swell is typically calculated by assuming an “active” zone, a depth of soil impacted by weather which predominantly affects surface movements due to soil swell. Expansive soils below the active zone are typically ignored as they are assumed to

be exposed to lower increases in moisture, experience higher overburden pressures, and have a less significant impact on the surface heave than the soils in the active zone.

“Deep-seated” soil movement is swelling of the clay soils below the active zone and above the equilibrium depth. The equilibrium depth is the depth at which the overburden pressure and clay swell pressure are equal. Deep-seated soil movement is caused by changes in moisture that are typically not related to weather or vegetation. They can be caused by man-made influences such as leaking deep water or sewer lines. They can be caused by natural influences such as fluctuations in soil moisture content or groundwater levels. They are notoriously hard to accurately predict ahead of their actual occurrence. Unless stated otherwise, we have not included the effects of deep-seated soil movement in our Potential Vertical Rise (PVR) calculation. The inclusion of deep-seated soil movement drastically increases the depth of build pad preparation required and may make a slab-on-grade target PVR of 1-inch theoretically unattainable. The inclusion or exclusion of deep-seated soil movement is a matter of professional opinion, on which there is no consensus among consultants. It is a matter of risk tolerance and cost, of which, the user of this report is being made aware.

As evidenced in this discussion, calculation of PVR is based on soil data, model assumptions, experience, and professional judgment. PVR is a calculated estimate and should not be construed to be an absolute number or a guarantee of performance. PVR can be higher or lower depending on actual site conditions. The PVR estimate we provide is our best estimate of what will be encountered and the user of this report with doubts is encouraged to get another professional opinion prior to using this report. However, based on this discussion, the reader understands variations between the model and reality can introduce significant differences in calculated PVR. The user of this report understands and accepts this risk. If this risk is intolerable, the user of this report should be prepared to utilize a structural slab suspended adequately above the subgrade surface and supported on deep foundations.

Maintaining a consistent moisture content in the soil is the key to minimizing both heave and shrinkage related structural problems. Therefore, control of water is paramount in the performance of a slab-on-grade.

PVR or Equivalent Calculations. The PVR or its equivalent can be estimated several ways. GES utilizes the TxDOT method, swell tests, and a Volflo analysis to provide the best possible understanding of expected PVR and its variability.

Calculated PVR using TxDOT Method Tex-124-E. PVR calculations were performed in general accordance with the Texas Department of Transportation (TxDOT) Method Tex-124-E. The Tex-124-E method is empirical and is based on the Atterberg limits and moisture content of the subsurface soils. The calculated PVR is an empirical estimate of a soil’s potential for swell based upon the soil’s plasticity index, applied loading (due to structures or overburden), and antecedent moisture condition. The PVR calculated using TxDOT Method Tex-124-E is about 1-inch assuming an average antecedent moisture condition. The

calculated PVR is consistent with soil moisture conditions at the time this investigation was conducted. An 8-foot zone of seasonal moisture variation was used in our analysis based on local experience.

Calculated PVR using Swell Test Results. The equivalent PVR based on the swell test results is about 1-inch. The PVR based on swell test results is dependent on the moisture conditions at the time of testing. An 8-foot zone of seasonal moisture variation was used in our analysis based on local experience.

Calculated PVR using Volflo Analysis. The equivalent PVR based on the Volflo analysis results is about 1-inch. The PVR based on swell test results is dependent on the moisture conditions at the time of testing. An 8-foot zone of seasonal moisture variation was used in our analysis based on local experience.

5.3 Soluble Sulfates

A total of twenty (20) soluble sulfate tests (TEX-145-E) were performed on the near surface collected soil samples. Based on the results of laboratory testing, the soluble sulfate content measured on the samples tested ranged from 113 ppm to 840 ppm indicating low potential of soluble sulfates. It should be noted that concentrations of soluble sulfates in soil are typically very localized and concentrations in other areas of the site could vary significantly.

5.4 Construction Excavations

Applicability. Recommendations in this section apply to short-term construction-related excavations for this project.

Sloped Excavations. All sloped short-term construction excavations on-site should be designed in accordance with Occupational Safety and Health Administration (OSHA) excavation standards. Borings from this investigation indicated that the soils may be classified per OSHA regulations as Type C from the ground surface to a depth of 10-feet bgs. Short-term construction excavations may be constructed with a maximum slope of 1.5:1, horizontal to vertical (H:V), to a depth of 10-feet bgs. If excavations are to be deeper than 10-feet, we should be contacted to evaluate the excavation. Recommendations provided herein are not valid for any long-term or permanent slopes on-site.

Shored Excavations. As an alternative to sloped excavations, vertical short-term construction excavations may be used in conjunction with trench boxes or other shoring systems. Shoring systems should be designed using an equivalent fluid weight of 65 pounds per cubic foot (pcf) above the groundwater table and 95 pcf below the groundwater table. Surcharge pressures at the ground surface due to dead and live loads should be added to the

lateral earth pressures where they may occur. Lateral surcharge pressures should be assumed to act as a uniform pressure along the upper 10-feet of the excavation based on a lateral earth coefficient of 0.5. Surcharge loads set back behind the excavation at a horizontal distance equal to or greater than the excavation depth may be ignored. We recommend that no more than 200-feet of unshored excavation should be open at any one time to prevent the possibility of failure and excessive ground movement to occur. We also recommend that unshored excavations do not remain open for a period of time longer than 24-hours.

Limitations. Recommendations provided herein assume there are no nearby structures or other improvements which might be detrimentally affected by the construction excavation. Before proceeding, we should be contacted to evaluate construction excavations with the potential to affect nearby structures or other improvements.

Excavation Monitoring. Excavations should be monitored to confirm site soil conditions consistent with those encountered in the borings drilled as part of this study. Discrepancies in soil conditions should be brought to the attention of GES for review and revision of recommendations, as appropriate.

5.5 Groundwater Control

Groundwater was initially encountered at depths as shallow as 13-feet bgs in borings during drilling and rose to depths as shallow as 6.8-feet within 15-minutes. If groundwater is encountered during excavation, dewatering to bring the groundwater below the bottom of excavations may be required. Dewatering could consist of standard sump pits and pumping procedures, which may be adequate to control seepage on a local basis during excavation. Supplemental dewatering will be required in areas where standard sump pits and pumping is not effective. Supplemental dewatering could include submersible pumps in slotted casings, well points, or eductors. The contractor should submit a groundwater control plan, prepared by a licensed engineer experienced in that type of work.

5.6 Earthwork

5.6.1 Site Preparation

In the area of improvements, all concrete, trees, stumps, brush, debris, septic tanks, abandoned structures, roots, vegetation, rubbish and any other undesirable matter should be removed and properly disposed. All vegetation should be removed and the exposed surface should be scarified to an additional depth of at least 6 inches. It is the intent of these recommendations to provide a loose surface with no features that would tend to prevent uniform compaction by the equipment to be used.

5.6.2 Proofroll

The proof-roll should conform to TxDOT Standard Specifications for Construction and Maintenance of Highways, Streets, And Bridges, Item 216 - Proof Rolling.

5.6.3 Wet Weather/Soft Subgrade

Soft and/or wet surface soils may be encountered during construction, especially following periods of wet weather. Wet or soft surface soils can present difficulties for compaction and other construction equipment. If specified compaction cannot be achieved due to soft or wet surface soils, one of the following corrective measures will be required:

1. Removal of the wet and/or soft soil and replacement with select fill,
2. Chemical treatment of the wet and/or soft soil to improve the subgrade stability, or
3. If allowed by the schedule, drying by natural means.

Chemical treatment is usually the most effective way to improve soft and/or wet surface soils. GES should be contacted for additional recommendations if chemical treatment is planned due to wet and/or soft soils.

5.6.4 Fill

Fill. The fill should meet TxDOT Standard Specifications, Item 132, Type A material. Type A material should consist of granular soil (passing sieve no. 200 more than 50 percent) and should have a Liquid Limit less than or equal to 45 and a Plasticity Index between 10 and 25. Fill should be placed on prepared surfaces in lifts not to exceed 8 inches in loose measure. The fill should be compacted to at least 98 percent or more of the Tex-114-E maximum dry density at a moisture content recommended in TxDOT Standard Specifications, Item 132.3 - Embankment, Construction. Prior to any filling operation, samples of the proposed borrow materials should be obtained for laboratory moisture-density testing. The tests will provide a basis for evaluation of fill compaction by in-place density testing.

Fill Restrictions. Fill should consist of those materials meeting the requirements stated. Select fill and general fill should not contain material greater than 4-inches in any direction, debris, vegetation, waste material, environmentally contaminated material, or any other unsuitable material.

Unsuitable Materials. Materials considered unsuitable for use as fill include low and high plasticity silt (ML and MH), silty clay (CL-ML), organic clay and silt (OH and OL) and highly organic soils such as peat (Pt). These soils may be used for site grading and restoration in unimproved areas as approved by the Geotechnical Engineer. Soil placed in unimproved areas should be placed in loose lifts not exceeding 10-inches and should be compacted to at

least 92 percent maximum dry density (per ASTM D-698) and at a moisture content within ± 4 percentage points of optimum.

5.6.5 Construction Monitoring

Required Testing and Inspections. Construction monitoring services must be provided for all construction activities according to TxDOT specifications. We recommended that at least three compaction tests (i.e. field density and moisture content test) be performed for every 1,000 linear feet of roadway per lift, per day.

5.7 Demolition Considerations

Applicability. Recommendations in this section apply to the removal of any existing foundations, utilities or pavement which may be present on this site.

General. Special care should be taken in the demolition and removal of existing floor slabs, foundations, utilities and pavements to minimize disturbance of the subgrade. Excessive disturbance of the subgrade resulting from demolition activities can have serious detrimental effects on planned foundation and paving elements.

Existing Foundations. Existing foundations are typically slabs, shallow footings, or drilled piers. If slab or shallow footings are encountered, they should be completely removed. If drilled piers are encountered, they should be cut off at an elevation at least 24-inches below proposed grade beams or the final subgrade elevation, whichever is deeper. The remainder of the drilled pier should remain in place. Foundation elements to remain in place should be surveyed and superimposed on the proposed development plans to determine the potential for obstructions to the planned construction. GES should be contacted if drilled piers are to be excavated and removed completely. Additional earthwork activities will be required to make the site suitable for new construction if the piers are to be removed completely.

Existing Utilities. Existing utilities and bedding to be abandoned should be completely removed. Existing utilities and bedding may be abandoned in place if they do not interfere with planned development. Utilities which are abandoned in place should be properly pressure-grouted to completely fill the utility.

Backfill. Excavations resulting from the excavation of existing foundations and utilities should be backfilled in accordance with Section 5.6.4 - Fill.

Other Buried Structures. Other types of buried structures (wells, cisterns, etc.) could be located on the site. If encountered, GES should be contacted to address these types of structures on a case-by-case basis.

5.8 Loading on Buried Structures

Uplift. Buried structures are subjected to uplift forces caused by differential water levels adjacent to and within the structure. Soils with any appreciable silt or sand content will likely become saturated during periods of heavy rainfall and the effective static water level will be at the ground surface. For design purposes, we recommend the groundwater level be assumed at the ground surface. Resistance to uplift pressure is provided by soil skin friction and the dead weight of the structure. Skin friction should be neglected for the upper 3 feet of soil. A skin friction of 200 pounds per square foot (psf) may be used below a depth of 3 feet.

Lateral Pressure. Buried structures should be designed for at-rest lateral earth pressures as provided in table below.

Lateral Earth Pressures		
Condition	Equivalent Fluid Pressure (pcf)	
	Above Assumed Groundwater Table	Below Assumed Groundwater Table
At-Rest (k_o)	75	100

The above lateral pressures include hydrostatic pressure but do not include surcharge loads. Groundwater levels should be assumed to be at the 100-year flood plain elevation. The lateral load produced by a surcharge may be computed as 50 percent of the vertical surcharge pressure applied as a constant pressure over the full depth of the buried structure. Surcharge loads set back behind the retaining structure at a horizontal distance equal to or greater than the structure height may be ignored.

Vertical Pressure. Vertical pressures on buried structures due to soil loading can be determined using an equivalent fluid weight of 125 pcf. This does not include surcharge loads. The vertical load produced by a surcharge may be computed as 100 percent of the vertical surcharge pressure applied as a constant pressure over the full width of the buried structure.

5.9 Buried Pipe

Applicability. Recommendations in this section are applicable to the design of buried piping placed by open cut methods associated with this project.

Pressure on Buried Pipe. Design recommendations provided in the "Loading on Buried Structures" section of this report apply to buried piping.

Thrust Restraints. Resistance to lateral forces at thrust blocks will be developed by friction developed along the base of the thrust block and passive earth pressure acting on the vertical face of the block. We recommend a coefficient of base friction of 0.25 (using a Factor of Safety of 2) along the base of the thrust block. Passive resistance on the vertical face of the thrust block may be calculated using the allowable passive earth pressures presented in the following table.

Allowable Passive Earth Pressure by Material Type	
Material	Allowable Passive Pressure (psf)
Sand	100 x Depth in Feet
Native Clay and Clayey Sand	2,000
Compacted Clay Fill	1,500
<u>Note:</u> Passive resistance should be neglected for any portion of the thrust block within 3 feet of the final site grade. The allowable passive resistance for native clays and clayey sand is based on the thrust block bearing directly against vertical, undisturbed cuts in these materials.	

Bedding and Backfill. Pipe bedding and pipe-zone backfill for the water and sanitary sewer piping should be in accordance with TxDOT standard specification Item 400.

Trench Backfill. Trench backfill should be in accordance with TxDOT standard specification Item 400.

Trench Settlement. Settlement of backfill should be anticipated. Even for properly compacted backfill, fills in excess of 8 to 10 feet are still subject to settlements over time of about 1 to 2 percent of the total fill thickness. This level of settlement can be significant for fills beneath streets. Therefore, close coordination and monitoring should be performed to reduce the potential for future movement.

5.10 Culverts

Pressure on Culverts. Design recommendations provided in the “Loading on Buried Structures” section of this report apply to buried culverts associated with this project.

Bedding and Backfill. Culverts bedding and backfill should be in accordance with TxDOT standard specification Item 400 - Excavation and Backfill for Structures.

Bearing Capacity. We understand that the box culverts may be supported on a seal slab foundation. Seal slab foundation can be proportioned using a net dead load plus sustained live load bearing pressure of 1,000 psf or a net total load bearing pressure of 1,500 psf, whichever condition results in a larger bearing surface.

Erosion Protection. Erosion protection recommendations are provided in Section 6.9 of this report.

5.11 Retaining Structures

Applicability. Recommendations provided in this section are applicable to retaining structures (e.g. headwalls and wingwalls). *It is imperative that global stability be reviewed by GES on any retaining structure in excess of 5- feet in height.*

Earth Pressure. Retaining structure walls should be designed for earth pressures at least to those indicated in this section. Earth pressures will be influenced by the structural design of the walls, conditions of wall restraint, methods of construction and/or compaction, and the strength of the materials being restrained. Two wall restraint conditions are considered. Active earth pressure is commonly used for design of free-standing cantilever retaining walls and assumes wall movement. The at-rest condition assumes no wall movement. The recommended design lateral earth pressures do not include a factor of safety for possible hydrostatic pressure on the walls.

Equivalent Fluid Density - Horizontal Ground Surface Extending Backward from the Top of the Wall				
Type of Backfill	Active Pressure (pcf) ^{(1) & (2)}		At-Rest Pressure (pcf) ^{(1) & (2)}	
	Above Water Table and Free Draining	Below Water Table or Not Free Draining	Above Water Table and Free Draining	Below Water Table or Not Free Draining
Onsite Clayey Soil Unit Weight 125 pcf and Friction Angle of 18° $k_a = 0.53$; $k_o = 0.69$; $k_p = 1.9$	66	95	87	106
Onsite Clayey Sand Unit Weight 125 pcf and Friction Angle of 25° $k_a = 0.4$; $k_o = 0.58$; $k_p = 2.5$	51	88	72	99
Onsite Silty Sand Unit Weight 115 pcf and Friction Angle of 30° $k_a = 0.33$; $k_o = 0.5$; $k_p = 3$	38	80	58	89
Granular Fill Unit Weight 115 pcf and Friction Angle of 32° $k_a = 0.31$; $k_o = 0.47$; $k_p = 3.25$	36	79	55	87
Notes: 1. The active earth pressure assumes that the top of the walls is not rigidly restrained, and the walls are not sensitive to some movement. The at-rest condition assumes the walls are restrained at top. 2. k_a = Coefficient of Active Earth Pressure; k_o = Coefficient of At-Rest Earth Pressure, k_p = Coefficient of Passive Earth Pressure.				

Applicable Conditions. For the above earth pressures to be valid, the following conditions apply:

- The equivalent fluid pressure values assume a horizontal ground surface extending backward from the top of the wall;
- For active earth pressure, the wall must rotate about the base, with top lateral movements of about 0.002H to 0.004H, where H is the wall height;
- For passive pressure to develop, the wall must move horizontally to mobilize resistance;
- Horizontal backfill is compacted between 95 and 98 percent of standard Proctor maximum dry density;
- Loading from heavy compaction equipment not included;
- No safety factor is included in the soil parameters.
- Appropriate safety factor should be utilized by the structural engineer for external stability analyses of the retaining structures.
- The above lateral pressures include hydrostatic pressure but do not include surcharge loads.
- The lateral load produced by a surcharge may be computed as 50 percent of the vertical surcharge pressure applied as a constant pressure over the full depth of retaining wall.
- Surcharge loads set back from the back face of the retaining wall at a horizontal distance equal to or greater than the pile height may be ignored.

Backfill. Backfill placed against retaining structures may be granular fill, onsite sandy or clayey soils. For the granular values to be valid, the granular backfill must extend out from the base of the wall at an angle of at least 45 degrees from vertical. Granular backfill should consist of a clean, non-plastic, relatively well-graded granular soil consisting of either a sand, or a sand and gravel mixture, with less than 5 percent finer than the No. 200 sieve size. To reduce surface water seepage into the free draining backfill, the top 1-ft of the backfill should consist of on-site clay type soil with a plasticity index of at least 15.

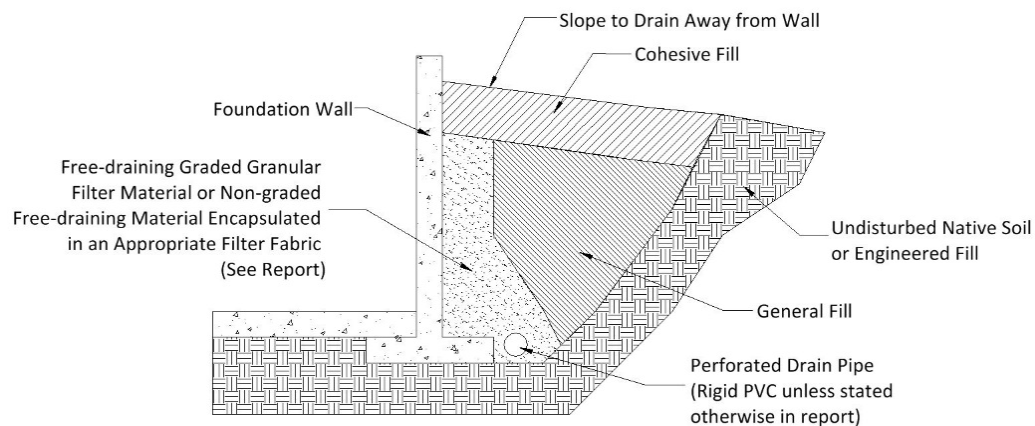
Passive Resistance. We recommend the following values for passive earth pressure (resistance) of adjacent soil in front of the wall.

Soil Type	Passive Resistance as Equivalent Fluid Density (pcf) ⁽¹⁾	
	Above Water Table and Free Draining	Below Water Table or Not Free Draining
Onsite Clayey Soil	237	181
Onsite Clayey Sand	308	217
Onsite Silty Sand	345	220
Granular Soils	374	233
Note: 1. Based on a Factor of Safety of 1.		

Sliding Resistance. We recommend the following sliding resistance values between the wall footing and underlying soil.

Material	Coefficient of Sliding Friction ⁽¹⁾
Onsite Clayey Soil	0.3
Onsite Clayey Sand	0.46
Onsite Silty Sand	0.57
Granular Soils	0.62
<u>Note:</u> 1. Based on a Factor of Safety of 1.	

Drainage. A perforated drain line installed behind the base of the walls that extend below adjacent grade is recommended to prevent hydrostatic loading on the walls as shown on the diagram below. The invert of the drain line around a below-grade exterior retaining wall should be placed near the foundation bearing level. The drain line should be sloped to provide positive gravity drainage to a sump pit and pump. The drain line should be surrounded by clean, free-draining granular material have less than 5 percent passing the No. 200 sieve. The free-draining aggregate should be encapsulated in a filter fabric. The granular fill should extend to within 2 feet of final grade, where it should be capped with cohesive fill to reduce infiltration of surface water into the drain system.



Alternative Drainage. As an alternative to free-draining granular fill, a prefabricated drainage system may be used. A prefabricated drainage system may be a plastic drainage core or mesh system which is covered with a filter fabric to prevent soil intrusion and is fastened to the wall prior to placing backfill.

Bearing Capacity. Assuming a minimum embedment depth of 24-inches, an allowable bearing capacity of 1,500 psf may be used for retaining structure footings (using a Factor of Safety of 3).

5.12 Embankment

Provided cross-sections by the client are presented in Appendix G - Cross Sections. Based on the provided cross-sections, we understand the critical embankment (highest and steepest) will be at station 33+00. Therefore, the slope stability analyses were performed for embankment slope at station 33+00. The embankment at station 33+00 will be approximately 13.53-feet in height with side slope of 3H:1V (horizontal:vertical). The highest and lowest elevation of the embankment at station 33+00 is 337.90-feet and 324.37-feet, respectively. Embankment fill should meet the requirement provided in section 5.6.4-Fill.

5.12.1 Embankment Slope Stability Analysis

General. The static stability analysis for the critical embankment slope was performed using GeoStudio 2019[®] software. Factors of safety was calculated using the Morgenstern-Price method, which satisfies both force and moment equilibrium.

Recommended Safety Factors. Based on our experience and TxDOT guidelines, the following safety factors were used in slope stability analyses:

Slope Stability Condition	Recommended Safety Factor
Short-Term Condition	1.3
Long-Term Condition	1.5

The slope stability analyses were performed with a surcharge load of 250 psf for long-term condition and 500 psf for short-term condition. Based on the provided information by the client, we understand the project is located out of the flood zone and will not be subject to rapid drawdown condition. In our analysis, we conservatively assumed the water table at the toe of the slope (elevation 324.37 at station 33+00).

Soil Properties for Slope Stability Analysis. Based on results of consolidated undrained (CU) and unconfined compressive shear strength tests, soil borings and our engineering judgment, we used the following shear strength parameters for slope stability of the critical embankment slope.

Soil Parameters for Slope Stability Analysis			
Soil Type	Total Unit Weight (pcf)	Angle of Internal Friction (degrees)	Cohesion (psf)
Lean to Fat Clay - Long Term	125	18	150
Lean to Fat Clay - Short Term	125	0	800
Clayey Sand - Long Term & Short Term	125	25	50
TxDOT Item 132 Type A Fill - Long Term & Short Term	125	27	50

Results of Slope Stability Analyses. Based on subsurface condition encountered, slope stability analyses were performed for two different native soil types (i.e., lean to fat clay and clayey sand) under the embankment fill. Results of our analyses are presented in Appendix H - Slope Stability Analysis Results and discussed in table below:

Results of Slope Stability Analyses for Lean to Fat Clay Native Soil		
Condition Analyzed	Surcharge (psf)	Minimum Safety Factor
Short-Term	500	1.89
Long-Term	250	1.74

Results of Slope Stability Analyses for Clayey Sand Native Soil		
Condition Analyzed	Surcharge (psf)	Minimum Safety Factor
Short-Term	500	1.70
Long-Term	250	1.78

Based on the results of our slope stability analyses, the critical embankment slope exceed the minimum safety factor requirements. Therefore, the embankment slopes should be stable with regards to slope failure with a maximum surcharge of 250 psf and 500 psf at the top of bank for long term and short term conditions, respectively. If the construction surcharge load is planned to exceed 500 psf, GES should be contacted for additional recommendations.

5.12.2 Embankment Settlement Analysis

General. We assumed Type A fill material per section 5.6.4 will be used as embankment fill on top of the native soil. Based on our experience, it is anticipated that properly compacted embankment fill material (Type A) will settle about 0.5 percent to 1 percent of the fill height. We believe that settlement of the fill material placed above the native soil will be substantially complete during construction.

Native Soil Settlement Under the Embankment Surcharge. Based on the subsurface conditions encountered in the borings and consolidation test results, we estimate the native soil under the embankment may be subjected to total settlements as discussed in the following

Native Soil Settlement under the Proposed Embankment	
Embankment Height 'H' (Feet) ⁽¹⁾	Estimated Total Settlement (inches) ⁽²⁾
H = 4.0	1 to 1.5
H = 9.0	1.5 to 2.0
H = 13.5	2.0 to 2.5

GES performed time-rate settlement analysis and the results are presented in the following tables.

Time Rate-of Native Soil Settlement under the Proposed Embankment	
Degree of Consolidation (%)	Time after Embankment Construction
30	1-month or less
60	1 to 4-months
90	6-months to 1.5-year

Based on our settlement analysis, about 60 percent of the total settlement will occur within 4 months and 90 percent of the total settlement will occur within 1.5 years after construction of the embankment.

6.0 GENERAL COMMENTS

Data Assumptions. By necessity, geotechnical engineering design recommendations are based on a limited amount of information about subsurface conditions. In the analysis, the geotechnical engineer must assume subsurface conditions are similar to those encountered in the borings. The analyses, conclusions and recommendations contained in this report are based on site conditions as they existed at the time of the field investigation and on the assumption that the exploratory borings are representative of the subsurface conditions throughout the site; that is, the subsurface conditions everywhere are not significantly different from those disclosed by the borings at the time they were completed.

Subsurface Anomalies. Anomalies in subsurface conditions are often revealed during construction. If during construction, different subsurface conditions from those encountered in our borings are observed, or appear to be present in excavations, we must be advised promptly so that we can review these conditions and reconsider our recommendations where necessary.

Change of Conditions. If there is a substantial lapse of time between submission of this report and the start of the work at the site, if conditions have changed due either to natural causes or to construction operations at or adjacent to the site, or if structure locations, structural loads or finish grades are changed, we should be promptly informed and retained

to review our report to determine the applicability of the conclusions and recommendations, considering the changed conditions and/or time lapse.

Design Review. GES, Inc. should be retained to review those portions of the plans and specifications for this particular project that pertain to earthwork and foundations as a means to determine whether the plans and specifications are consistent with the recommendations contained in this report.

Construction Materials Testing and Inspection. GES should be retained to observe earthwork and foundation installation and perform materials evaluation and testing during the construction phase of the project. This enables GES's geotechnical engineer to stay abreast of the project and to be readily available to evaluate unanticipated conditions, to conduct additional tests if required and, when necessary, to recommend alternative solutions to unanticipated conditions. It is proposed that construction phase observation and materials testing commence by the project geotechnical engineer (GES) at the outset of the project. Experience has shown that the most suitable method for procuring these services is for the owner to contact directly with the project geotechnical engineer. This results in a clear, direct line of communication between the owner and the owner's design engineers and the geotechnical engineer.

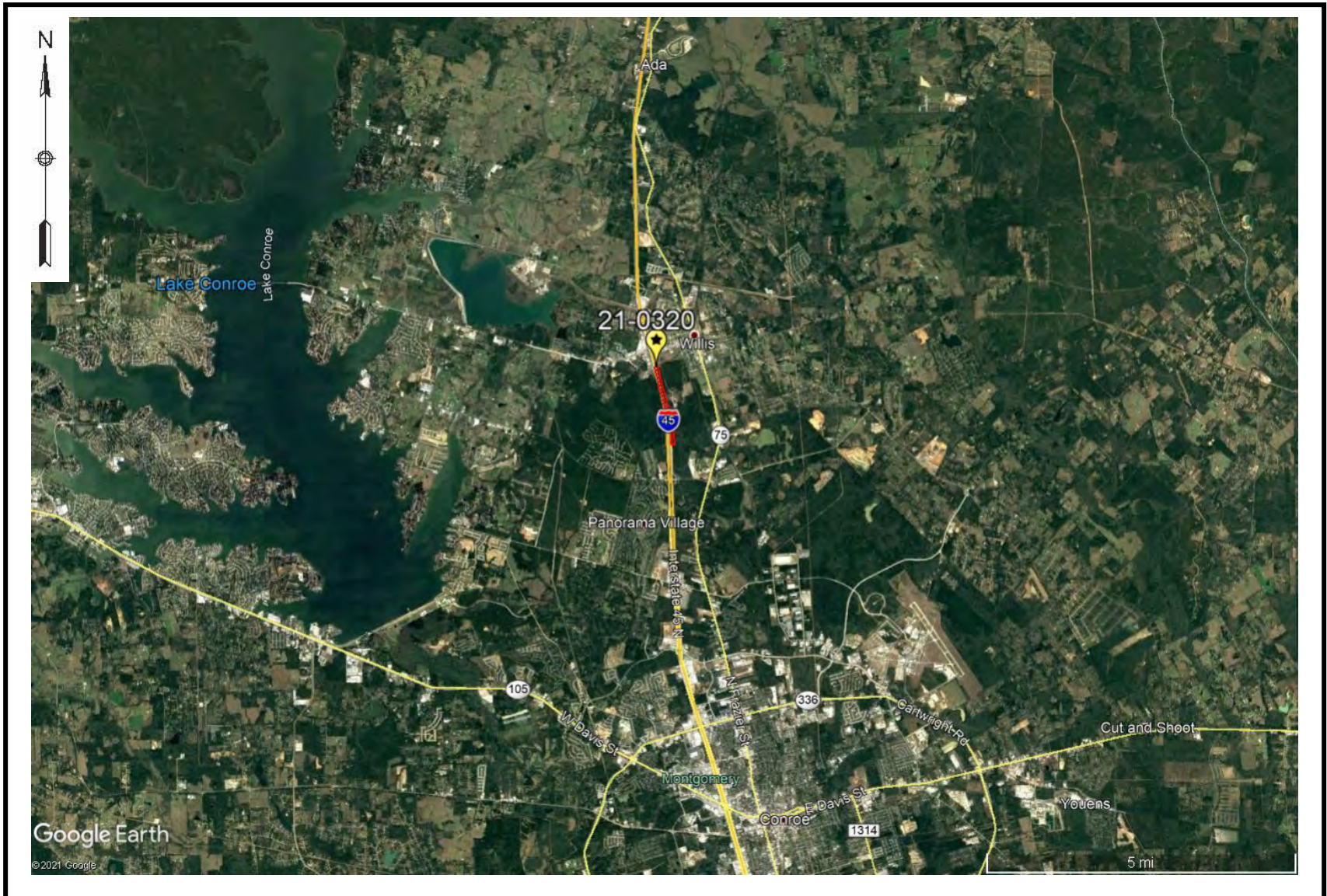
Report Recommendations are Preliminary. Until the recommended construction phase services are performed by GES, the recommendations contained in this report on such items as final foundation bearing elevations, final depth of undercut of expansive soils for non-expansive earth fill pads and other such subsurface-related recommendations should be considered as preliminary.

Liability Limitation. GES cannot assume responsibility or liability for recommendations provided in this report if construction inspection and/or testing recommended herein is performed by another party.

Warranty. This report has been prepared for the exclusive use of the Client and their designated agents for specific application to design of this project. We have used that degree of care and skill ordinarily exercised under similar conditions by reputable members of our profession practicing in the same or similar locality. No other warranty, expressed or implied, is made or intended.

Appendix A - Project Location Diagrams

PROJECT LOCATION DIAGRAM - GENERAL

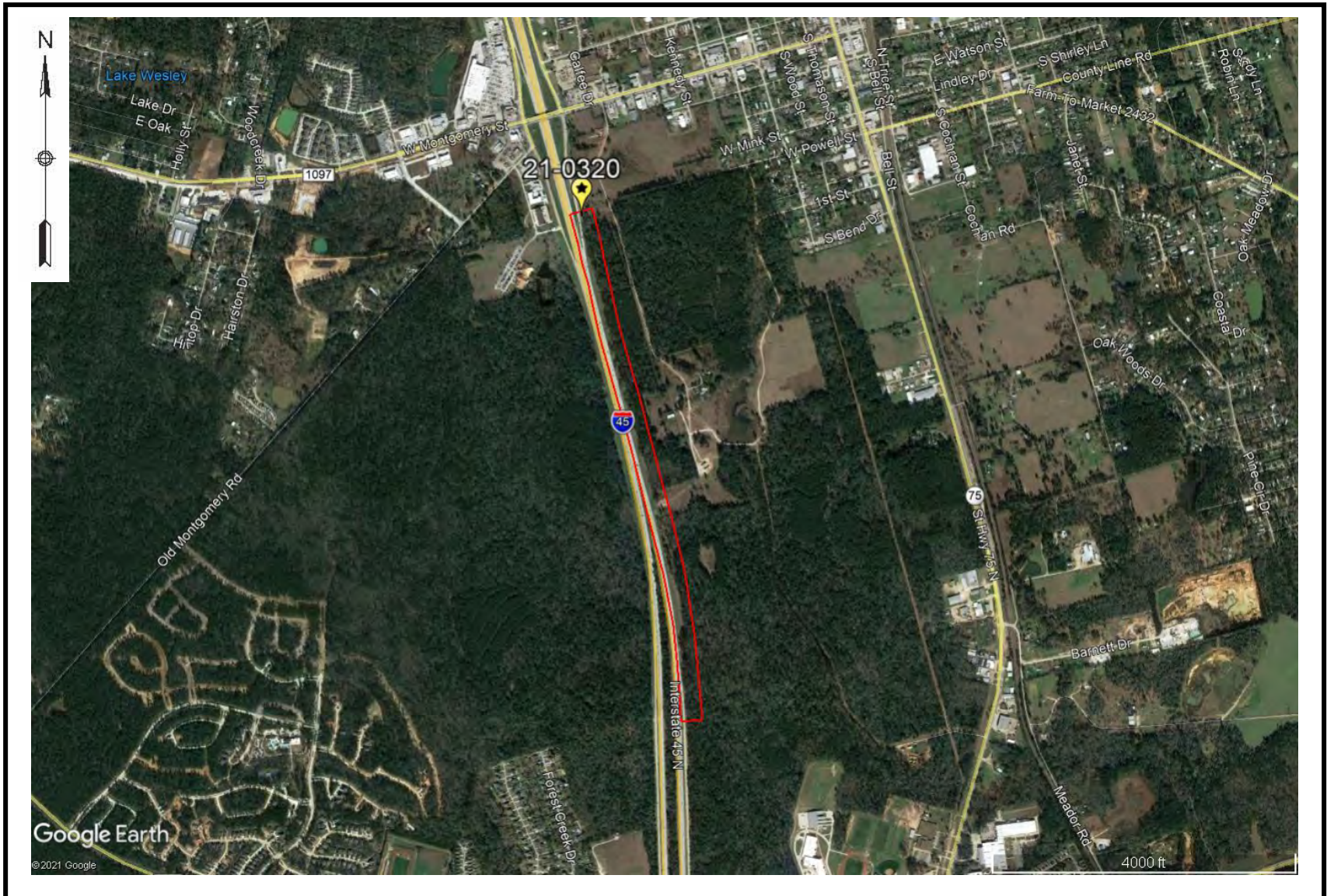


GES Project No. 21-0320

I-45 Frontage Road Extension
CSJ No. 0675-08-114



PROJECT LOCATION DIAGRAM - LOCAL



GES Project No. 21-0320

I-45 Frontage Road Extension
CSJ No. 0675-08-114



Appendix B - Boring Location Diagram

BORING LOCATION DIAGRAM



GES Project No. 21-0320

I-45 Frontage Road Extension
CSJ No. 0675-08-114



Appendix C - Boring Logs and Laboratory Results

BORING LOGS



DRILLING LOG

1 of 1

WinCore
Version 3.3

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-01
Structure Storm Sewer
Station 21+69.59
Offset 27.27R

District Houston
Date 8/24/21
Grnd. Elev. 340.10 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
338.1			SAND, clayey, light gray and light brown. (SC)			11	36	22		PP = 4.50, -#200 = 48%
			CLAY, sandy, stiff, light gray and reddish brown, with iron nodules. (CL)			15				PP = 4.50+
5		15 (6) 12 (6)				17	46	29		PP = 4.50, -#200 = 52%
334.1			CLAY, sandy, stiff, light gray and reddish brown, slickensided. (CH)			23				PP = 4.50
10		18 (6) 22 (6)		0	33	25			125.1	PP = 4.50
325.1		16 (6) 18 (6)				22	58	40		PP = 2.50, -#200 = 66%
			SAND, silty, clayey, compact, light brown. (SC-SM)							
320.1		23 (6) 29 (6)				6	19	6		N = 10-11-12(23), -#200 = 26%

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10142504.81 & Easting: 3824278.98.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



DRILLING LOG

1 of 1

WinCore
Version 3.3

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-02
Structure Storm Sewer
Station 27+03.03
Offset 1.41L

District Houston
Date 8/24/21
Grnd. Elev. 345.98 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
342.			SAND, silty, clayey, gray and reddish brown. (SC-SM)			10				PP = 2.50
						8	16	7		PP = 2.50, -#200 = 26%
		19 (6) 20 (6)	SAND, clayey, slightly compact to compact, light gray and reddish brown. (SC)			13				PP = 4.50
						13	29	17		PP = 3.00, -#200 = 39%
5		20 (6) 22 (6)		0	27	15			132.6	PP = 4.50
10										
15		31 (6) 28 (6)				15	33	18		PP = 4.50, -#200 = 38%
331.			CLAY, stiff, light gray and reddish brown, slickensided. (CH)							
20		14 (6) 18 (6)				31	101	69		PP = 3.50, -#200 = 95%
326.										

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10143035.54 & Easting: 3824217.89.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



WinCore
Version 3.3

DRILLING LOG

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County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-03
Structure Storm Sewer
Station 31+41.80
Offset 27.97R

District Houston
Date 8/24/21
Grnd. Elev. 331.92 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
329.9			SAND, silty, light gray and light brown. (SM)							
						9				PP = 1.00, #200 = 21%
			SAND, clayey, compact, light gray and light brown. (SC)							
						14	25	12		PP = 4.00, #200 = 43%
						14	27	10		PP = 3.00, #200 = 32%
5		23 (6) 20 (6)								
321.9										
10		23 (6) 24 (6)								
			SAND, silty, slightly compact to compact, light brown. (SM)							
15		18 (6) 22 (6)								
311.9										
20		23 (6) 26 (6)								
						6				N = 6-7-7(14)

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10143475.71 & Easting: 3824201.49.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



DRILLING LOG

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WinCore
Version 3.3

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-04
Structure Storm Sewer
Station 34+44.04
Offset 41.52R

District Houston
Date 8/24/21
Grnd. Elev. 329.38 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
327.4			SAND, clayey, gray and brown. (SC)			13	34	21		PP = 3.50, -#200 = 48%
325.4			SILT, sandy, brown and gray, with clay pockets. (ML)			14				PP = 1.50, -#200 = 60%
5		6 (6) 6 (6)	CLAY, sandy, soft, light gray and light brown. (CL)			10				
321.4						16	46	31		PP = 4.50, -#200 = 60%
319.4		19 (6) 18 (6)	SAND, clayey, slightly compact, light gray and light brown. (SC)			14	23	12		PP = 2.50, -#200 = 26%
309.4			SAND, silty, compact, brown, with gravel. (SM)							
15		23 (6) 21 (6)				11				N = 9-12-12(24), -#200 = 16%
309.4		24 (6) 33 (6)				9				N = 8-13-13(26)

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10143777.83 & Easting: 3824168.19.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



DRILLING LOG

1 of 1

WinCore
Version 3.3

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-05
Structure Storm Sewer
Station 38+30.20
Offset 20.78R

District Houston
Date 8/24/21
Grnd. Elev. 340.46 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
5		23 (6) 22 (6)	CLAY, sandy, very stiff, reddish brown and brown. (CL)			29	43	24		PP = 4.50, -#200 = 53%
						31				PP = 4.50
						11				PP = 4.50+
						20				PP = 4.50, -#200 = 64%
332.5			CLAY, stiff to very stiff, light gray and reddish brown. (CH)							
10		14 (6) 10 (6)		0	61	18			128.3	PP = 4.50
325.5		30 (6) 24 (6)	SAND, poorly graded with silt, compact, light brown. (SP-SM)			29	87	59		PP = 4.50, -#200 = 96%
320.5		37 (6) 32 (6)				4				N = 13-12-14(26), -#200 = 7%

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10144153.61 & Easting: 3824070.39.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



WinCore
Version 3.3

DRILLING LOG

1 of 1

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-06
Structure Storm Sewer
Station 44+73.47
Offset 13.49R

District Houston
Date 8/24/21
Grnd. Elev. 338.78 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
334.8	5	26 (6) 44 (6)	SAND, silty, gray and brown. (SM)			25				PP = 1.00, -#200 = 19%
						7				
		50 (4.5) 50 (4.5)	SAND, clayey, compact to dense, light brown, with clay pockets. (SC)			8				-#200 = 24%
						7				
328.8	10	50 (4.5) 50 (4.5)	SAND, silty, clayey, dense, light brown, with gravel. (SC-SM)			10	28	12		N = 16-20-20(40)
		50 (4.5) 50 (3)								N = 17-21-32(53), -#200 = 30%
318.8	15	50 (4.5) 50 (3)	SAND, silty, clayey, dense, light brown, with gravel. (SC-SM)			8	21	7		N = 26-28-35(63), -#200 = 17%
	50 (5) 50 (5)								N = 19-20-24(44), -#200 = 15%	
318.8	20	50 (5) 50 (5)				10	19	6		

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10144772.66 & Easting: 3823890.68.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



DRILLING LOG

1 of 1

WinCore
Version 3.3

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-07
Structure Storm Sewer
Station 49+63.91
Offset 2.36L

District Houston
Date 8/25/21
Grnd. Elev. 335.26 ft
GW Elev. 321.76 ft

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
333.3			SAND, clayey, gray and brown. (SC)							
						11	21	10		PP = 1.50, #200 = 38%
			SAND, silty, slightly compact to compact, light gray. (SM)							
						6				N = 3-4-4(8)
						6				N = 4-5-7(12), #200 = 18%
5		12 (6) 13 (6)								
325.3										
						10				N = 5-7-7(14)
						8				N = 5-6-8(14)
			SAND, clayey, slightly compact to compact, reddish brown. (SC)							
10		28 (6) 37 (6)								
315.3										
						11	20	10		N = 12-16-23(39), #200 = 21%
						18				N = 9-9-9(18)
15		28 (6) 26 (6)								
315.3		13 (6) 18 (6)								
20										

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10145232.90 & Easting: 3823720.21.

Any ground water elevation information provided on this boring log is representative of conditions existing on the day and for the specific location where this information was collected. The actual groundwater elevation may fluctuate due to time, climatic conditions, and/or construction activity.

Driller: GES

Logger: H.S.

Organization: GES



DRILLING LOG

1 of 1

WinCore
Version 3.3

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-08
Structure Storm Sewer
Station 51+67.90
Offset 16.84L

District Houston
Date 8/25/21
Grnd. Elev. 333.94 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
5		34 (6) 50 (5)	SAND, silty, dense, gray and brown. (SM)			7				PP = 2.00
						7				PP = 1.50, -#200 = 27%
						9				PP = 1.00
327.9			SAND, clayey, compact, light gray and light brown. (SC)			8	22	11		N = 20-22-22(44), -#200 = 18%
323.9	10	35 (6) 36 (6)				8				N = 18-21-24(45)
			SAND, silty, clayey, slightly compact, light gray and light brown. (SC-SM)							
318.9	15	16 (6) 14 (6)				11	19	7		PP = 2.00, -#200 = 20%
			SAND, silty, slightly compact, reddish brown. (SM)							
313.9	20	16 (6) 17 (6)				15				N = 5-9-20(29), -#200 = 17%

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10,145,421.75 & Easting: 3823641.75.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



DRILLING LOG

1 of 1

WinCore
Version 3.3

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-09
Structure Storm Sewer
Station 55+83.00
Offset 3.17R

District Houston
Date 8/25/21
Grnd. Elev. 337.71 ft
GW Elev. 324.71 ft

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
5		13 (6) 15 (6)	SAND, silty, slightly compact, light gray and light brown, with clay pockets. (SM)			9				#200 = 23%
						5				N = 3-4-4(8)
						5				N = 2-3-5(8), #200 = 18%
327.7 10		16 (6) 22 (6)	SAND, silty, slightly compact, light gray and light brown, with clay pockets. (SM)							
						13				
										N = 5-7-8(15)
						10				N = 6-7-9(16), #200 = 15%
15		46 (6) 38 (6)	SAND, clayey, slightly compact to dense, reddish brown, with gravel. (SC)							
						8	23	11		
										N = 19-17-16(33), #200 = 18%
317.7 20		16 (6) 19 (6)				16				N = 8-11-15(26)

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10145821.75 & Easting: 3823529.00.

Any ground water elevation information provided on this boring log is representative of conditions existing on the day and for the specific location where this information was collected. The actual groundwater elevation may fluctuate due to time, climatic conditions, and/or construction activity.

Driller: GES

Logger: H.S.

Organization: GES



DRILLING LOG

1 of 1

WinCore
Version 3.3

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-10
Structure Storm Sewer
Station 60+60.9
Offset 7.34R

District Houston
Date 8/25/21
Grnd. Elev. 354.12 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
352.1			SAND, silty, light gray and light brown. (SM)			7				-#200 = 21%
			SAND, clayey, compact to dense, light gray and reddish brown. (SC)			21				PP = 4.50+
5		30 (6) 25 (6)				23				PP = 4.00
						12	30	14		PP = 4.50+, -#200 = 40%
10		26 (6) 28 (6)				12				PP = 4.00
15		50 (4.5) 50 (3.25)				11	27	10		PP = 4.50, -#200 = 29%
334.1 20		50 (4) 50 (3)				11	25	10		N = 23-28-36(64), -#200 = 20%

Remarks: Auger 0- to 30-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10146276.40 & Easting: 3823382.02.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



WinCore
Version 3.3

DRILLING LOG

1 of 1

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-11
Structure Storm Sewer
Station 64+98.09
Offset 15.38R

District Houston
Date 8/25/21
Grnd. Elev. 359.51 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
5		8 (6) 10 (6)	SAND, silty, loose, light gray and light brown. (SM)			7				N = 3-5-5(10), #200 = 20% N = 4-5-7(12)
						13				
						12				
353.5			CLAY, sandy, stiff, light gray and reddish brown, with iron nodules. (CL)			15				PP = 4.50
349.5	10	20 (6) 20 (6)	SAND, clayey, slightly compact to compact, reddish brown, with iron nodules. (SC)			14	36	20		PP = 4.50, #200 = 52%
15		18 (6) 22 (6)				14	33	18		PP = 3.50, #200 = 44%
339.5	20	20 (6) 22 (6)				14				PP = 1.50

Remarks: Auger 0- to 30-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10146696.05 & Easting: 3823259.32.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



WinCore
Version 3.3

DRILLING LOG

1 of 1

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-12
Structure Storm Sewer
Station 69+72.11
Offset 26.41R

District Houston
Date 8/25/21
Grnd. Elev. 365.40 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
363.4			SAND, silty, gray. (SM)			8				-#200 = 23%
			CLAY, sandy, very stiff to hard, light gray and reddish brown, with iron nodules. (CL)			21				PP = 4.50+
5		40 (6) 40 (6)				22				PP = 4.50+
						13	42	24		PP = 4.00, -#200 = 53%
355.4	10	38 (6) 46 (6)				14				PP = 4.50+
			SAND, clayey, dense, reddish brown, with iron nodules. (SC)							
15		50 (5) 50 (3)				10	29	14		PP = 4.50+, -#200 = 32%
345.4	20	50 (3.5) 50 (2)				18				N = 20-25-30(55)

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10147151.82 & Easting: 3823128.60.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



DRILLING LOG

1 of 1

WinCore
Version 3.3

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-13
Structure Storm Sewer
Station 74+80.63
Offset 8.08L

District Houston
Date 8/26/21
Grnd. Elev. 368.29 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
364.3 5		15 (6) 10 (6)	SAND, silty, gray, light brown, with calcareous nodules. (SM)			9				PP = 1.50
						9				PP = 2.00, -#200 = 31%
			SAND, clayey, slightly compact, light gray and light brown, with iron nodules. (SC)			12				PP = 3.50
360.3						10	27	14		-#200 = 48%
358.3		19 (6) 21 (6)	CLAY, sandy, stiff, light gray and reddish brown, with iron nodules. (CL)			13	44	25		PP = 4.50+, -#200 = 60%
353.3		24 (6) 22 (6)	CLAY, very stiff, reddish brown. (CH)			20	51	35		PP = 4.50+, -#200 = 86%
348.3		50 (2.5) 50 (2)	CLAY, with sand, very hard, reddish brown. (CL)			8	29	15		PP = 1.00, -#200 = 71%

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10147626.96 & Easting: 3822944.14.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



DRILLING LOG

1 of 1

WinCore
Version 3.3

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-14
Structure Storm Sewer
Station 79++94.96
Offset 4.56R

District Houston
Date 8/26/21
Grnd. Elev. 368.50 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
5		4 (6) 4 (6)	SAND, clayey, very loose, light gray and light brown, with gravel and calcareous nodules. (SC)			8	28	15		PP = 4.50, #200 = 32%
						14				PP = 1.00
						10				PP = 4.50
358.5 10		3 (6) 3 (6)	SAND, clayey, very loose, light gray and light brown, with gravel and calcareous nodules. (SC)			16				PP = 2.50
						10	19	8		PP = 2.50, #200 = 31%
15		19 (6) 28 (6)	CLAY, sandy, stiff to very stiff, light gray and reddish brown. (CL)			20	47	30		PP = 1.50, #200 = 54%
348.5 20		12 (6) 12 (6)		0	13	15			131.8	PP = 4.50+

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10148121.89 & Easting: 3822803.86.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES



DRILLING LOG

1 of 1

WinCore
Version 3.3

County MONTGOMERY
Highway I-45
CSJ 0675-08-114

Hole B-15
Structure Storm Sewer
Station 84+82.60
Offset 11.28R

District Houston
Date 8/26/21
Grnd. Elev. 369.95 ft
GW Elev. N/A

Elev. (ft)	LOG	Texas Cone Penetrometer	Strata Description	Triaxial Test		Properties				Additional Remarks
				Lateral Press. (psi)	Deviator Stress (psi)	MC	LL	PI	Wet Den. (pcf)	
368.			SAND, clayey, gray and brown, with calcareous nodules. (SC)			7	24	12		PP = 1.50, -#200 = 35%
			SAND, silty, loose, gray. (SM)							
						9				N = 3-4-4(8), -#200 = 31%
5		5 (6) 6 (6)				10				N = 4-5-4(9)
364.			SAND, clayey, compact, light gray and light brown, with gravel and iron nodules. (SC)							
						13	34	20		PP = 4.50, -#200 = 40%
360.	10	28 (6) 30 (6)				14	30	14		PP = 3.00, -#200 = 37%
			CLAY, sandy, very stiff, light brown and reddish brown, with iron nodules. (CL)							
355.	15	16 (6) 33 (6)				19	44	29		PP = 4.50+, -#200 = 53%
			SAND, clayey, compact, reddish brown. (SC)							
350.	20	33 (6) 34 (6)				14	30	19		PP = 2.00, -#200 = 27%

Remarks: Auger 0- to 20-feet. SPT tests were performed with a 170-lb hammer dropped 24-inches. Northing: 10148592.94 & Easting: 3822677.78.

The ground water elevation was not determined during the course of this boring.

Driller: GES

Logger: H.S.

Organization: GES

SWELL TEST RESULTS

ABSORPTION SWELL TEST (ASTM D4546) RESULTS

Boring No.	B-01	B-01	B-02	B-02	B-03	B-03
Average Sample Depth (ft)	1	4.5	3	7	1	5
Sample Height (in)	0.8	0.8	0.8	0.8	0.8	0.8
Sample Diameter (in)	2.5	2.5	2.5	2.5	2.5	2.5
Initial Sample Volume (cu in)	3.93	3.93	3.93	3.93	3.93	3.93
Initial Sample Weight (gr)	142.6	134.0	138.4	138.2	135.0	135.0
Initial Moisture (%)	13	19	10	14	11	15
Final Moisture (%)	16	19	10	14	14	16
Initial Wet Unit Weight (pcf)	138	130	134	134	131	131
Initial Dry Unit Weight (pcf)	123	109	122	118	118	113
Applied Over Burden (psi)	0.8	3.8	2.5	5.8	0.8	4.2
Initial Dial Reading (in)	0.0559	0.0595	0.0411	0.0552	0.0567	0.0575
Final Dial Reading (in)	0.0604	0.0596	0.0411	0.0552	0.0467	0.0575
Swell (%)	0.56	0.01	0.00	0.00	0.00	0.00

ABSORPTION SWELL TEST (ASTM D4546) RESULTS

Boring No.	B-04	B-04	B-05	B-05	B-07	B-10
Average Sample Depth (ft)	3	7	1	4	1	7
Sample Height (in)	0.8	0.8	0.8	0.8	0.8	0.8
Sample Diameter (in)	2.5	2.5	2.5	2.5	2.5	2.5
Initial Sample Volume (cu in)	3.93	3.93	3.93	3.93	3.93	3.93
Initial Sample Weight (gr)	121.5	134.5	138.0	138.0	133.4	139.8
Initial Moisture (%)	13	20	34	16	11	13
Final Moisture (%)	18	20	34	17	15	14
Initial Wet Unit Weight (pcf)	118	130	116	134	129	135
Initial Dry Unit Weight (pcf)	104	109	86	116	117	120
Applied Over Burden (psi)	2.5	5.8	0.8	3.3	0.8	5.8
Initial Dial Reading (in)	0.0427	0.042	0.0361	0.035	0.0459	0.0362
Final Dial Reading (in)	0.0427	0.042	0.0361	0.035	0.0459	0.0362
Swell (%)	0.00	0.00	0.00	0.00	0.00	0.00

ABSORPTION SWELL TEST (ASTM D4546) RESULTS

Boring No.	B-11	B-12	B-13	B-13	B-14	B-14
Average Sample Depth (ft)	9	7	3	7	1	7
Sample Height (in)	0.8	0.8	0.8	0.8	0.8	0.8
Sample Diameter (in)	2.5	2.5	2.5	2.5	2.5	2.5
Initial Sample Volume (cu in)	3.93	3.93	3.93	3.93	3.93	3.93
Initial Sample Weight (gr)	137.0	126.8	141.0	127.7	136.9	137.9
Initial Moisture (%)	16	13	10	10	8	13
Final Moisture (%)	17	17	12	13	11	13
Initial Wet Unit Weight (pcf)	133	123	137	124	133	134
Initial Dry Unit Weight (pcf)	114	109	124	113	123	119
Applied Over Burden (psi)	7.5	5.8	2.5	5.8	0.8	5.8
Initial Dial Reading (in)	0.021	0.0343	0.0403	0.0366	0.023	0.0568
Final Dial Reading (in)	0.0349	0.0343	0.0403	0.0366	0.023	0.0568
Swell (%)	1.74	0.00	0.00	0.00	0.00	0.00

ABSORPTION SWELL TEST (ASTM D4546) RESULTS

Boring No.	B-15					
Average Sample Depth (ft)	9					
Sample Height (in)	0.8					
Sample Diameter (in)	2.5					
Initial Sample Volume (cu in)	3.93					
Initial Sample Weight (gr)	133.7					
Initial Moisture (%)	14					
Final Moisture (%)	15					
Initial Wet Unit Weight (pcf)	130					
Initial Dry Unit Weight (pcf)	114					
Applied Over Burden (psi)	7.5					
Initial Dial Reading (in)	0.0242					
Final Dial Reading (in)	0.0242					
Swell (%)	0.00					

SULFATE CONTENT TEST RESULTS



Gorrondona Engineering Services, Inc.

Soil Laboratory Sulfate Content (Tex-145-E)

Project Name: I-45 Frontage Road Extension, CSJ No. 0675-08-114

Project No.: 21-0320

Date: 9/9/2021

Boring No. B-01		Sample No. 2		Sample Depth (ft.) 2-4	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	9.0	Average Reading	8.0	Average Reading	8.0
Sulfate Content	180	Sulfate Content	160	Sulfate Content	160
Average Sulfate Content:		167	ppm		

Boring No. B-01		Sample No. 4		Sample Depth (ft.) 6-8	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	17.0	Average Reading	18.0	Average Reading	18.0
Sulfate Content	340	Sulfate Content	360	Sulfate Content	360
Average Sulfate Content:		353	ppm		

Boring No. B-02		Sample No. 1		Sample Depth (ft.) 0-2	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	33.0	Average Reading	32.0	Average Reading	30.0
Sulfate Content	660	Sulfate Content	640	Sulfate Content	600
Average Sulfate Content:		633	ppm		

Boring No. B-02		Sample No. 3		Sample Depth (ft.) 4-6	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	8.0	Average Reading	7.0	Average Reading	7.0
Sulfate Content	160	Sulfate Content	140	Sulfate Content	140
Average Sulfate Content:		147	ppm		

Boring No. B-03		Sample No. 2		Sample Depth (ft.) 2-4	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	7.0	Average Reading	8.0	Average Reading	9.0
Sulfate Content	140	Sulfate Content	160	Sulfate Content	180
Average Sulfate Content:		160	ppm		

Boring No. B-04		Sample No. 3		Sample Depth (ft.) 4-6	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	26.0	Average Reading	26.0	Average Reading	27.0
Sulfate Content	520	Sulfate Content	520	Sulfate Content	540
Average Sulfate Content:		527	ppm		



Gorrondona Engineering Services, Inc.

Soil Laboratory Sulfate Content (Tex-145-E)

Project Name: I-45 Frontage Road Extension, CSJ No. 0675-08-114

Project No.: 21-0320

Date: 9/9/2021

Boring No. B-05		Sample No. 2		Sample Depth (ft.) 2-4	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	7.0	Average Reading	9.0	Average Reading	7.0
Sulfate Content	140	Sulfate Content	180	Sulfate Content	140
Average Sulfate Content:		153	ppm		

Boring No. B-05		Sample No. 4		Sample Depth (ft.) 6-8	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	10.0	Average Reading	11.0	Average Reading	10.0
Sulfate Content	200	Sulfate Content	220	Sulfate Content	200
Average Sulfate Content:		207	ppm		

Boring No. B-06		Sample No. 1		Sample Depth (ft.) 0-2	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	17.0	Average Reading	18.0	Average Reading	18.0
Sulfate Content	340	Sulfate Content	360	Sulfate Content	360
Average Sulfate Content:		353	ppm		

Boring No. B-06		Sample No. 4		Sample Depth (ft.) 6-8	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	10.0	Average Reading	9.0	Average Reading	9.0
Sulfate Content	200	Sulfate Content	180	Sulfate Content	180
Average Sulfate Content:		187	ppm		

Boring No. B-07		Sample No. 2		Sample Depth (ft.) 2-4	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	22.0	Average Reading	22.0	Average Reading	22.0
Sulfate Content	440	Sulfate Content	440	Sulfate Content	440
Average Sulfate Content:		440	ppm		

Boring No. B-07		Sample No. 5		Sample Depth (ft.) 8-10	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	22.0	Average Reading	21.0	Average Reading	21.0
Sulfate Content	440	Sulfate Content	420	Sulfate Content	420
Average Sulfate Content:		427	ppm		



Gorrondona Engineering Services, Inc.

Soil Laboratory Sulfate Content (Tex-145-E)

Project Name: I-45 Frontage Road Extension, CSJ No. 0675-08-114

Project No.: 21-0320

Date: 9/9/2021

Boring No. B-08		Sample No. 3		Sample Depth (ft.) 4-6	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	43.0	Average Reading	40.0	Average Reading	43.0
Sulfate Content	860	Sulfate Content	800	Sulfate Content	860
Average Sulfate Content:		840 ppm			

Boring No. B-09		Sample No. 2		Sample Depth (ft.) 2-4	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	25.0	Average Reading	25.0	Average Reading	24.0
Sulfate Content	500	Sulfate Content	500	Sulfate Content	480
Average Sulfate Content:		493 ppm			

Boring No. B-11		Sample No. 1		Sample Depth (ft.) 0-2	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	21.0	Average Reading	21.0	Average Reading	20.0
Sulfate Content	420	Sulfate Content	420	Sulfate Content	400
Average Sulfate Content:		413 ppm			

Boring No. B-12		Sample No. 2		Sample Depth (ft.) 2-4	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	6.0	Average Reading	6.0	Average Reading	5.0
Sulfate Content	120	Sulfate Content	120	Sulfate Content	100
Average Sulfate Content:		113 ppm			

Boring No. B-13		Sample No. 1		Sample Depth (ft.) 0-2	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	25.0	Average Reading	25.0	Average Reading	23.0
Sulfate Content	500	Sulfate Content	500	Sulfate Content	460
Average Sulfate Content:		487 ppm			

Boring No. B-15		Sample No. 1		Sample Depth (ft.) 0-2	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	25.0	Average Reading	26.0	Average Reading	27.0
Sulfate Content	500	Sulfate Content	520	Sulfate Content	540
Average Sulfate Content:		520 ppm			



Gorrondona Engineering Services, Inc.

Soil Laboratory Sulfate Content (Tex-145-E)

Project Name: I-45 Frontage Road Extension, CSJ No. 0675-08-114

Project No.: 21-0320

Date: 9/9/2021

Boring No. B-15		Sample No. 2		Sample Depth (ft.) 2-4	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	25.0	Average Reading	27.0	Average Reading	25.0
Sulfate Content	500	Sulfate Content	540	Sulfate Content	500
Average Sulfate Content:		513	ppm		

Boring No. B-15		Sample No. 3		Sample Depth (ft.) 4-6	
Dilution Ratio Used	1:20	Dilution Ratio Used	1:20	Dilution Ratio Used	1:20
Average Reading	26.0	Average Reading	23.0	Average Reading	24.0
Sulfate Content	520	Sulfate Content	460	Sulfate Content	480
Average Sulfate Content:		487	ppm		

GRAIN SIZE DISTRIBUTION RESULTS

Particle Size Distribution Report



Particle Size Distribution Report



	% +3"		% Gravel		% Sand			% Fines	
			Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
☐	0.0		0.0	3.1	5.1	49.6	35.2	7.0	

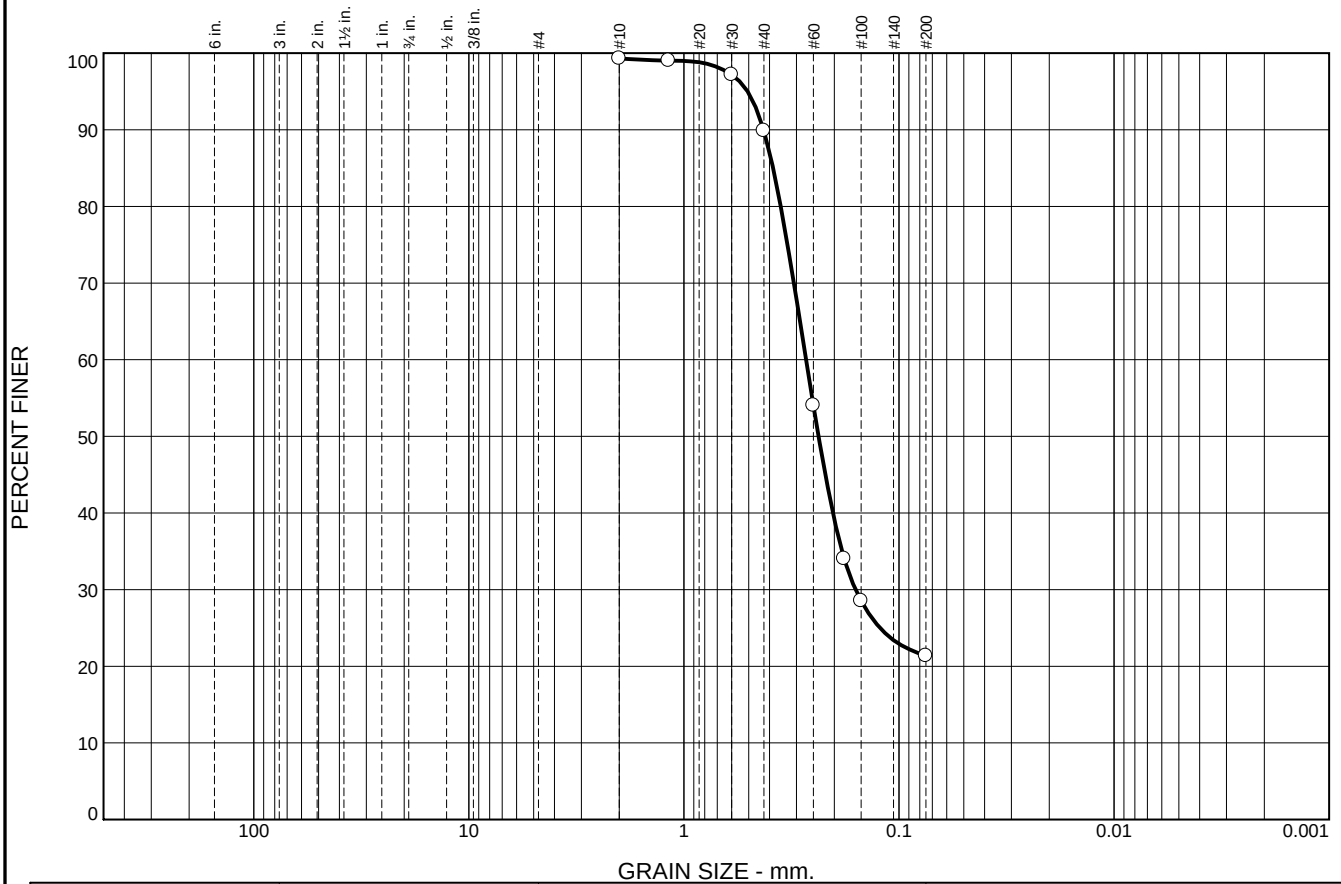
☒	LL	PL	D ₈₅	D ₆₀	D ₅₀	D ₃₀	D ₁₅	D ₁₀	C _c	C _u
☐			1.1463	0.6043	0.4982	0.3141	0.2022	0.1361	1.20	4.44

Material Description								USCS	AASHTO
☐									

Project No. 21-0320 Client: L Squared Engineering Project: I-45 Frontage Road Extension ☐ Location: Boring B-05 Depth: 18' - 20' Sample Number: 7	Remarks:
Gorrondonga Engineering Services Houston, Texas	

Tested By: L.R. Checked By: A.G.

Particle Size Distribution Report



	% +3"		% Gravel		% Sand			% Fines		
			Coarse	Fine	Coarse	Medium	Fine	Silt		Clay
☐						9.4	68.5	21.4		
☐										
☐										
☒	LL	PL	D ₈₅	D ₆₀	D ₅₀	D ₃₀	D ₁₅	D ₁₀	C _c	C _u
☐			0.3850	0.2704	0.2365	0.1593				
☐										

Material Description								USCS	AASHTO
☐									

Project No. 21-0320 Client: L Squared Engineering Project: I-45 Frontage Road Extension ☐ Location: Boring B-06 Depth: 6' - 8' Sample Number: 4	Remarks:
Gorrondonga Engineering Services Houston, Texas	

Tested By: L.R. Checked By: A.G.

Figure

Particle Size Distribution Report

Grain Size - mm.	
6 in.	3 in.
2 in.	1½ in.
1 in.	¾ in.
½ in.	⅜ in.
#4	#10
#20	#30
#40	#60
#100	#140
#200	

% +3"	% Gravel		% Sand			% Fines			
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay		
0.0	0.0	9.7	8.8	29.6	36.3	15.6			
LL	PL	D ₈₅	D ₆₀	D ₅₀	D ₃₀	D ₁₅	D ₁₀	C _c	C _u
		2.8414	0.5405	0.4015	0.2189				
Material Description								USCS	AASHTO

Project No. 21-0320 **Client:** L Squared Engineering
Project: I-45 Frontage Road Extension

Location: Boring B-06 **Depth:** 18' - 20' **Sample Number:** 7

Gorrondona Engineering Services
Houston, Texas

Remarks:

Figure

Tested By: L.R. Checked By: A.G.

Particle Size Distribution Report



	% +3"		% Gravel		% Sand			% Fines		
			Coarse	Fine	Coarse	Medium	Fine	Silt		Clay
☐	0.0		10.6	3.2	2.5	25.9	41.6	16.2		
☐										
☐										
☒	LL	PL	D ₈₅	D ₆₀	D ₅₀	D ₃₀	D ₁₅	D ₁₀	C _c	C _u
☐			3.2156	0.4450	0.3699	0.2553				
☐										

Material Description								USCS	AASHTO
☐									

Project No. 21-0320 Client: L Squared Engineering Project: I-45 Frontage Road Extension ☐ Location: Boring B-07 Depth: 6' - 8' Sample Number: 4	Remarks:
Gorronadona Engineering Services Houston, Texas	

Tested By: L.R. Checked By: A.G.

Figure

Particle Size Distribution Report



	% +3"		% Gravel		% Sand			% Fines	
			Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
☐	0.0		0.0	6.7	4.3	30.0	32.7	26.3	

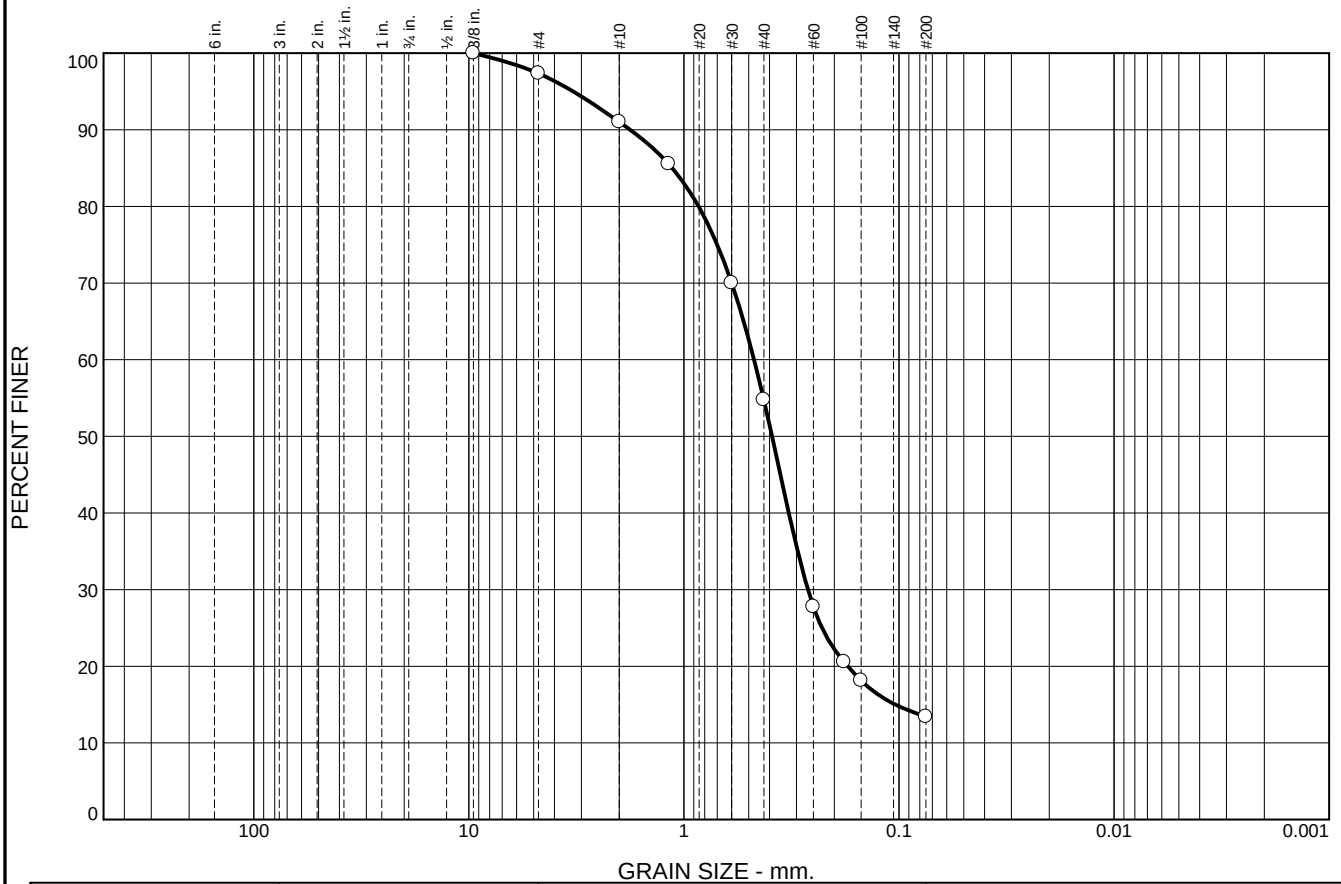
☒	LL	PL	D ₈₅	D ₆₀	D ₅₀	D ₃₀	D ₁₅	D ₁₀	C _c	C _u
☐			1.2104	0.4364	0.3386	0.1399				

Material Description								USCS	AASHTO
☐									

Project No. 21-0320 Client: L Squared Engineering Project: I-45 Frontage Road Extension ☐ Location: Boring B-07 Depth: 18' - 20' Sample Number: 7	Remarks:
Gorrondona Engineering Services Houston, Texas	

Tested By: L.R. Checked By: A.G.

Particle Size Distribution Report



	% +3"		% Gravel		% Sand			% Fines	
			Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
☐	0.0		0.0	2.6	6.4	36.2	41.4	13.4	

☒	LL	PL	D ₈₅	D ₆₀	D ₅₀	D ₃₀	D ₁₅	D ₁₀	C _c	C _u
☐			1.1347	0.4720	0.3893	0.2651	0.1040			

Material Description								USCS	AASHTO
☐									

Project No. 21-0320 Client: L Squared Engineering Project: I-45 Frontage Road Extension ☐ Location: Boring B-09 Depth: 6' - 8' Sample Number: 4	Remarks:
Gorronadona Engineering Services Houston, Texas	

Tested By: L.R. Checked By: A.G.

Particle Size Distribution Report

The graph plots Percent Finer (Y-axis, 0 to 100) against Grain Size in mm (X-axis, logarithmic scale from 100 to 0.001). The curve shows a soil that is nearly 100% finer than 4.75 mm, with a sharp drop between 0.85 mm and 0.425 mm, and a gradual decrease below 0.425 mm.

Grain Size (mm)	Percent Finer (%)
60	100
30	100
15	100
7.5	100
4.75	100
2.5	100
1.18	100
0.85	98
0.425	85
0.25	65
0.15	35
0.075	21

% +3"		% Gravel		% Sand			% Fines	
		Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
	0.0	0.0	0.1	1.7	33.9	43.8	20.5	

LL	PL	D85	D60	D50	D30	D15	D10	Cc	Cu
		0.6147	0.3972	0.3383	0.2172				

Material Description		USCS	AASHTO

Project No. 21-0320 **Client:** L Squared Engineering

Project: I-45 Frontage Road Extension

Location: Boring B-10 **Depth:** 18' - 20' **Sample Number:** 7

Gorrondona Engineering Services

Houston, Texas

Remarks:

Figure

Tested By: L.R. Checked By: A.G.

Particle Size Distribution Report



	% +3"		% Gravel		% Sand			% Fines		
			Coarse	Fine	Coarse	Medium	Fine	Silt		Clay
○	0.0		0.0	0.2	0.0	4.5	77.1	18.2		
×	LL	PL	D ₈₅	D ₆₀	D ₅₀	D ₃₀	D ₁₅	D ₁₀	C _c	C _u
○			0.3299	0.2289	0.2008	0.1336				

Material Description							USCS	AASHTO
○								

Project No. 21-0320 Client: L Squared Engineering Project: I-45 Frontage Road Extension ○ Location: Boring B-11 Depth: 4' - 6' Sample Number: 3	Remarks: Figure
Gorrondonga Engineering Services Houston, Texas	

Tested By: L.R. Checked By: A.G.

Particle Size Distribution Report



	% +3"	% Gravel		% Sand			% Fines	
		Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
☐	0.0	0.0	0.0	0.1	1.9	70.6	27.4	

☒	LL	PL	D ₈₅	D ₆₀	D ₅₀	D ₃₀	D ₁₅	D ₁₀	C _c	C _u
☐			0.2861	0.2105	0.1880	0.1142				

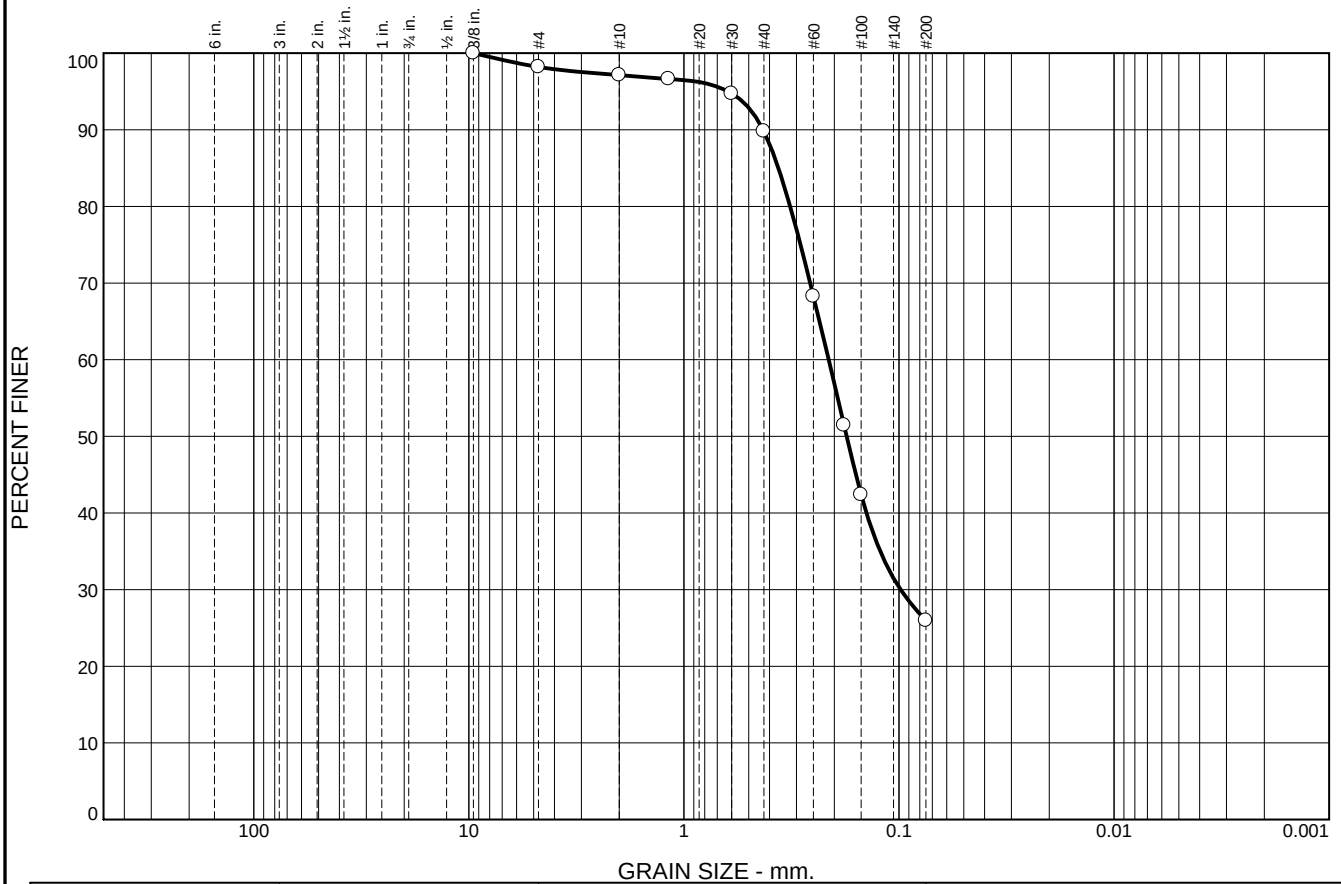
Material Description								USCS	AASHTO
☐									

Project No. 21-0320 Client: L Squared Engineering Project: I-45 Frontage Road Extension ☐ Location: Boring B-12 Depth: 18' - 20' Sample Number: 7	Remarks:
Gorrondonga Engineering Services Houston, Texas	

Tested By: L.R. Checked By: A.G.

Figure

Particle Size Distribution Report



	% +3"	% Gravel		% Sand			% Fines	
		Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
☐	0.0	0.0	1.8	1.1	7.3	63.8	26.0	

☒	LL	PL	D ₈₅	D ₆₀	D ₅₀	D ₃₀	D ₁₅	D ₁₀	C _c	C _u
☐			0.3637	0.2123	0.1751	0.0983				

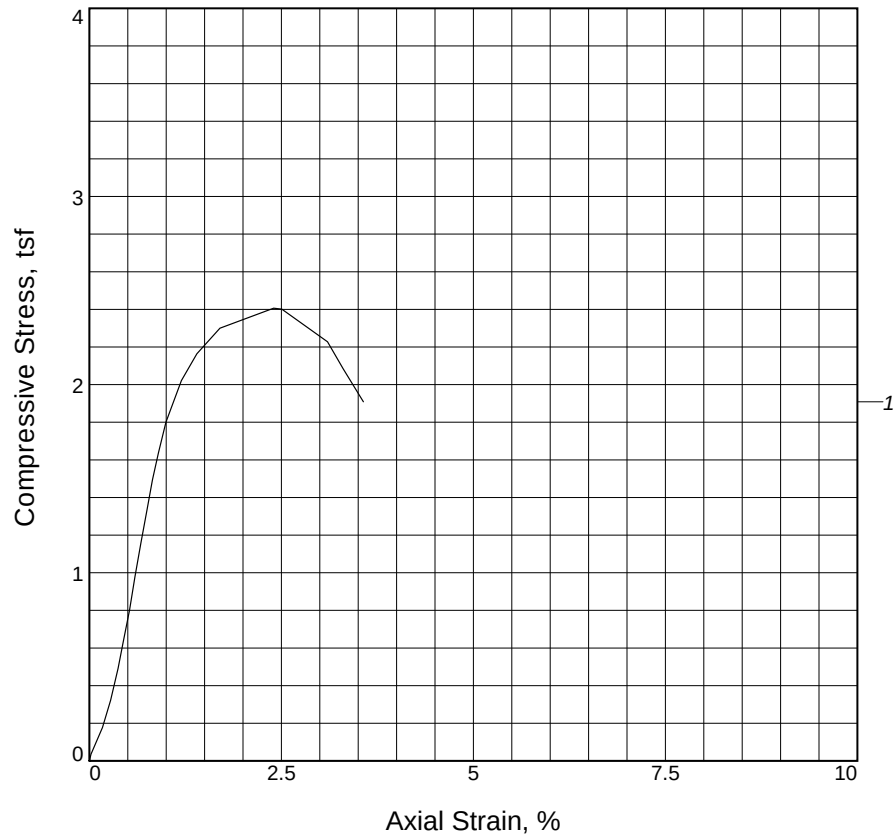
Material Description								USCS	AASHTO
☐									

Project No. 21-0320 Client: L Squared Engineering Project: I-45 Frontage Road Extension ☐ Location: Boring B-15 Depth: 4' - 6' Sample Number: 3	Remarks:
Gorronadona Engineering Services Houston, Texas	

Tested By: L.R. Checked By: A.G.

TRIAXIAL TEST RESULTS

UNCONFINED COMPRESSION TEST



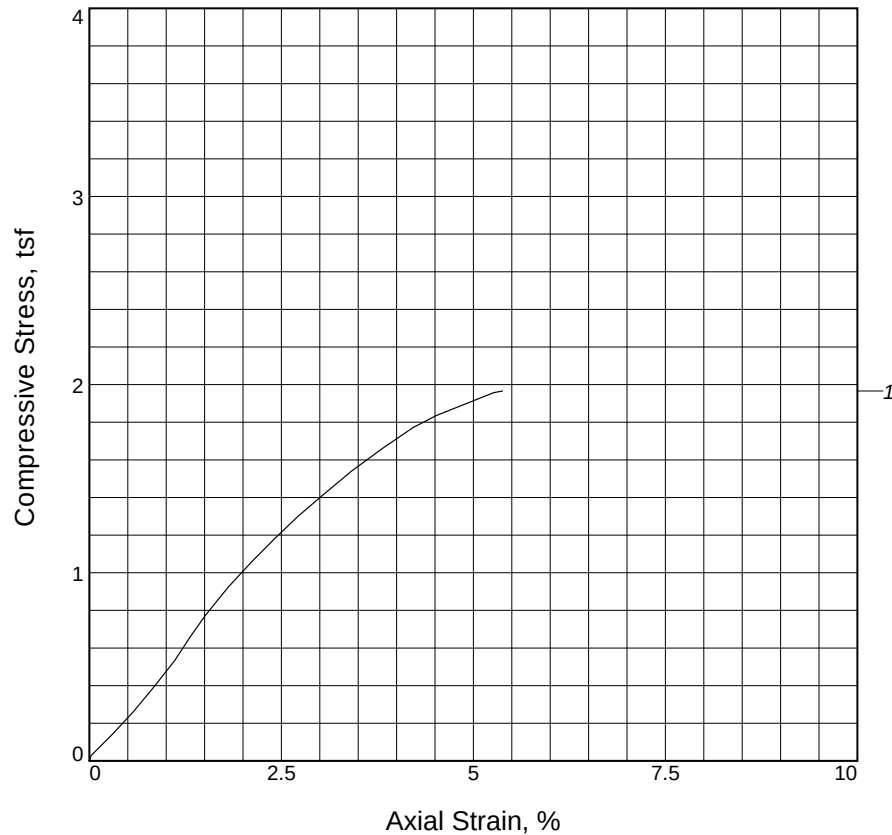
Sample No.	1			
Unconfined strength, tsf	2.407			
Undrained shear strength, tsf	1.203			
Failure strain, %	2.4			
Strain rate, %/min.	1.00			
Water content, %	25.1			
Wet density, pcf	125.1			
Dry density, pcf	100.0			
Saturation, %	97.7			
Void ratio	0.6984			
Specimen diameter, in.	2.74			
Specimen height, in.	5.75			
Height/diameter ratio	2.10			

Description:

LL =	PL =	PI =	Assumed GS= 2.72	Type: Shelby Tube
Project No.: 21-0320 Date Sampled: 8/24/2021 Remarks: Shear Plane Failure			Client: L Squared Engineering Project: I-45 Frontage Road Extension, CSJ No. 0675-08-114 Location: Boring B-01 Sample Number: 5 Depth: 8' - 10' UNCONFINED COMPRESSION TEST Gorronona Engineering Services Houston, Texas	
Figure _____				

Tested By: L.R. Checked By: A.G.

UNCONFINED COMPRESSION TEST



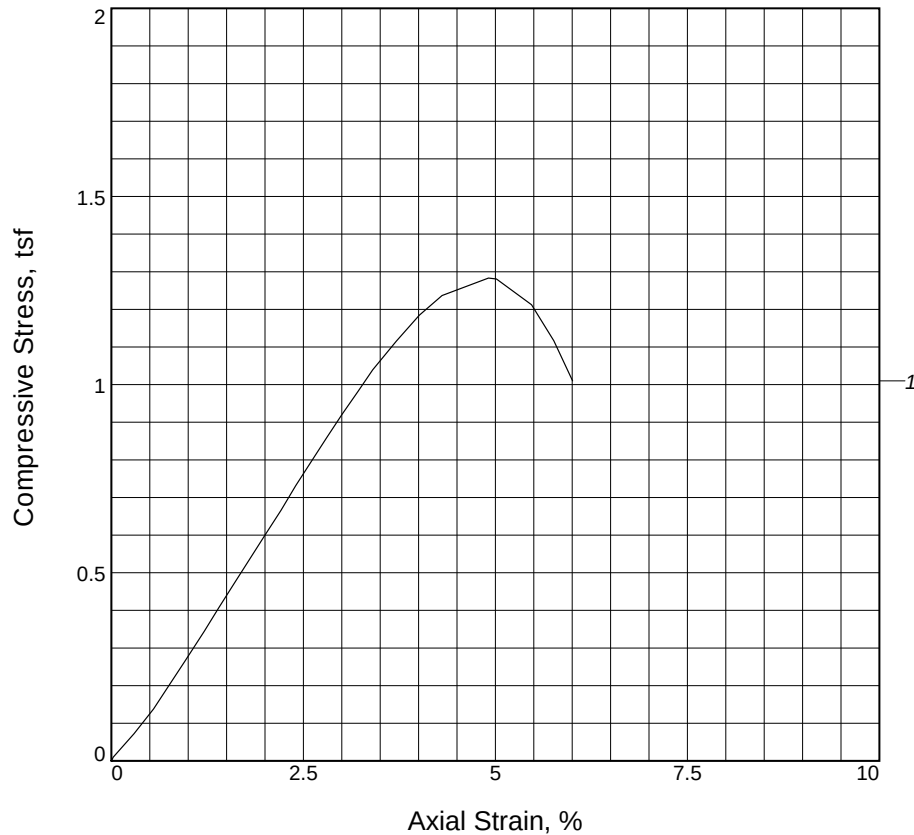
Sample No.	1			
Unconfined strength, tsf	1.966			
Undrained shear strength, tsf	0.983			
Failure strain, %	5.4			
Strain rate, %/min.	1.00			
Water content, %	15.5			
Wet density, pcf	132.6			
Dry density, pcf	114.8			
Saturation, %	87.9			
Void ratio	0.4788			
Specimen diameter, in.	2.71			
Specimen height, in.	5.75			
Height/diameter ratio	2.12			

Description:

LL =	PL =	PI =	Assumed GS= 2.72	Type: Shelby Tube
Project No.: 21-0320 Date Sampled: 8/24/2021 Remarks: Bulge Failure			Client: L Squared Engineering Project: I-45 Frontage Road Extension, CSJ No. 0675-08-114 Location: Boring B-02 Sample Number: 5 Depth: 8' - 10' UNCONFINED COMPRESSION TEST Gorrondona Engineering Services Houston, Texas	
Figure _____				

Tested By: L.R. Checked By: A.G.

UNCONFINED COMPRESSION TEST



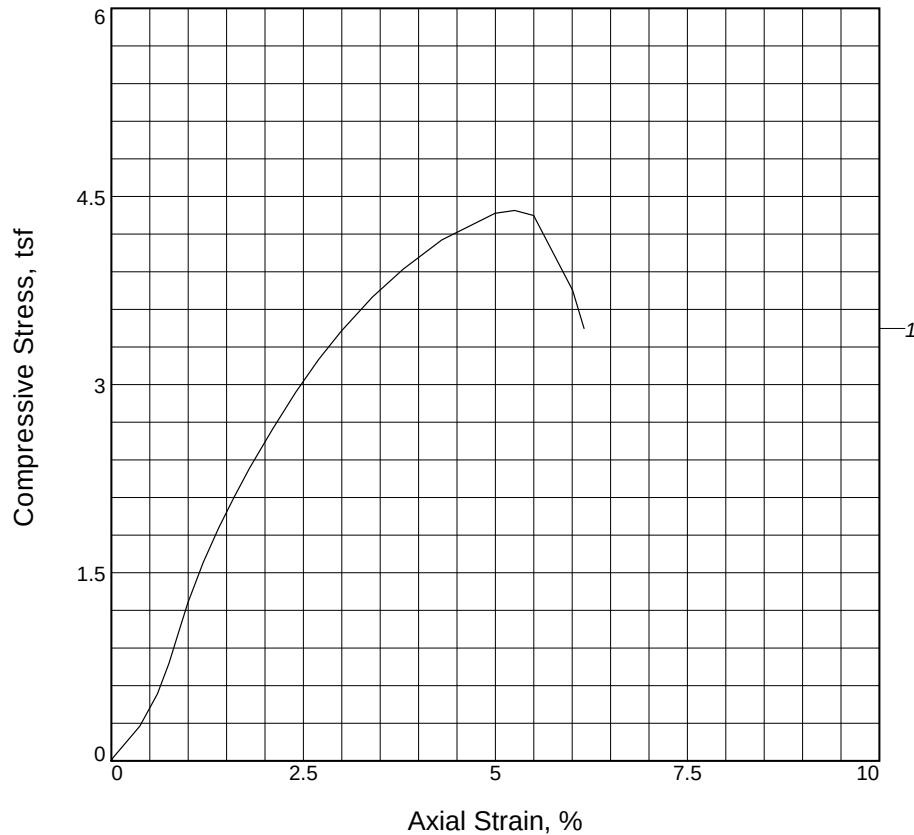
Sample No.	1			
Unconfined strength, tsf	1.283			
Undrained shear strength, tsf	0.642			
Failure strain, %	4.9			
Strain rate, %/min.	1.00			
Water content, %	15.7			
Wet density, pcf	134.0			
Dry density, pcf	115.8			
Saturation, %	91.5			
Void ratio	0.4663			
Specimen diameter, in.	2.71			
Specimen height, in.	5.75			
Height/diameter ratio	2.12			

Description:

LL =	PL =	PI =	Assumed GS= 2.72	Type: Shelby Tube
Project No.: 21-0320 Date Sampled: 8/24/2021 Remarks: Bulge Failure			Client: L Squared Engineering Project: I-45 Frontage Road Extension, CSJ No. 0675-08-114 Location: Boring B-03 Sample Number: 4 Depth: 6' - 8' UNCONFINED COMPRESSION TEST Gorronona Engineering Services Houston, Texas	
Figure _____				

Tested By: L.R. Checked By: A.G.

UNCONFINED COMPRESSION TEST



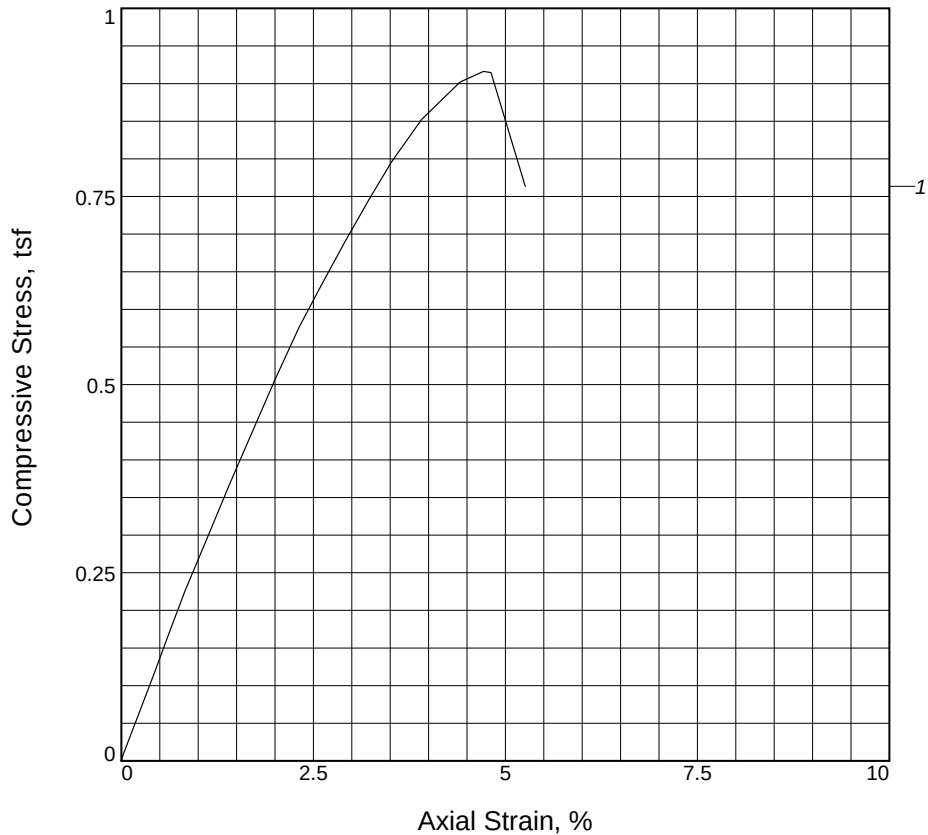
Sample No.	1			
Unconfined strength, tsf	4.388			
Undrained shear strength, tsf	2.194			
Failure strain, %	5.3			
Strain rate, %/min.	1.00			
Water content, %	18.3			
Wet density, pcf	128.3			
Dry density, pcf	108.5			
Saturation, %	88.0			
Void ratio	0.5656			
Specimen diameter, in.	2.72			
Specimen height, in.	5.75			
Height/diameter ratio	2.11			

Description:

LL =	PL =	PI =	Assumed GS= 2.72	Type: Shelby Tube
Project No.: 21-0320 Date Sampled: 8/24/2021 Remarks: Bulge Failure			Client: L Squared Engineering Project: I-45 Frontage Road Extension, CSJ No. 0675-08-114 Location: Boring B-05 Sample Number: 5 Depth: 8' - 10' UNCONFINED COMPRESSION TEST Gorronona Engineering Services Houston, Texas	
Figure _____				

Tested By: L.R. Checked By: A.G.

UNCONFINED COMPRESSION TEST

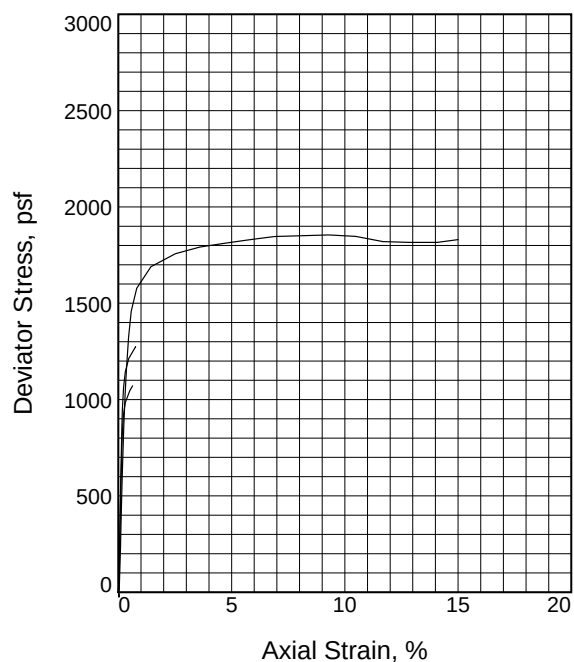
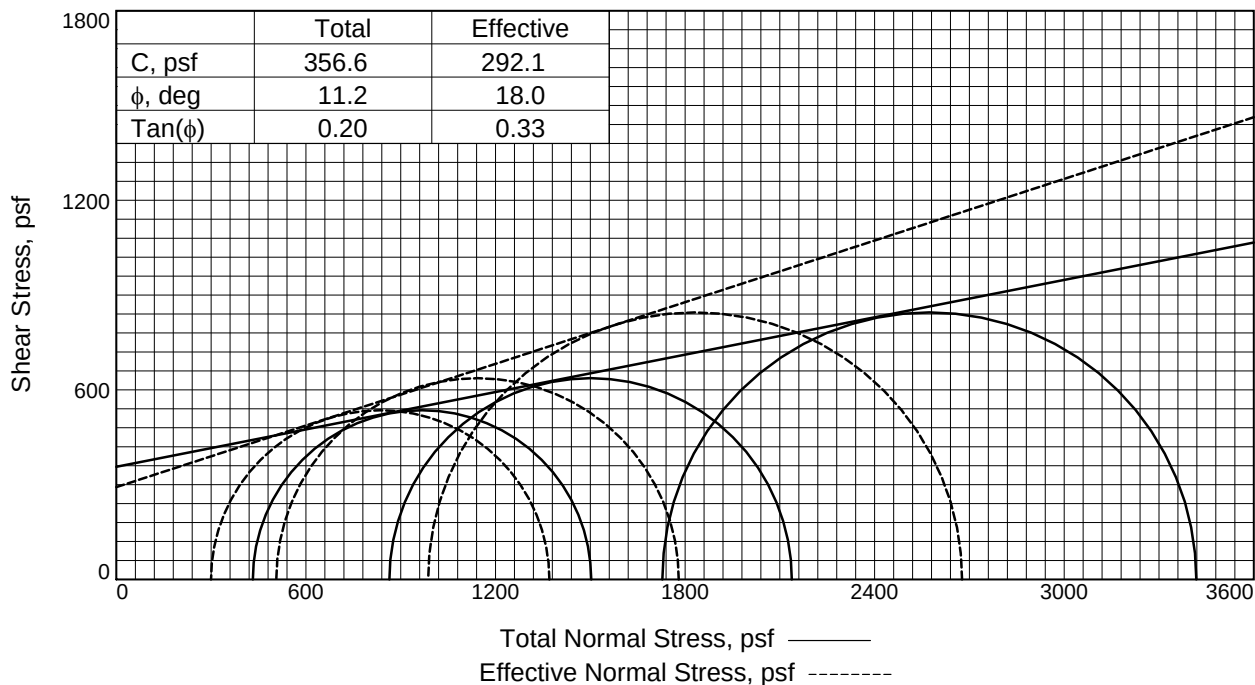


Sample No.	1			
Unconfined strength, tsf	0.916			
Undrained shear strength, tsf	0.458			
Failure strain, %	4.7			
Strain rate, %/min.	1.00			
Water content, %	14.7			
Wet density, pcf	131.8			
Dry density, pcf	115.0			
Saturation, %	83.7			
Void ratio	0.4770			
Specimen diameter, in.	2.75			
Specimen height, in.	5.75			
Height/diameter ratio	2.09			

Description:

LL =	PL =	PI =	Assumed GS= 2.72	Type: Shelby Tube
Project No.: 21-0320 Date Sampled: 8/26/2021 Remarks: Bulge Failure			Client: L Squared Engineering Project: I-45 Frontage Road Extension, CSJ No. 0675-08-114 Location: Boring B-14 Sample Number: 7 Depth: 18' -20' UNCONFINED COMPRESSION TEST Gorrondona Engineering Services Houston, Texas	
Figure _____				

Tested By: L.R. Checked By: A.G.



Sample No.		1	2	3
Initial	Water Content, %	23.5	23.5	23.5
	Dry Density, pcf	99.9	99.9	99.9
	Saturation, %	90.2	90.2	90.2
	Void Ratio	0.7131	0.7131	0.7131
	Diameter, in.	2.727	2.727	2.727
	Height, in.	5.652	5.652	5.652
At Test	Water Content, %	25.4	25.4	25.4
	Dry Density, pcf	100.9	100.9	100.9
	Saturation, %	100.0	100.0	100.0
	Void Ratio	0.6962	0.6961	0.6958
	Diameter, in.	2.724	2.742	2.991
	Height, in.	5.610	5.535	4.652
Strain rate, %/min.		0.0010	0.0010	0.0010
Back Pressure, psi		18.000	18.000	18.000
Cell Pressure, psi		21.000	24.000	30.000
Fail. Stress, psf		1070.9	1273.9	1690.1
Total Pore Pr., psf		2724.5	2949.9	3333.4
Ult. Stress, psf		1070.9	1273.9	1690.1
Total Pore Pr., psf		2724.5	2949.9	3333.4
$\bar{\sigma}_1$ Failure, psf		1370.4	1780.0	2676.7
$\bar{\sigma}_3$ Failure, psf		299.5	506.1	986.6

Type of Test:

CU with Pore Pressures

Sample Type: One Undisturbed Sample

Description: Fat Clay (CH)

-200=89.8%

LL= 80

PL= 25

PI= 55

Specific Gravity= 2.741

Remarks: Tested by: ER

Date: 10/08/21

Checked by: JH

Date: 10/13/21

Figure 4

Client: Gorrondona Engineering Services, Inc.

Project: I-45 Frontage Road Extension,

Consolidation Test, GES Job No. 21-0320

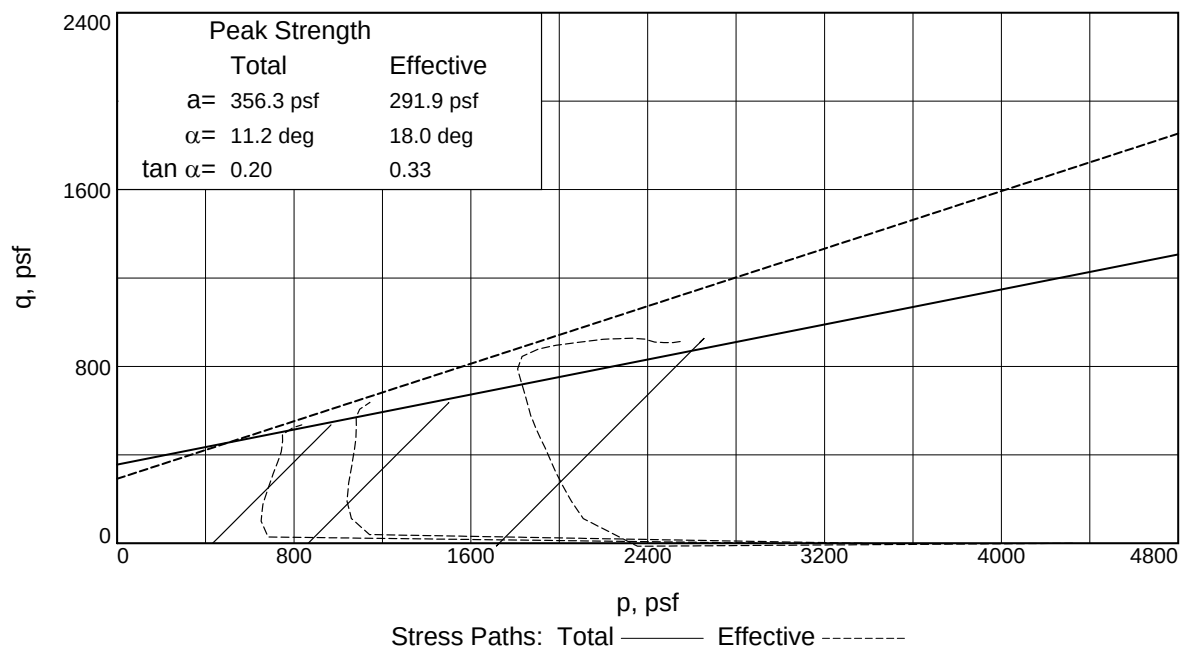
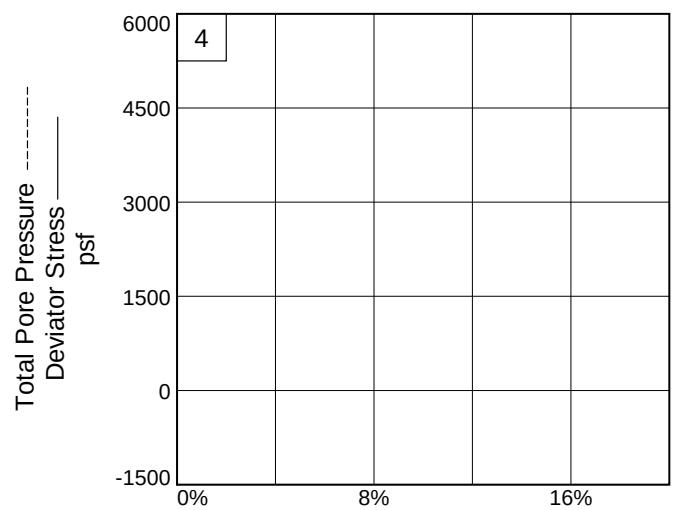
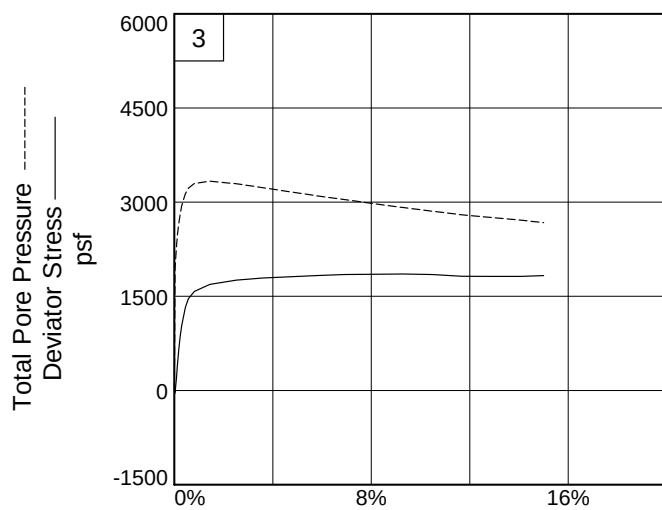
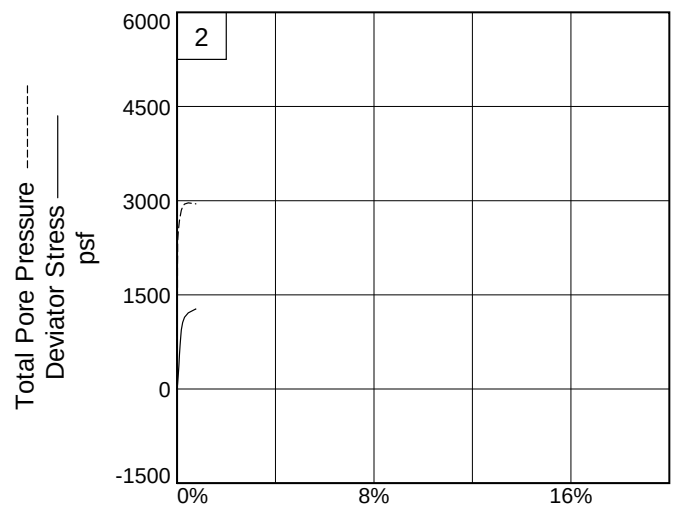
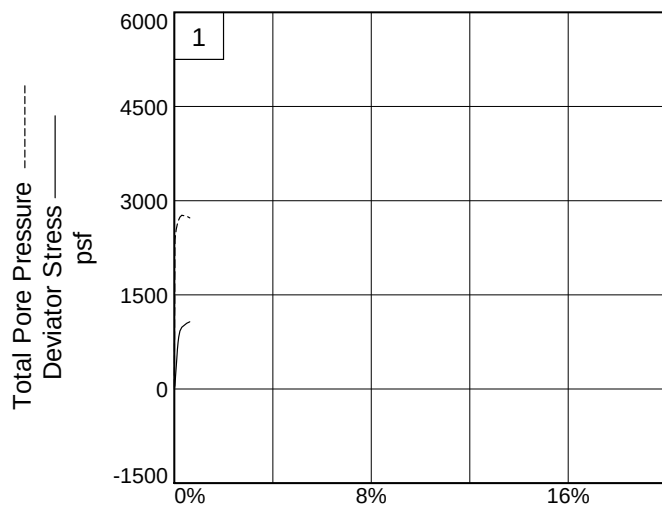
Sample Number: B-01

Depth: 6'-8'

Proj. No.: 21-S-430

Date Sampled:





Client: Gorrondona Engineering Services, Inc.

Project: I-45 Frontage Road Extension,

Depth: 6'-8' **Sample Number:** B-01

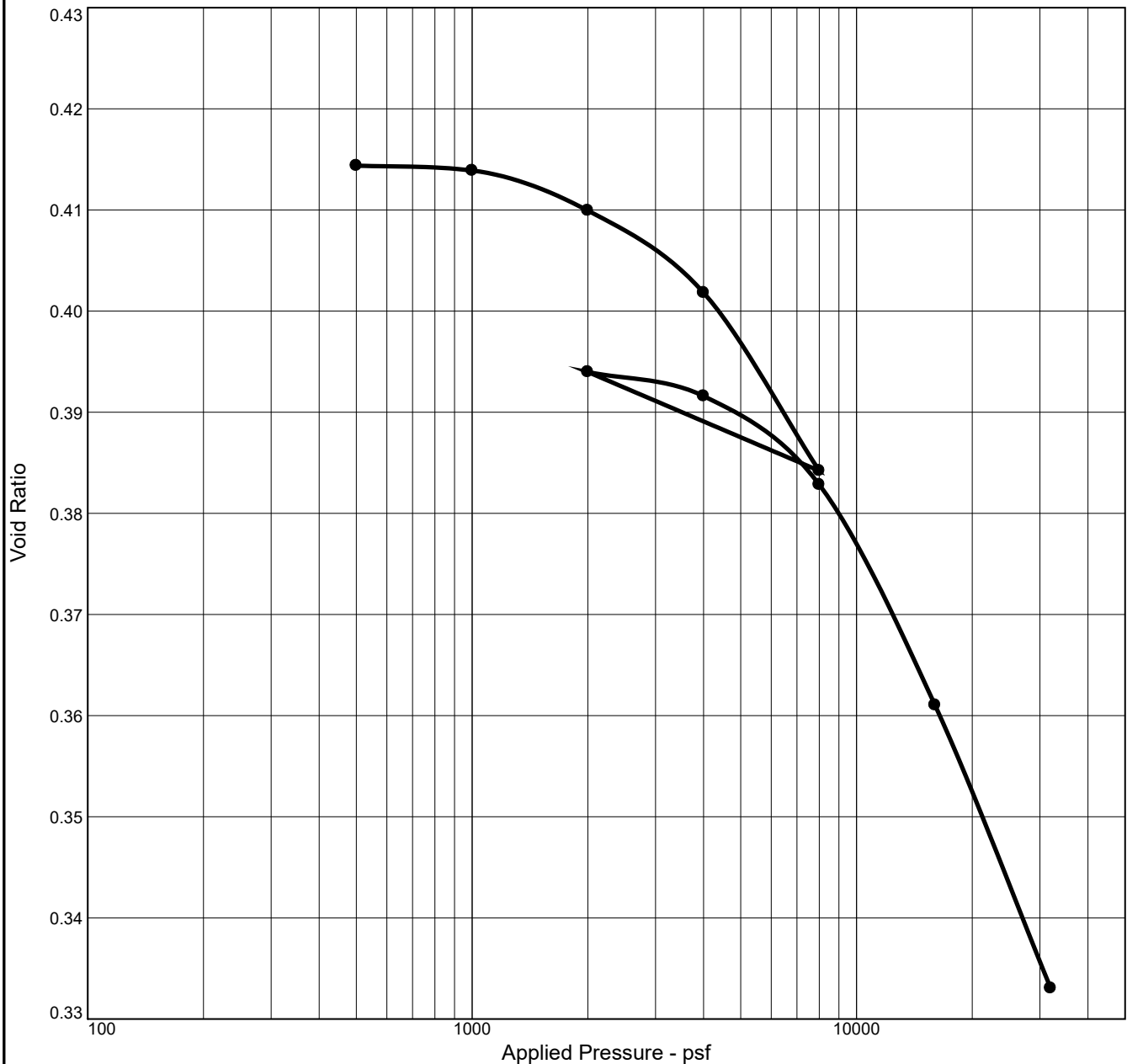
Project No.: 21-S-430

Figure 4A

HTS, Inc.

CONSOLIDATION TEST RESULTS

CONSOLIDATION TEST REPORT



Natural		Dry Dens. (pcf)	LL	PI	Sp. Gr.	Overburden (psf)	P _c (psf)	C _c	C _r	Initial Void Ratio
Saturation	Moisture									
90.8 %	14.1 %	115.6	58	40	2.60		4900	0.093	0.018	0.404

MATERIAL DESCRIPTION								USCS	AASHTO
Sandy Fat Clay (CH)								CH	N/A

Project No. 21-S-430		Client: Gorrondona Engineering Services, Inc.		Remarks: Tested by: ER Date: 09/25/21 Checked by: JH Date: 09/27/21 -200=66%
Project: I-45 Frontage Road Extension, Consolidation Test, GES Job No. 21-0320				
Depth: 13'-15'		Sample Number: B-01		
				Figure 1

Figure 1

The graph illustrates the relationship between Void Ratio and Applied Pressure (psf) for a normally consolidated clay. The y-axis represents the Void Ratio, ranging from 0.450 to 0.600. The x-axis represents the Applied Pressure in psf, on a logarithmic scale from 100 to 10,000. Two curves are plotted: a compression curve (upper) and a recompression curve (lower). The compression curve starts at approximately (500, 0.584) and ends at (20,000, 0.465). The recompression curve starts at approximately (2,000, 0.550) and ends at (8,000, 0.540).

Applied Pressure (psf)	Void Ratio	Curve Type
500	0.584	Compression
1,000	0.578	Compression
2,000	0.572	Compression
4,000	0.560	Compression
8,000	0.540	Compression
20,000	0.465	Compression
2,000	0.550	Recompression
4,000	0.545	Recompression
8,000	0.540	Recompression

Natural		Dry Dens. (pcf)	LL	PI	Sp. Gr.	Overburden (psf)	P _c (psf)	C _c	C _r	Initial Void Ratio
Saturation	Moisture									
91.5 %	20.5 %	102.5	47	30	2.60		6810	0.150	0.025	0.583

MATERIAL DESCRIPTION	USCS	AASHTO
Sandy Lean Clay (CL)	CL	N/A



Tested by: ER
Date: 9/27/21
Checked by: JH
Date: 9/30/21
-200=54%

Figure 3

Appendix D - Site Photographs

SITE PHOTOGRAPHS



Facing South at Boring B-01



Facing West at Boring B-02



Facing East at Boring B-03



Facing North at Boring B-04

SITE PHOTOGRAPHS



Facing South at Boring B-05



Facing North at Boring B-07



Facing South at boring B-08



Facing Northwest at Boring B-10

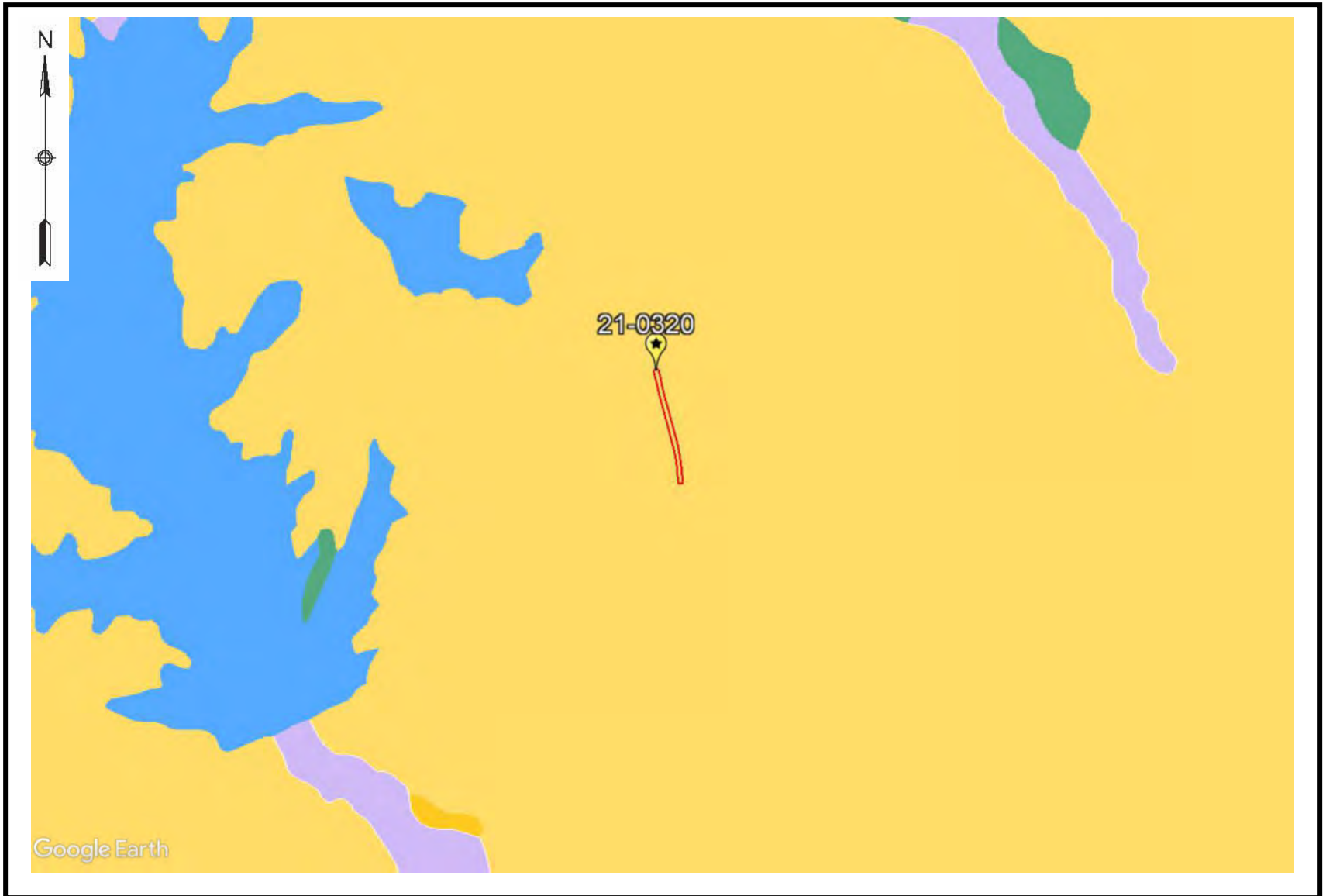
SITE PHOTOGRAPHS



Facing North at Boring B-12

Appendix E - Geologic Information

GEOLOGIC ATLAS



GES Project No. 21-0320

I-45 Frontage Road Extension
CSJ No. 0675-08-114





Mineral Resources On-Line Spatial Data

[Mineral Resources](#) > [Online Spatial Data](#) > [Geology](#) > [by state](#) > [Texas](#)

Willis Formation

Willis Formation

State [Texas](#)

Name Willis Formation

Geologic age Phanerozoic | Cenozoic | Tertiary | Pliocene

Original map label Pow

Comments Clay, silt, sand, siliceous granule to pebble gravel, some petrified wood; sand coarser than younger units, noncalcareous; deeply weathered, locally cemented by iron oxide; fluvatile; forms north-facing scarp. On Seguin Sheet (1974) thickness 100+- ft. On Austin 4 x 6-degree sheet (Moore and Wermund, 1993) unit consists of 1) channel facies--alluvial pebble gravel and sand, lt. gray to orange-brown, orange, gravelly coarse to fine sand which lenses of red, sandy silt and white to gray clay 10-60 m thick, pebbles mostly quartz, some chert and petrified wood, and 2) overbank facies--alluvial silt and clay made of brown, yellow, orange, fine silt and clay are intermixed and interbedded. 5-50+m thick.

Primary rock type [clay or mud](#)

Secondary rock type [silt](#)

Other rock types [sand](#); [gravel](#)

Lithologic constituents Major

Unconsolidated > Fine-detrital (*Alluvial*)

Unconsolidated > Coarse-detrital (*Alluvial*)

Minor

Unconsolidated > Coarse-detrital > Gravel (*Bed*)

Map references Bureau of Economic Geology, 1992, Geologic Map of Texas: University of Texas at Austin, Virgil E. Barnes, project supervisor, Hartmann, B.M. and Scranton, D.F., cartography, scale 1:500,000

Unit references Moore, D.W. and Wermund, E.G., Jr., 1993a, Quaternary geologic map of the Austin 4 x 6 degree quadrangle, United States: U.S. Geological Survey Miscellaneous Investigations Series Map I-1420 (NH-14), scale 1:1,000,000.

[[http://pubs.er.usgs.gov/publication/i1420\(NH14\)](http://pubs.er.usgs.gov/publication/i1420(NH14))]

Geographic coverage [Austin](#) - [Bastrop](#) - [Colorado](#) - [DeWitt](#) - [Fayette](#) - [Gonzales](#) - [Grimes](#) - [Guadalupe](#) - [Hays](#) - [Jasper](#) - [Lavaca](#) - [Liberty](#) - [Montgomery](#) - [Newton](#) - [Polk](#) - [Sabine](#) - [San Jacinto](#) - [Tyler](#) - [Victoria](#) - [Walker](#) - [Waller](#) - [Washington](#)

Show this information as [[XML](#)] - [[JSON](#)]

[U.S. Department of the Interior](#) | [U.S. Geological Survey](#)

URL: <http://mrdata.usgs.gov/geology/state/sgmc-unit.php?unit=TXPOw;0>

Page Contact Information: [Peter Schweitzer](#)

Appendix F - Unified Soil Classification System

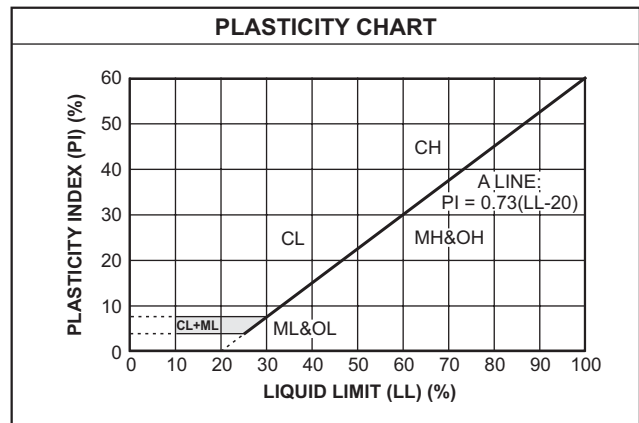
UNIFIED SOIL CLASSIFICATION SYSTEM

UNIFIED SOIL CLASSIFICATION AND SYMBOL CHART		
COARSE-GRAINED SOILS (more than 50% of material is larger than No. 200 sieve size.)		
GRAVELS More than 50% of coarse fraction larger than No. 4 sieve size	Clean Gravels (Less than 5% fines)	
	GW	Well-graded gravels, gravel-sand mixtures, little or no fines
	GP	Poorly-graded gravels, gravel-sand mixtures, little or no fines
	Gravels with fines (More than 12% fines)	
	GM	Silty gravels, gravel-sand-silt mixtures
	GC	Clayey gravels, gravel-sand-clay mixtures
SANDS 50% or more of coarse fraction smaller than No. 4 sieve size	Clean Sands (Less than 5% fines)	
	SW	Well-graded sands, gravelly sands, little or no fines
	SP	Poorly graded sands, gravelly sands, little or no fines
	Sands with fines (More than 12% fines)	
	SM	Silty sands, sand-silt mixtures
	SC	Clayey sands, sand-clay mixtures
FINE-GRAINED SOILS (50% or more of material is smaller than No. 200 sieve size.)		
SILTS AND CLAYS Liquid limit less than 50%	ML	Inorganic silts and very fine sands, rock flour, silty of clayey fine sands or clayey silts with slight plasticity
	CL	Inorganic clays of low to medium plasticity, gravelly clays, sandy clays, silty clays, lean clays
	OL	Organic silts and organic silty clays of low plasticity
SILTS AND CLAYS Liquid limit 50% or greater	MH	Inorganic silts, micaceous or diatomaceous fine sandy or silty soils, elastic silts
	CH	Inorganic clays of high plasticity, fat clays
	OH	Organic clays of medium to high plasticity, organic silts
HIGHLY ORGANIC SOILS	PT	Peat and other highly organic soils

LABORATORY CLASSIFICATION CRITERIA		
GW	$C_u = \frac{D_{60}}{D_{10}}$ greater than 4; $C_c = \frac{D_{30}}{D_{10} \times D_{60}}$ between 1 and 3	
GP	Not meeting all gradation requirements for GW	
GM	Atterberg limits below "A" line or P.I. less than 4	Above "A" line with P.I. between 4 and 7 are borderline cases requiring use of dual symbols
GC	Atterberg limits above "A" line with P.I. greater than 7	
SW	$C_u = \frac{D_{60}}{D_{10}}$ greater than 4; $C_c = \frac{D_{30}}{D_{10} \times D_{60}}$ between 1 and 3	
SP	Not meeting all gradation requirements for GW	
SM	Atterberg limits below "A" line or P.I. less than 4	Limits plotting in shaded zone with P.I. between 4 and 7 are borderline cases requiring use of dual symbols.
SC	Atterberg limits above "A" line with P.I. greater than 7	

Determine percentages of sand and gravel from grain-size curve. Depending on percentage of fines (fraction smaller than No. 200 sieve size), coarse-grained soils are classified as follows:

Less than 5 percent GW, GP, SW, SP
 More than 12 percent GM, GC, SM, SC
 5 to 12 percent Borderline cases requiring dual symbols



TERMS DESCRIBING SOIL CONSISTENCY				
Fine Grained Soils		Coarse Grained Soils		
Description	Penetrometer Reading (tsf)	Penetration Resistance (blows/ft)	Description	Relative Density
Soft	0.0 to 1.0	0 to 4	Very Loose	0 to 20%
Firm	1.0 to 1.5	4 to 10	Loose	20 to 40%
Stiff	1.5 to 3.0	10 to 30	Medium Dense	40 to 70%
Very Stiff	3.0 to 4.5	30 to 50	Dense	70 to 90%
Hard	4.5+	Over 50	Very Dense	90 to 100%

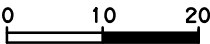
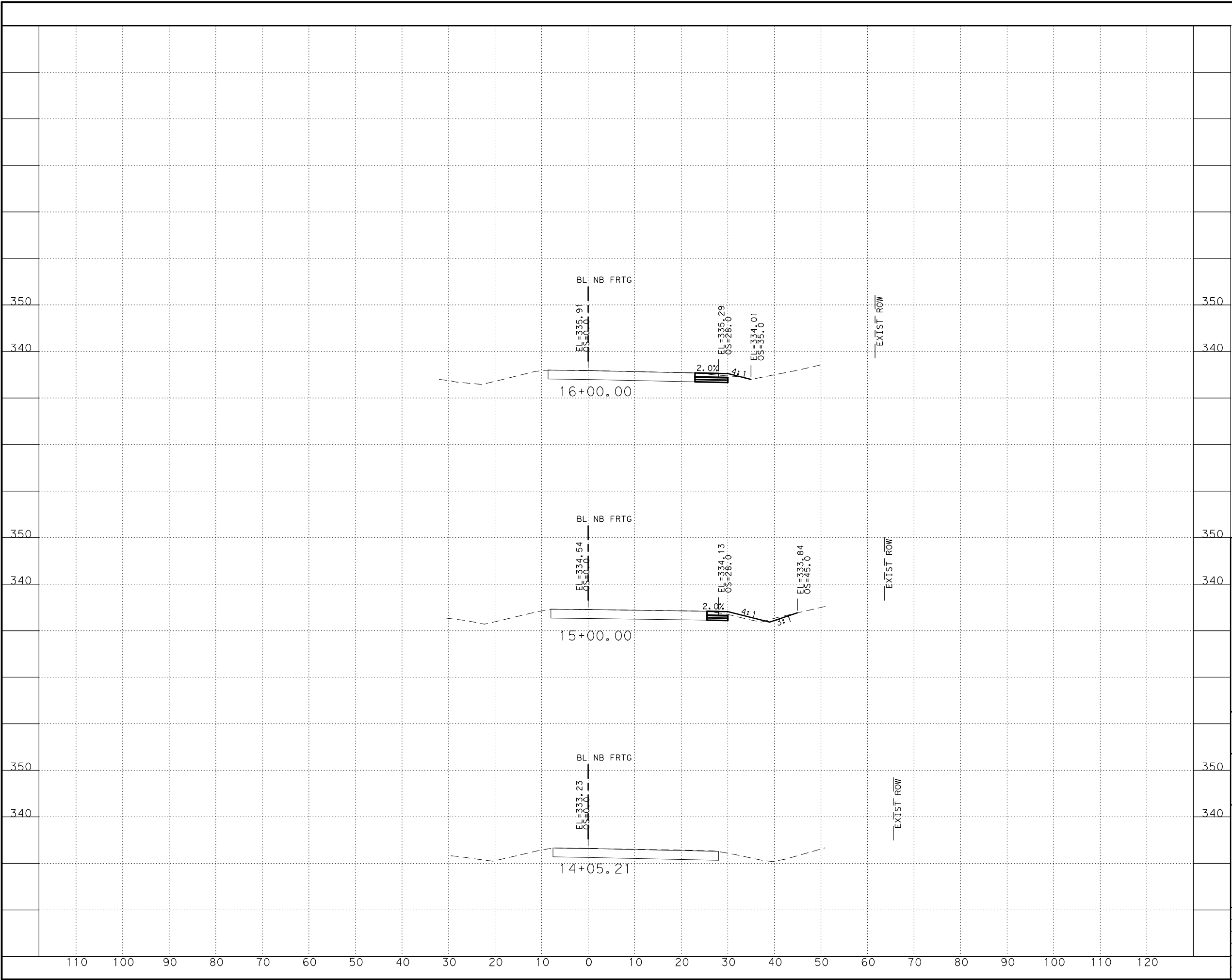
Appendix G - Cross Sections

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STATE	DIST	COUNTY	
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CONT	SECT	JOB	HIGHWAY NO
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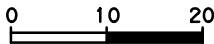
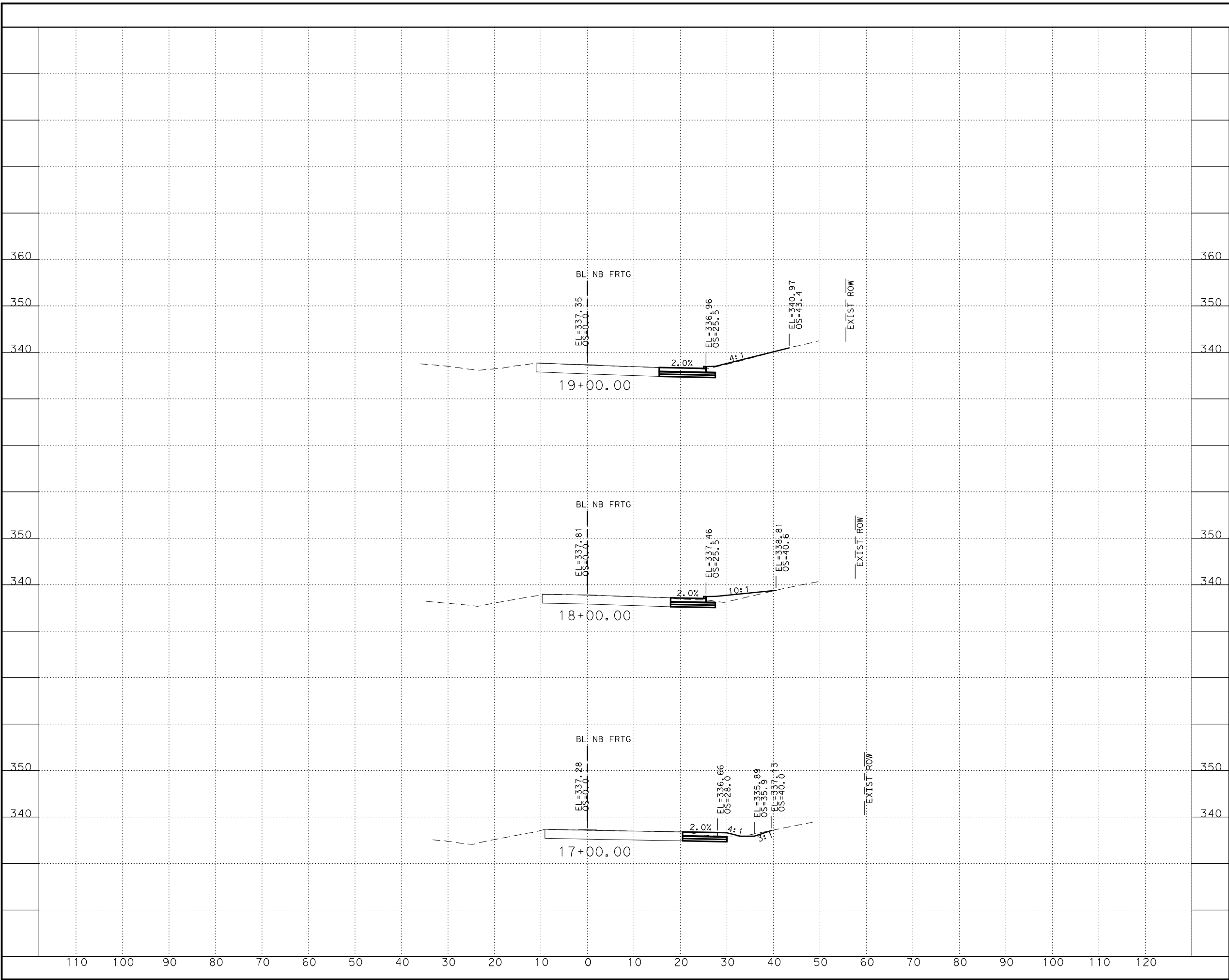
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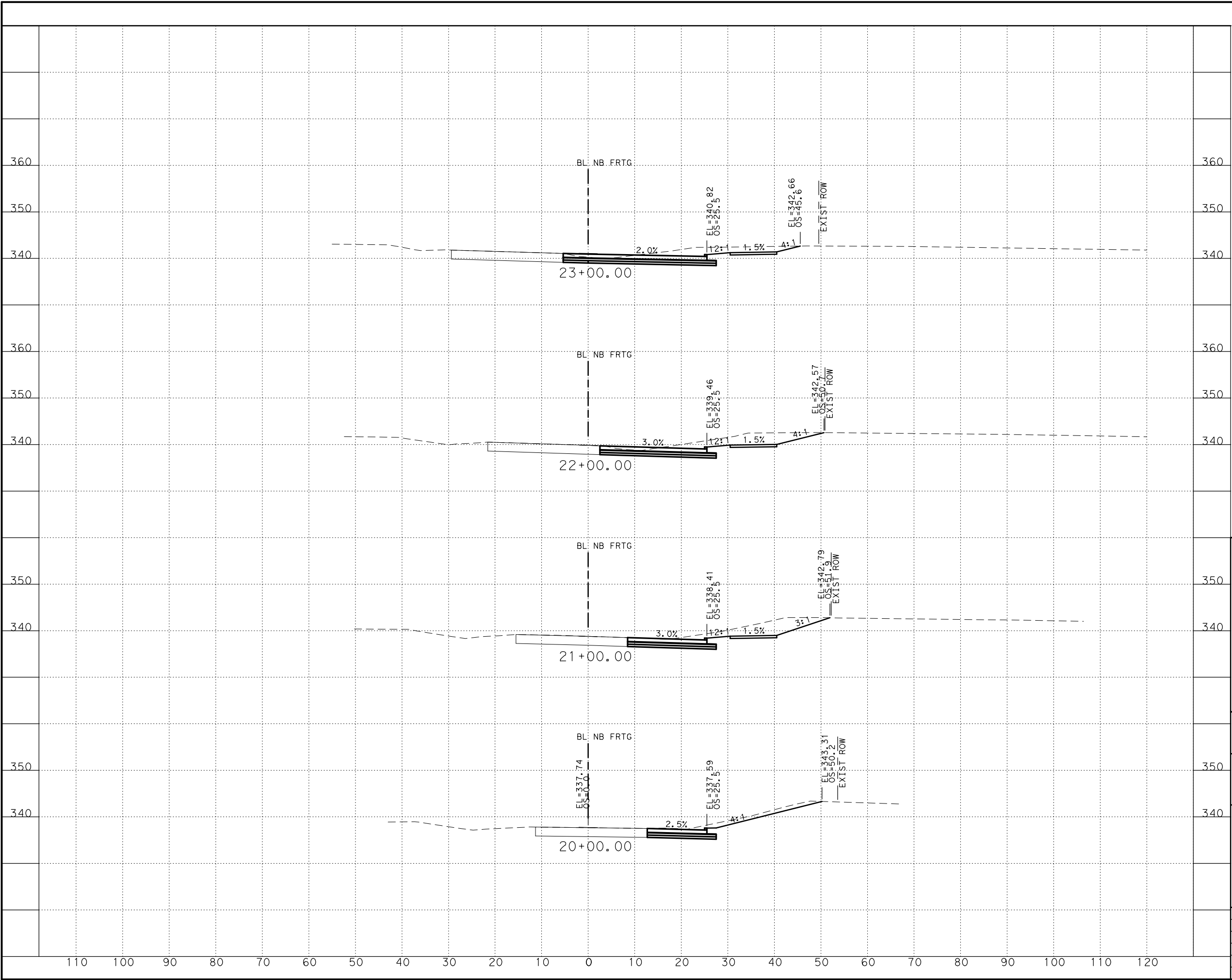
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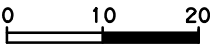
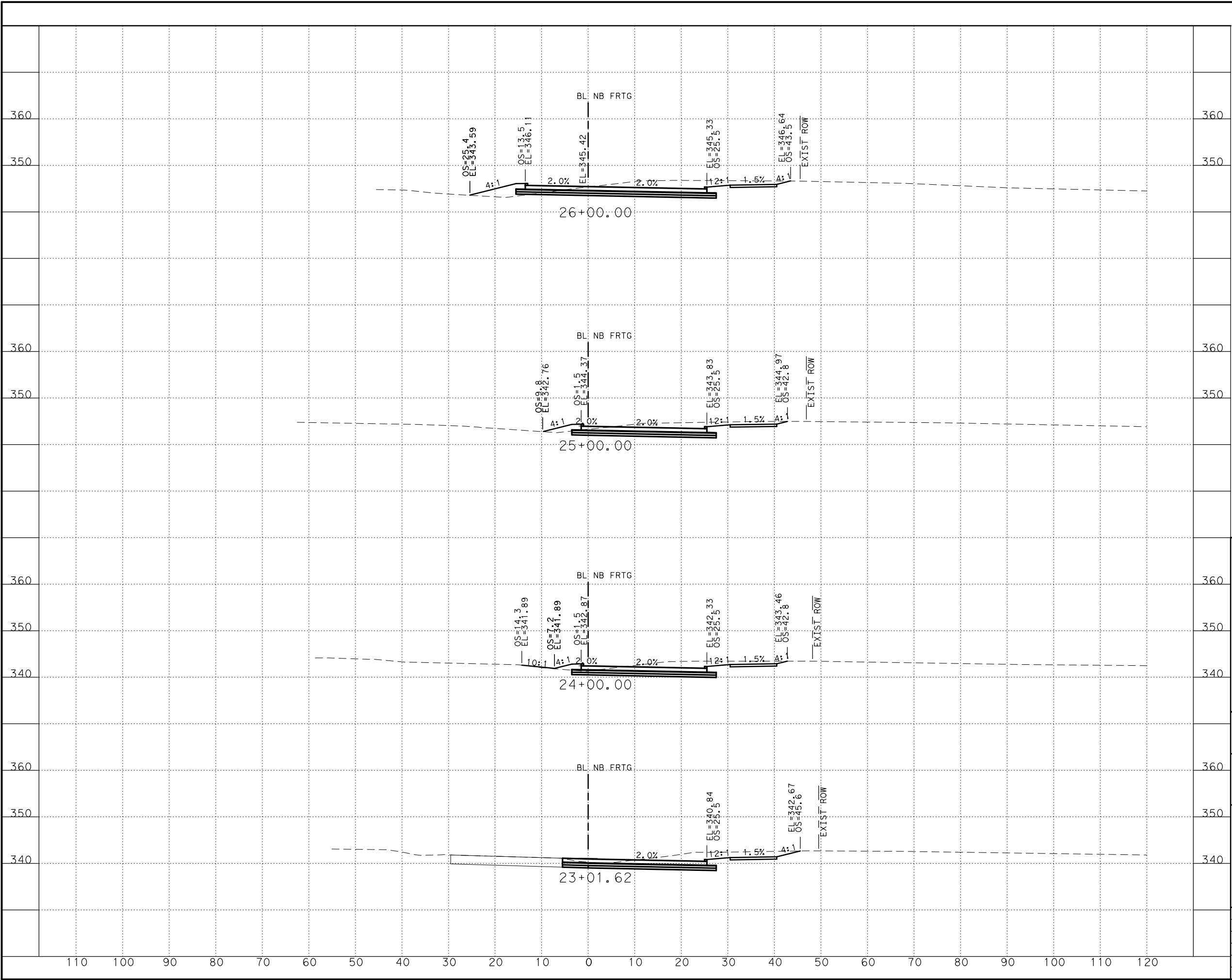
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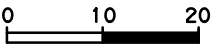
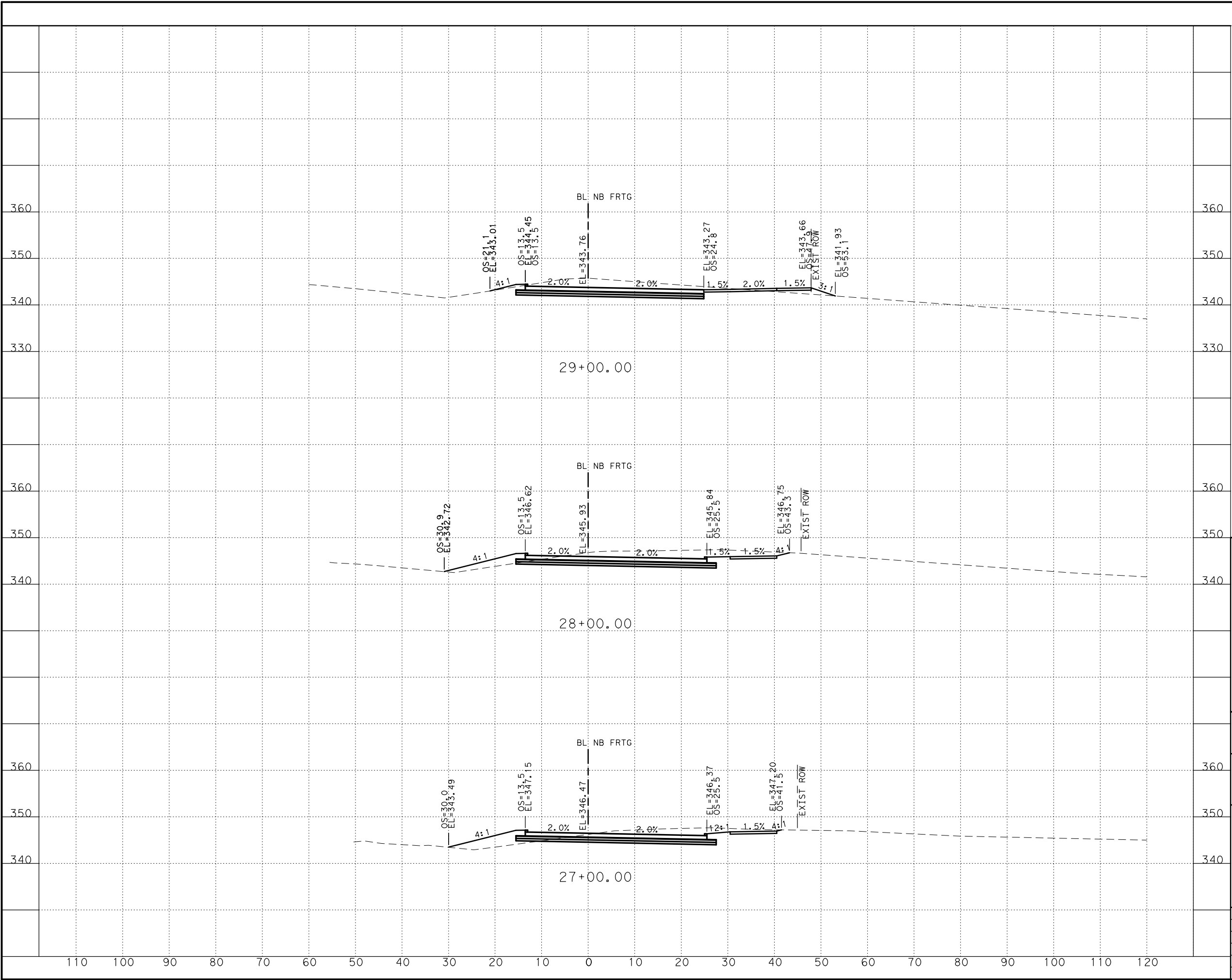
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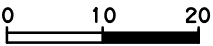
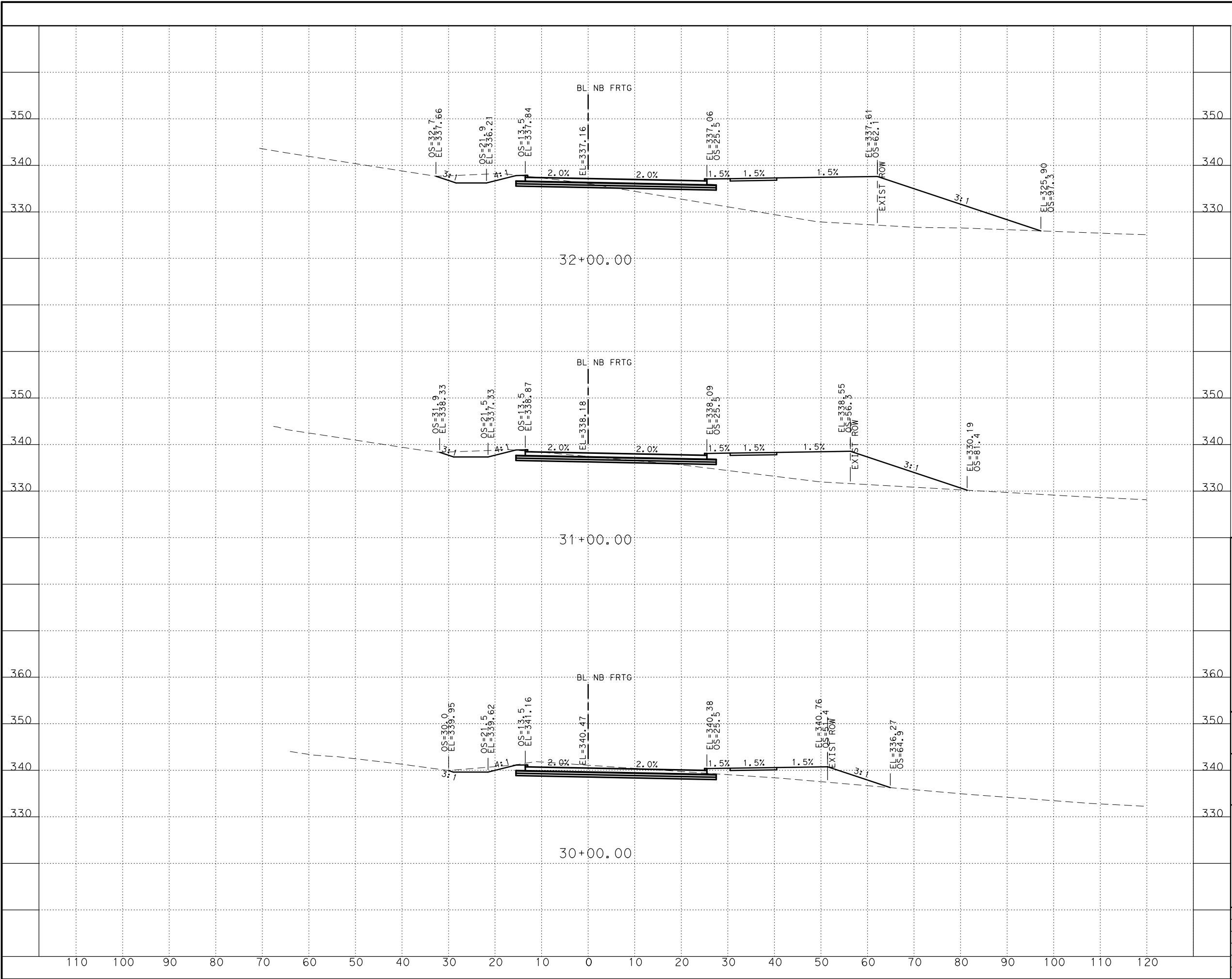
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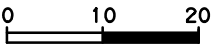
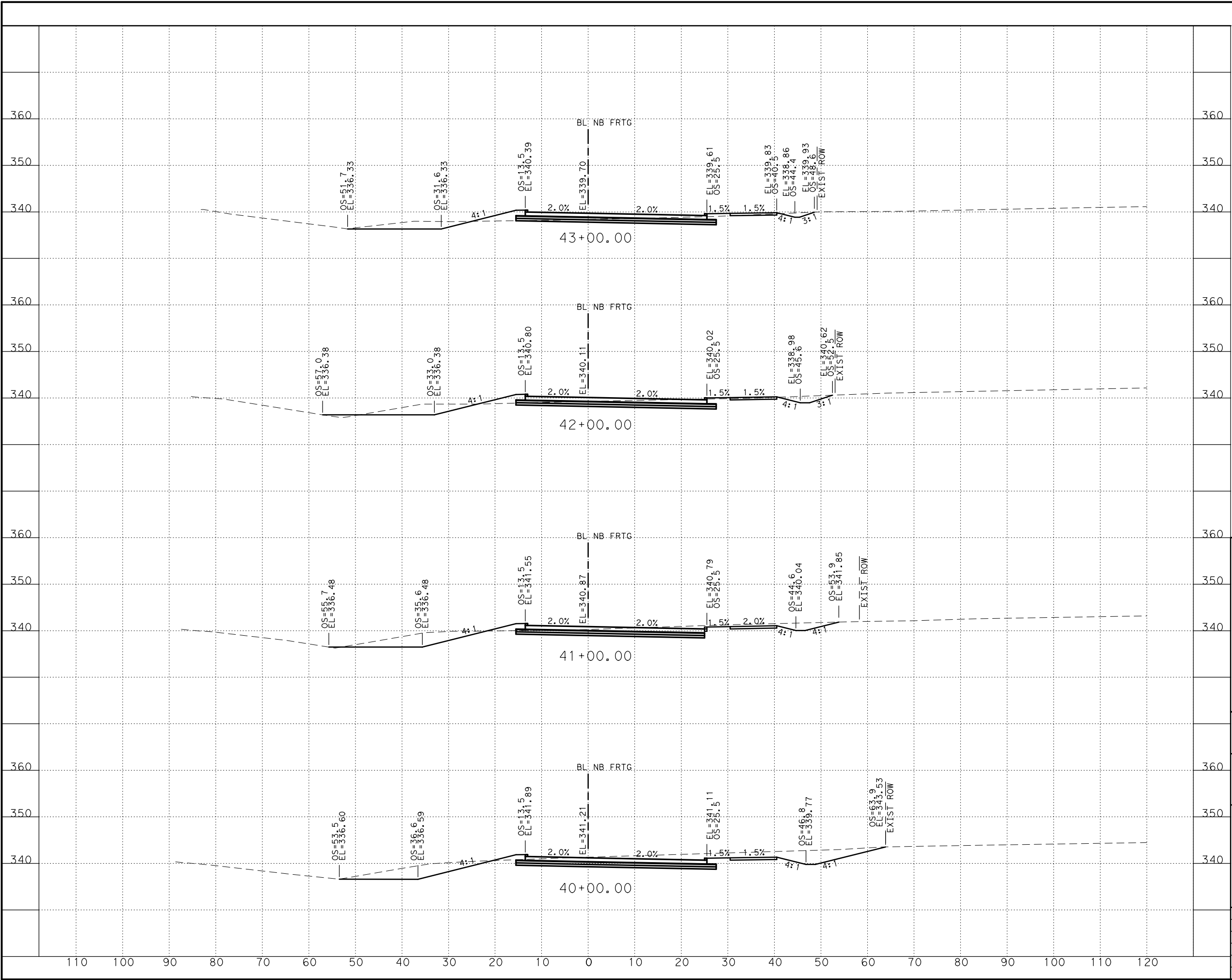
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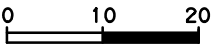
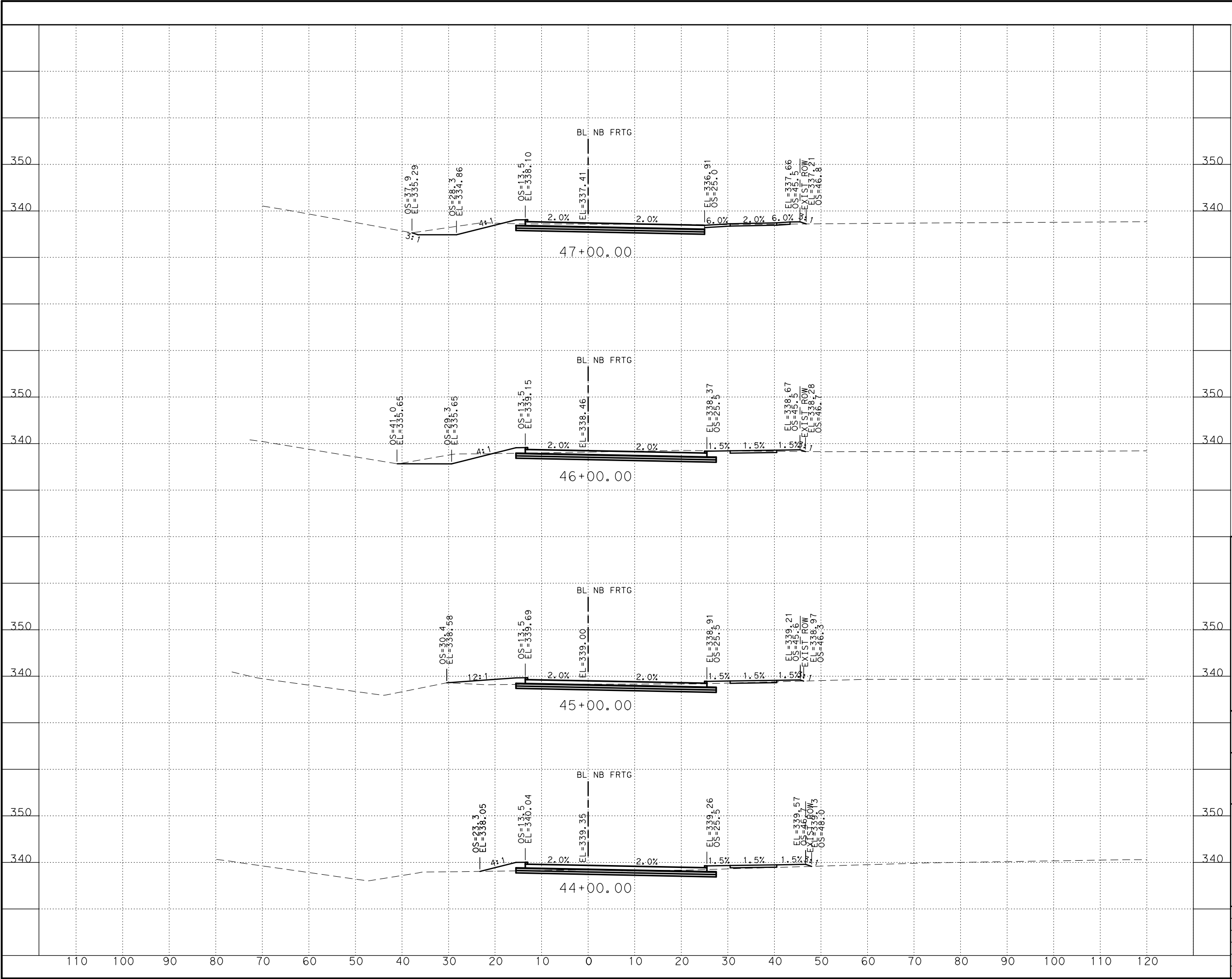
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STATE	DIST	COUNTY	
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CONT	SECT	JOB	HIGHWAY NO
0675	08	114	IH 45

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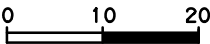
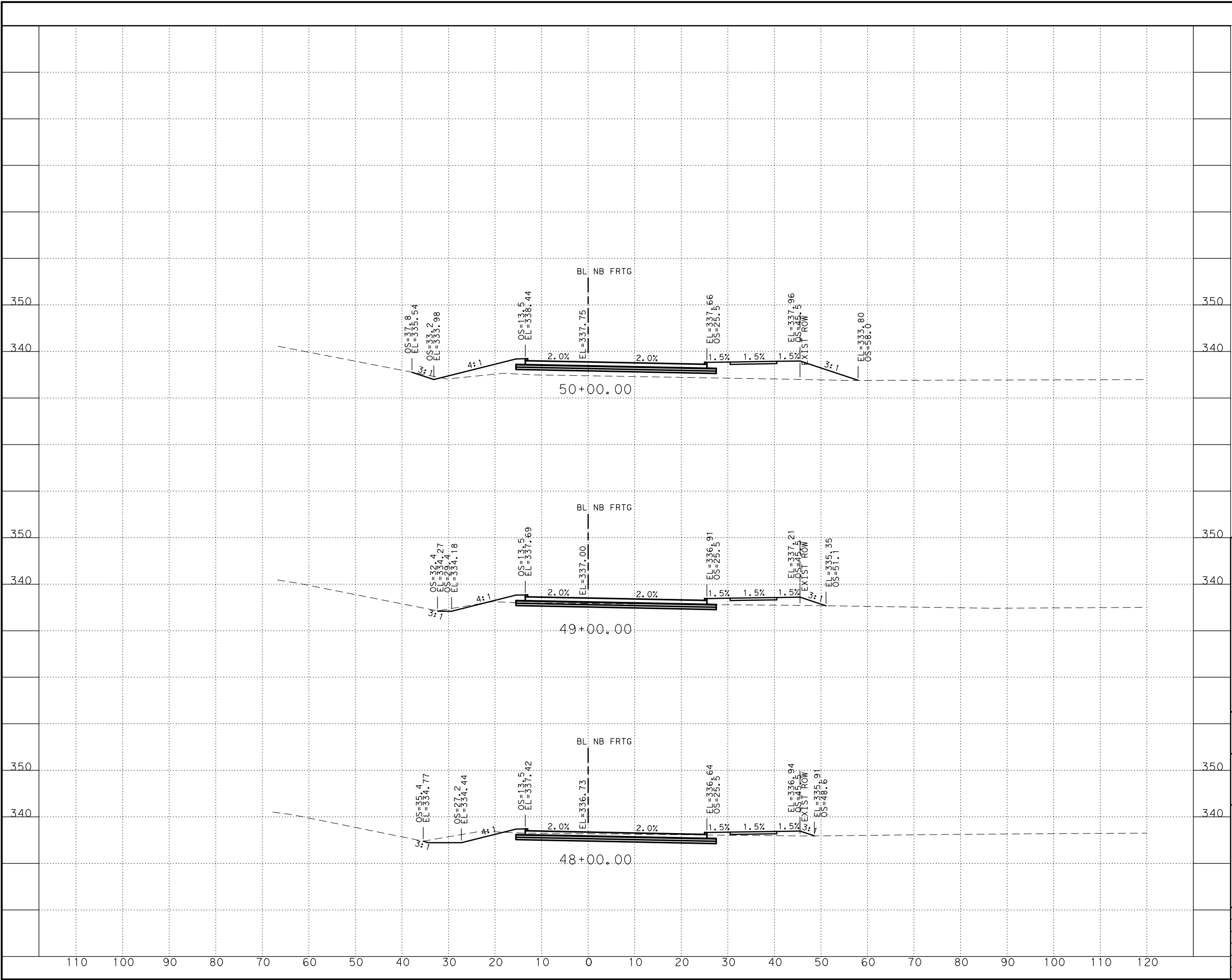
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STATE	DIST	COUNTY	
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CONT	SECT	JOB	HIGHWAY NO
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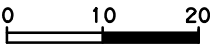
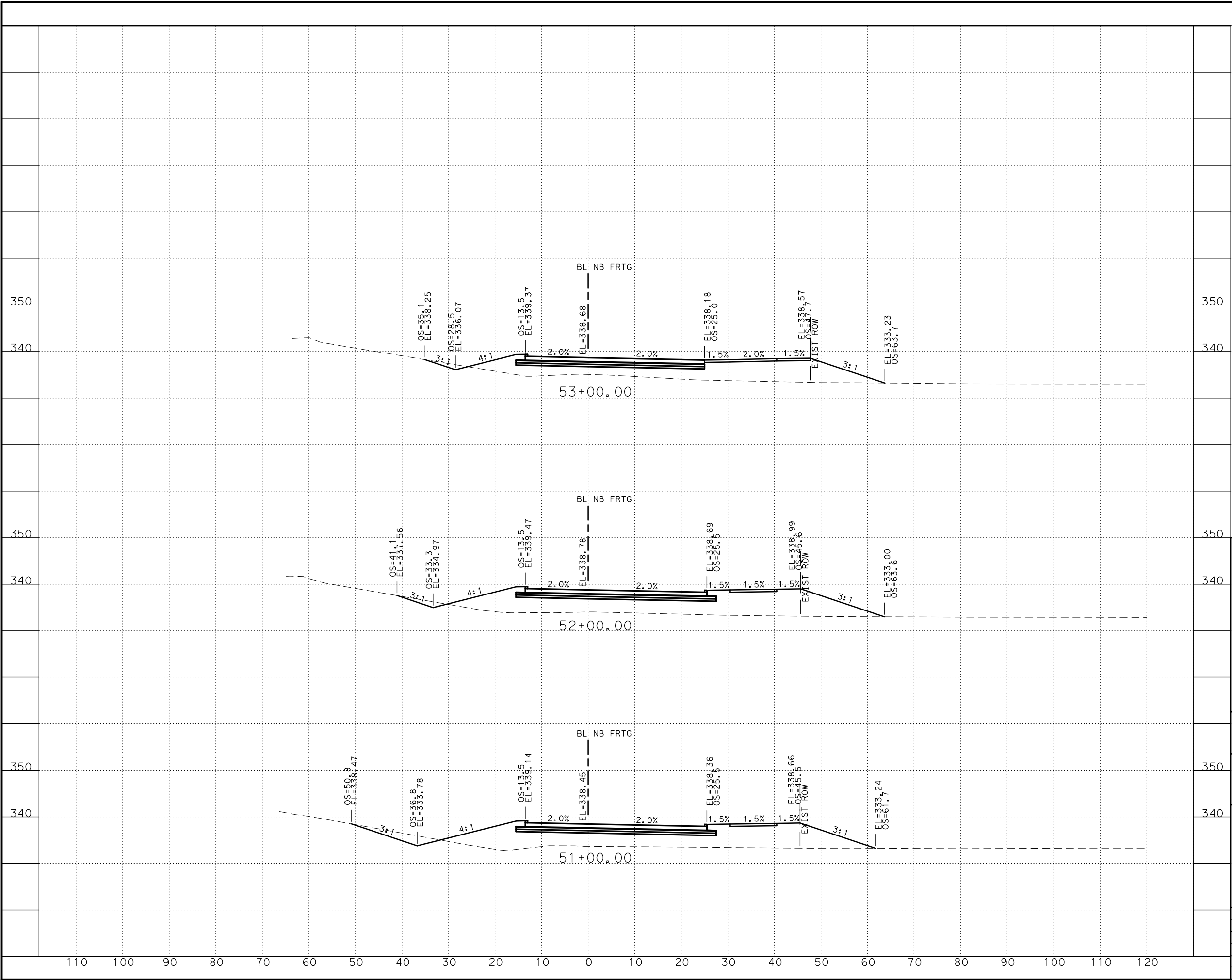
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ROADWAY CROSS SECTIONS

SHEET 12 OF 23

FED RD DIV NO	PROJECT NO		SHEET NO
6			
STATE	DIST	COUNTY	
TEXAS	HOU	MONTGOMERY	
CONT	SECT	JOB	HIGHWAY NO
0675	08	114	IH 45

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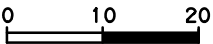
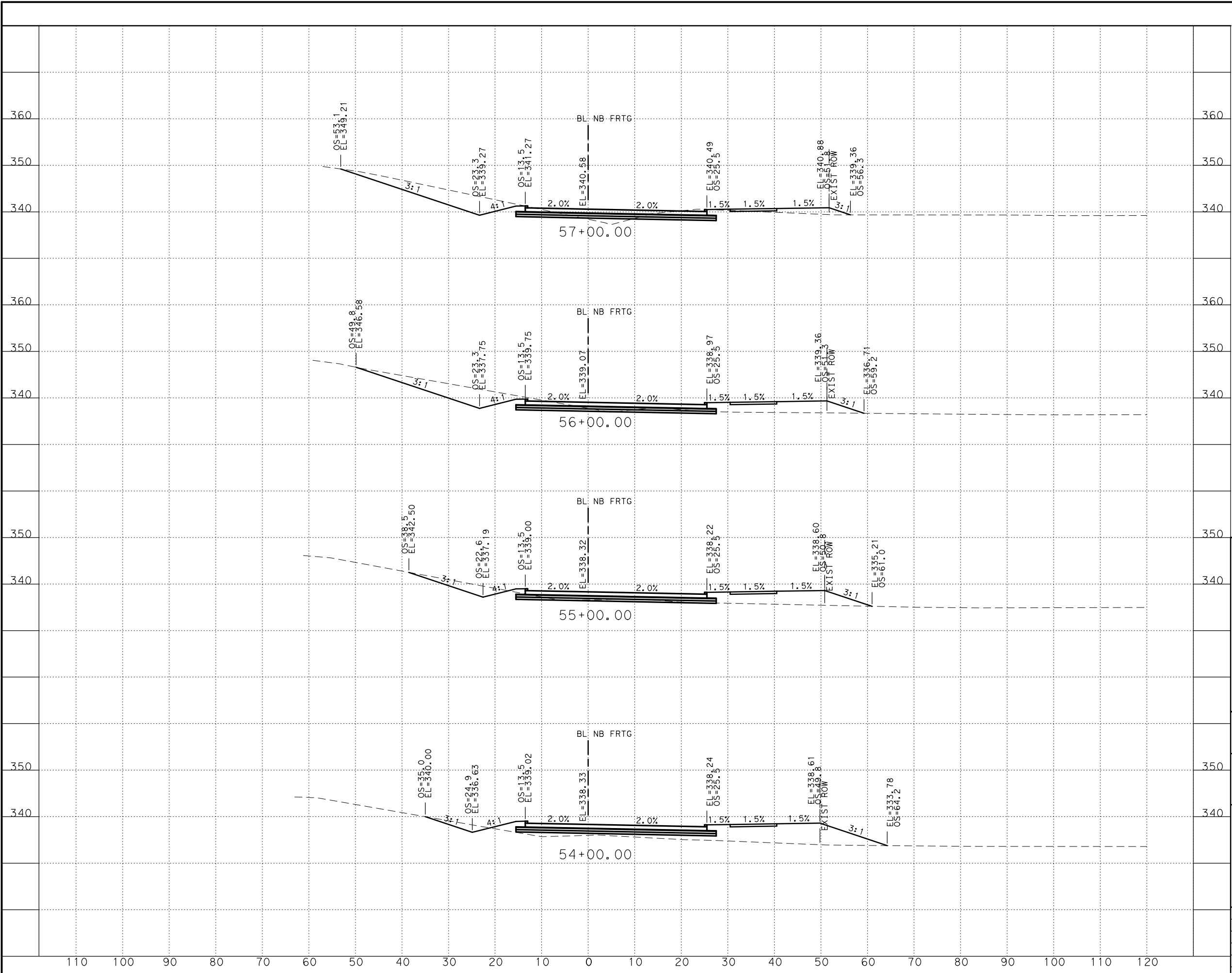
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SHEET 13 OF 23

FED RD DIV NO	PROJECT NO		SHEET NO
6			
STATE	DIST	COUNTY	
TEXAS	HOU	MONTGOMERY	
CONT	SECT	JOB	HIGHWAY NO
0675	08	114	IH 45

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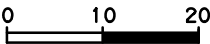
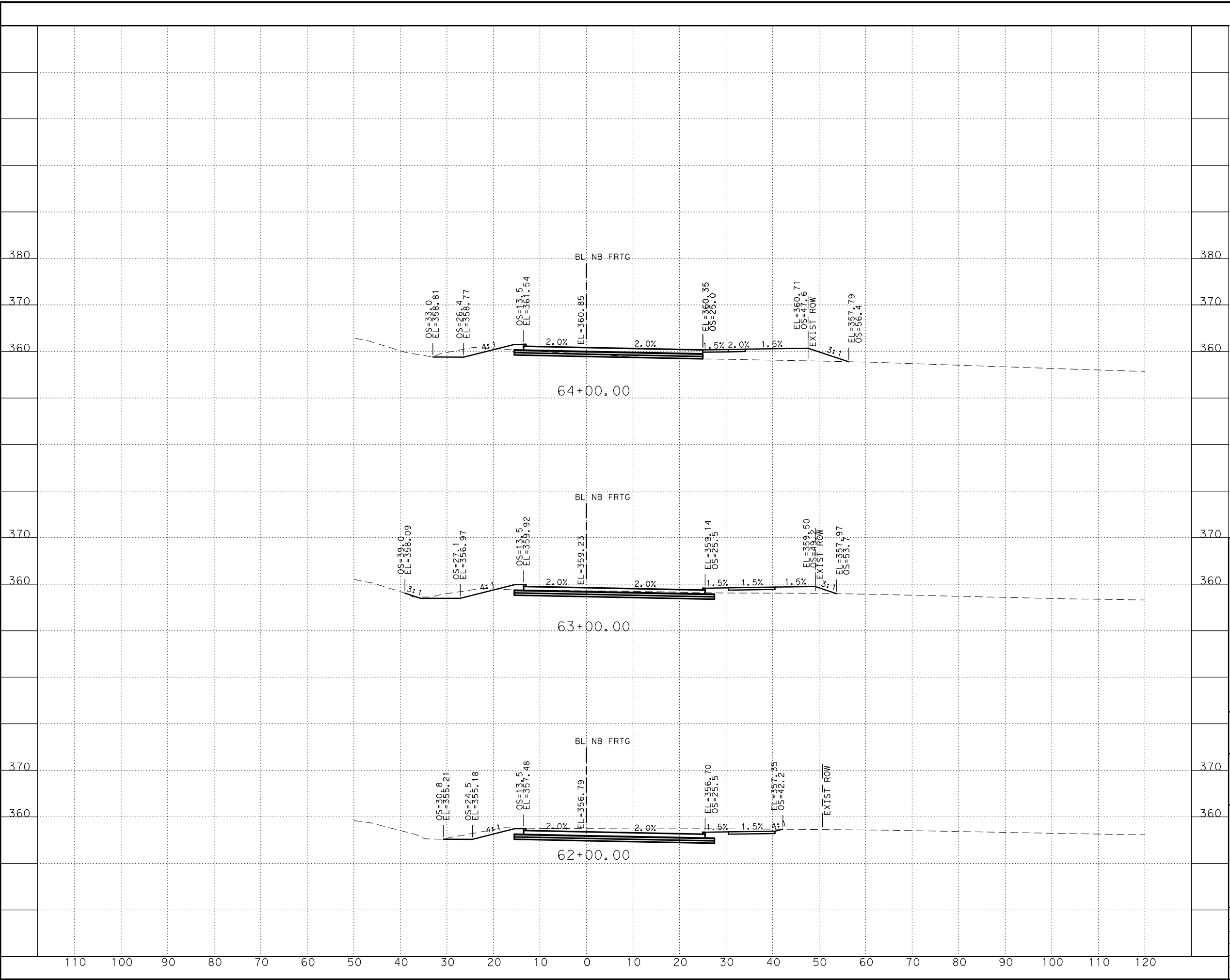
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SHEET 15 OF 23

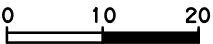
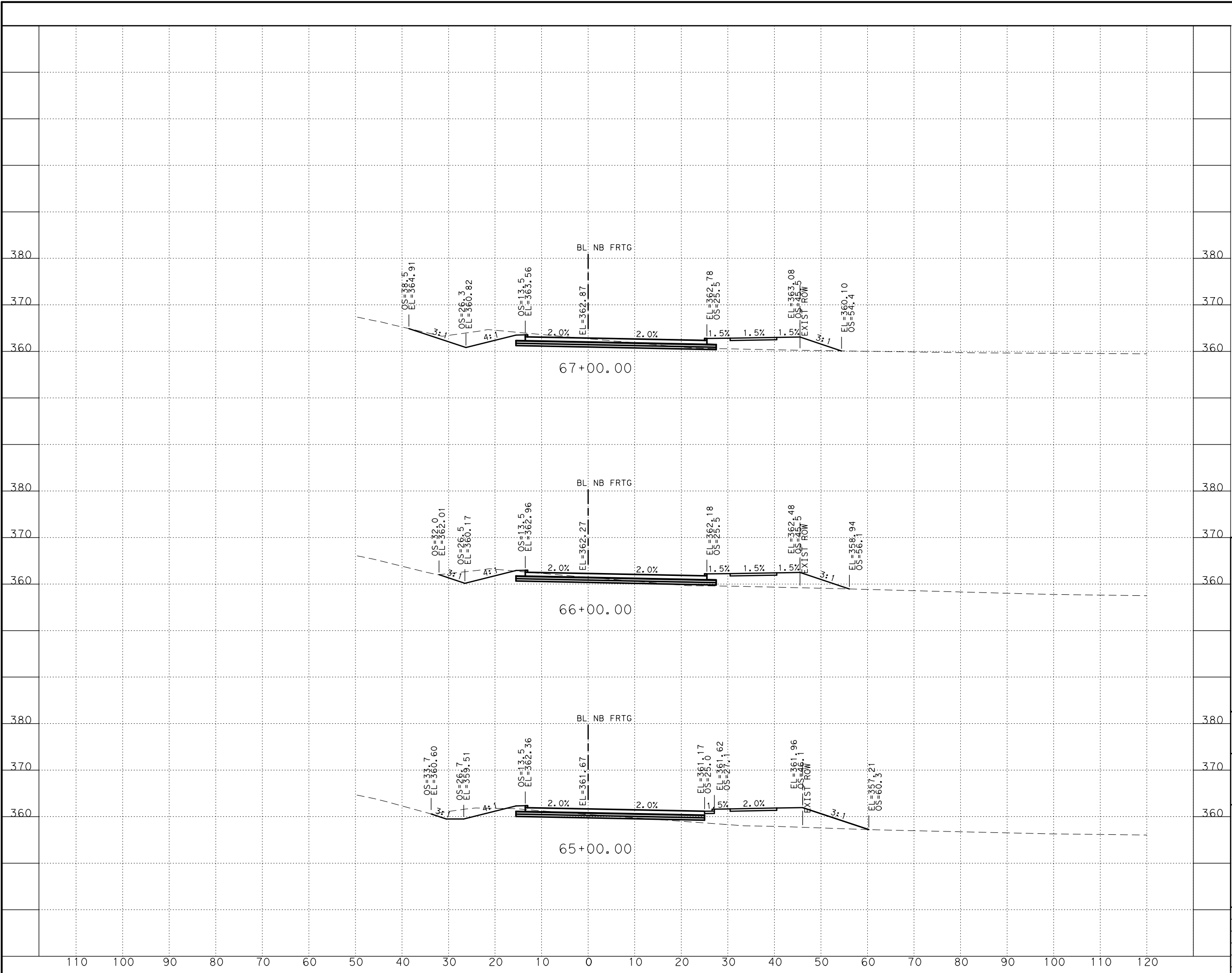
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SHEET 16 OF 23

FED RD DIV NO	PROJECT NO		SHEET NO
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STATE	DIST	COUNTY	
TEXAS	HOU	MONTGOMERY	
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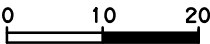
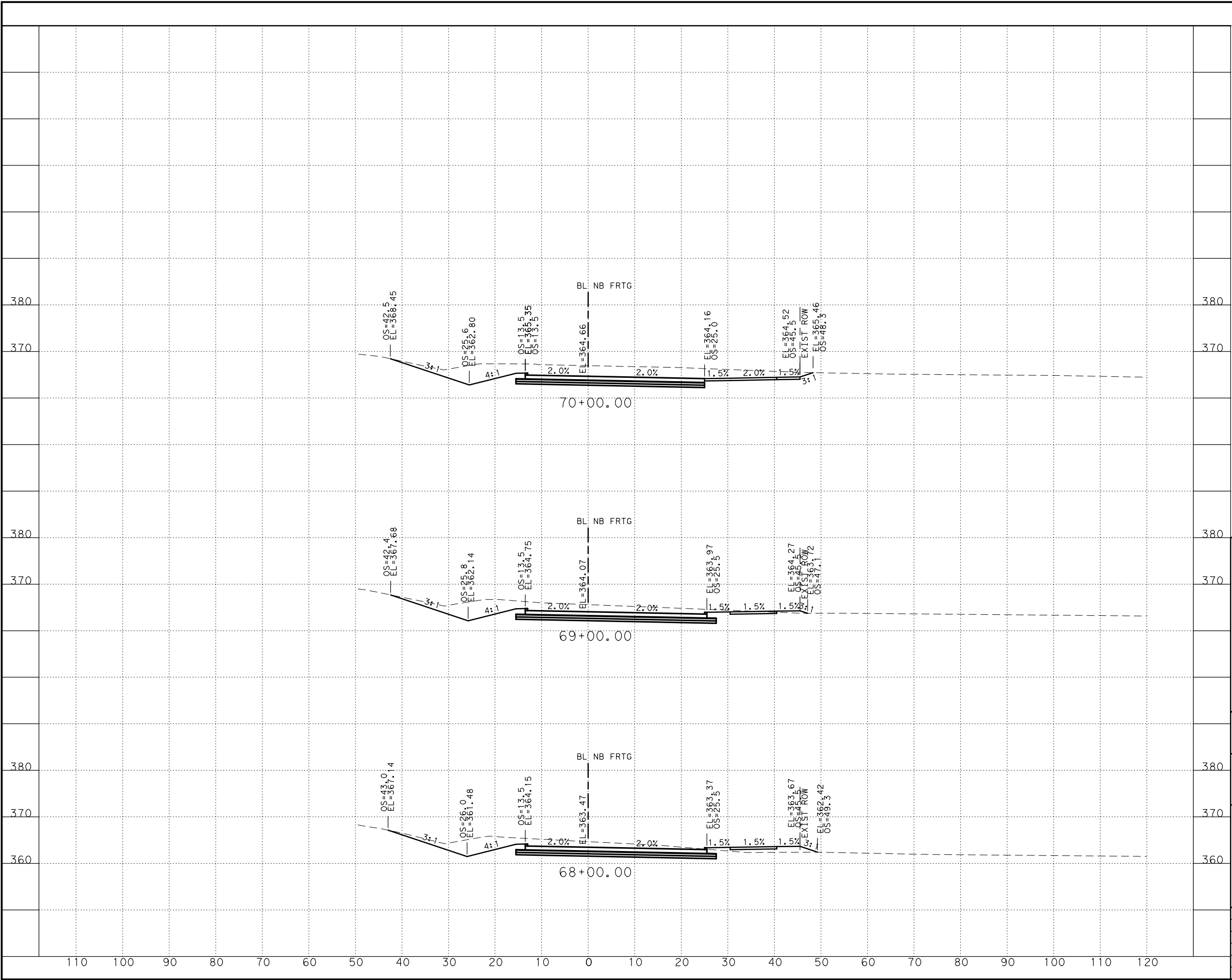
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SHEET 17 OF 23

FED RD DIV NO	PROJECT NO		SHEET NO
6			
STATE	DIST	COUNTY	
TEXAS	HOU	MONTGOMERY	
CONT	SECT	JOB	HIGHWAY NO
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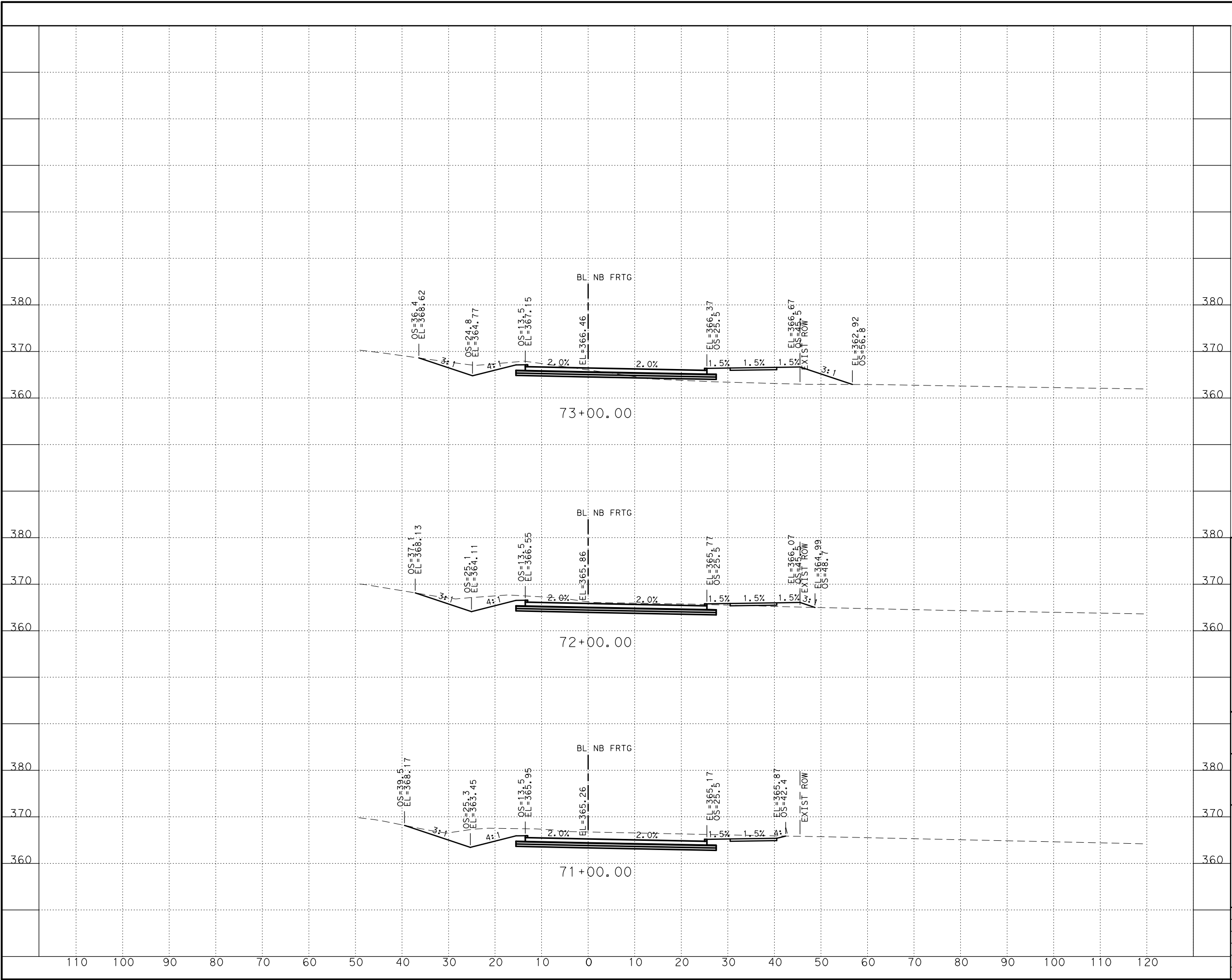
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FED RD DIV NO	PROJECT NO		SHEET NO
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STATE	DIST	COUNTY	
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CONT	SECT	JOB	HIGHWAY NO
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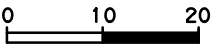
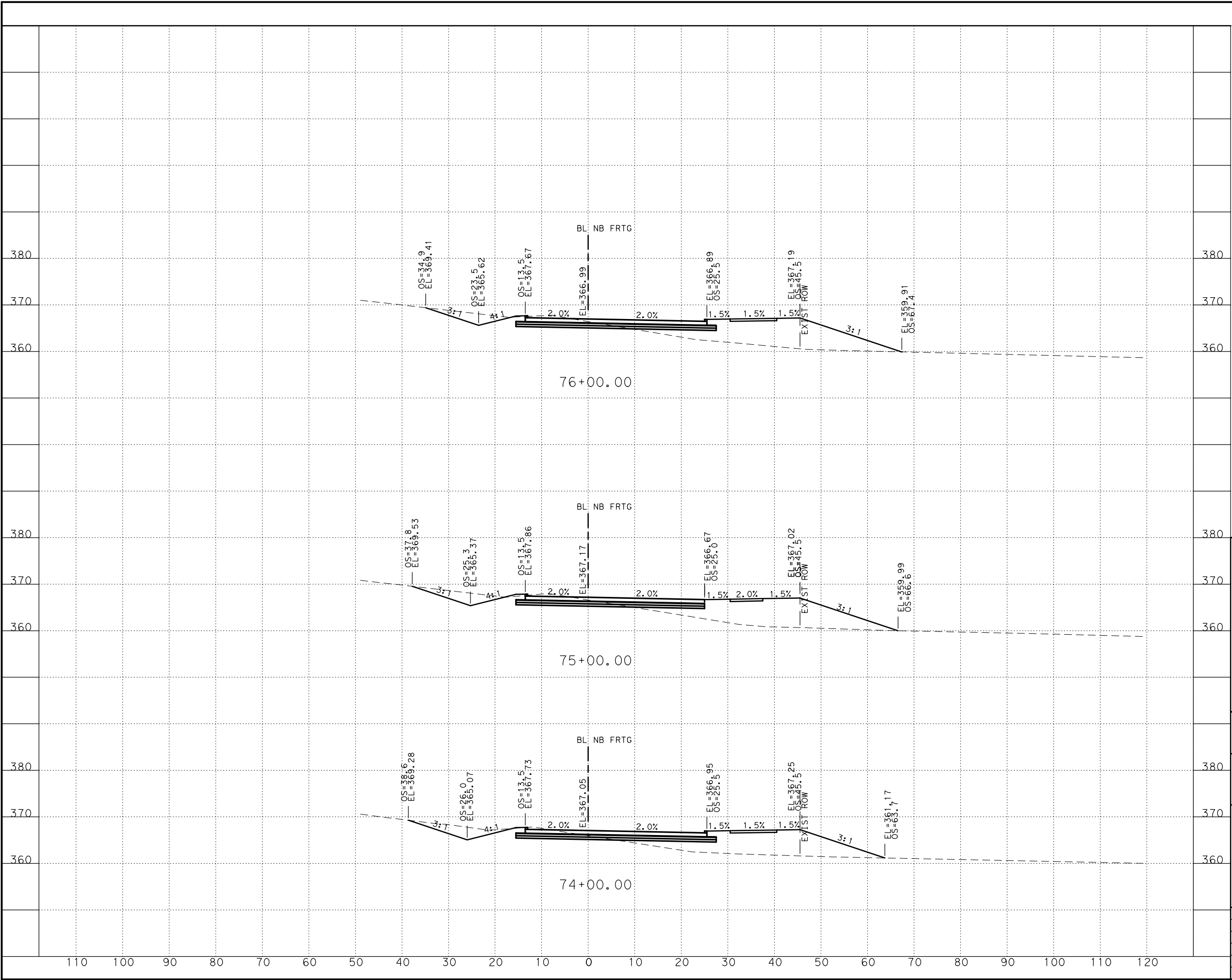
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FED RD DIV NO	PROJECT NO		SHEET NO
6			
STATE	DIST	COUNTY	
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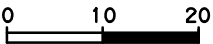
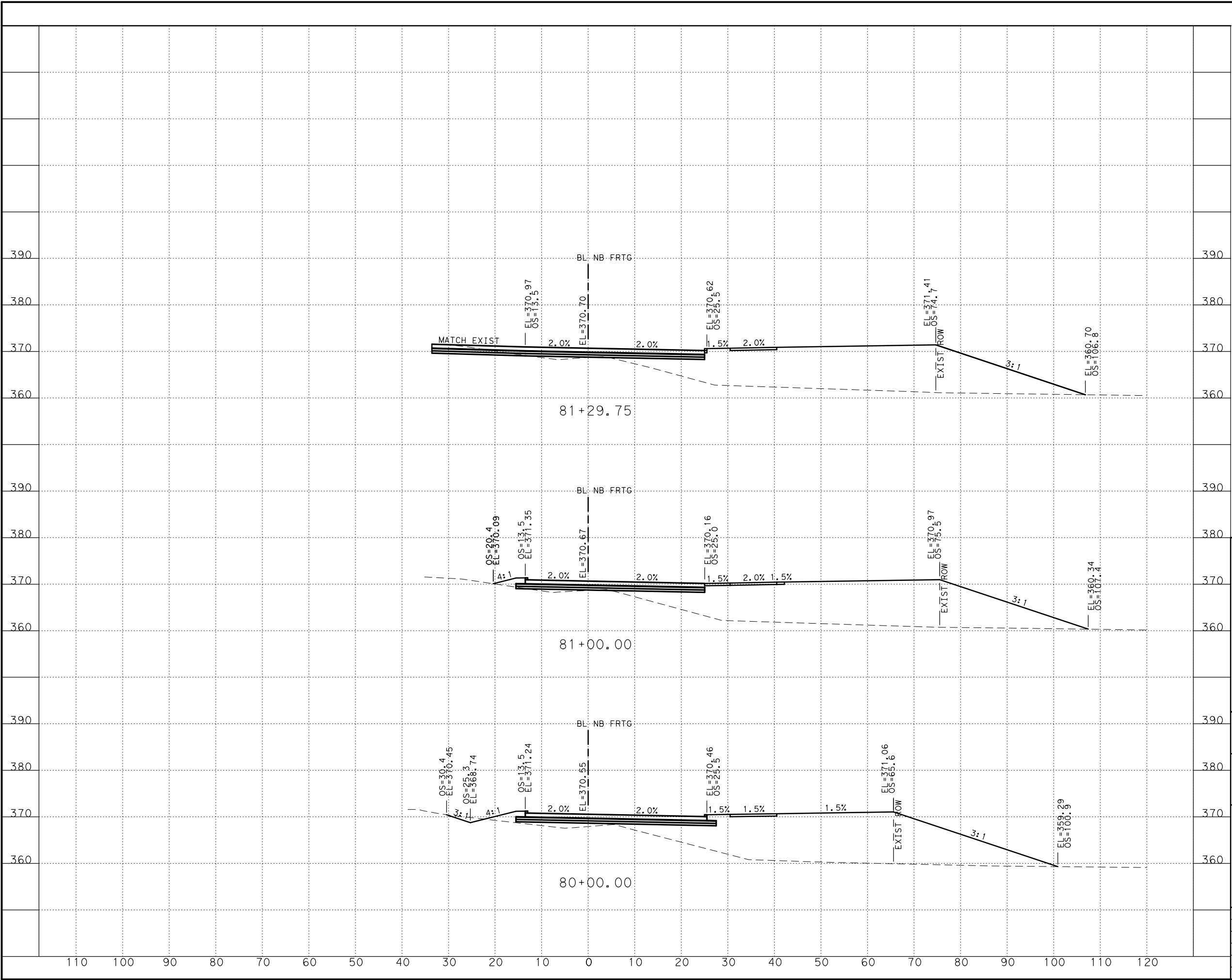
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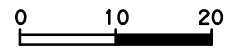
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ROADWAY CROSS SECTIONS

SHEET 22 OF 23

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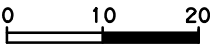
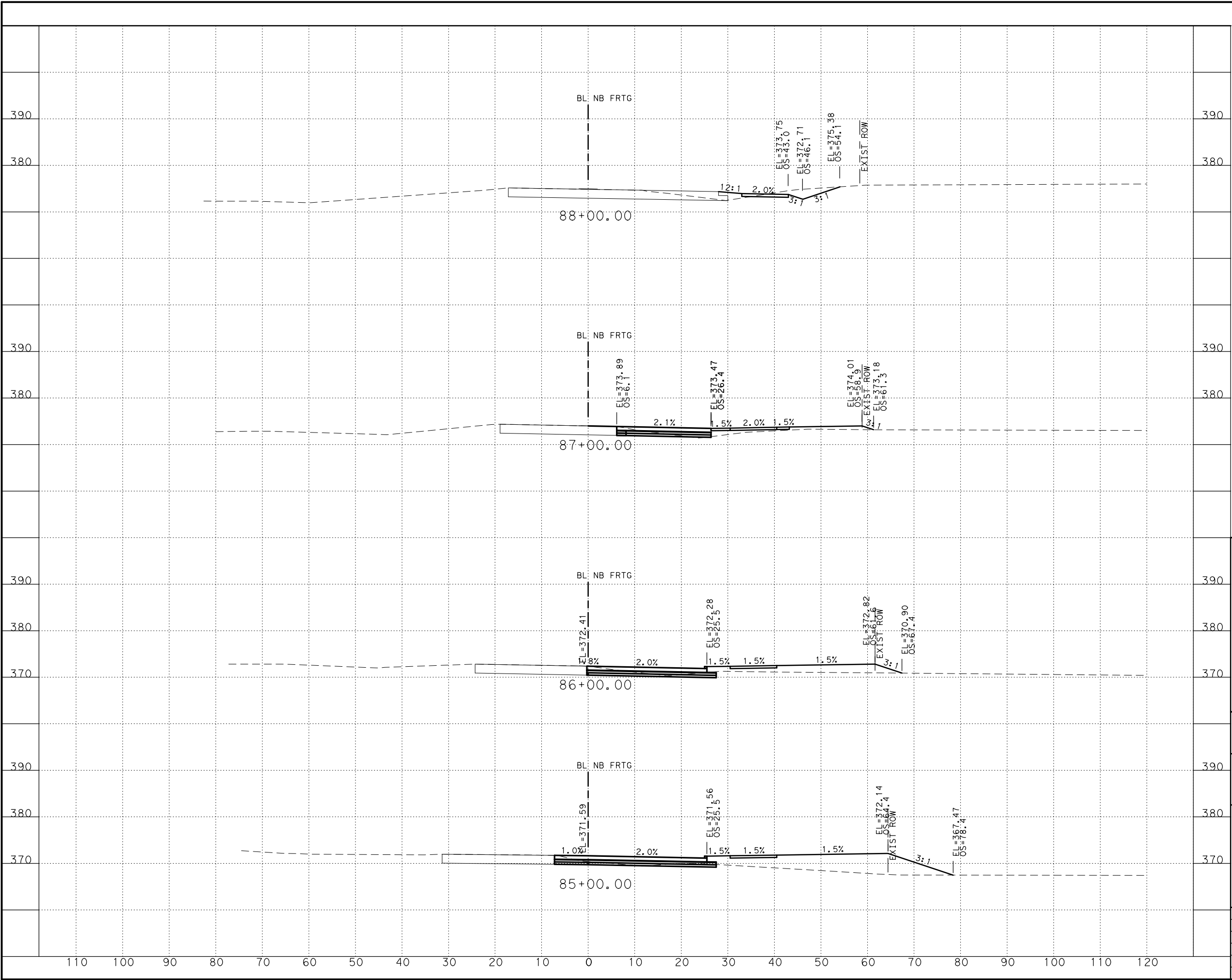
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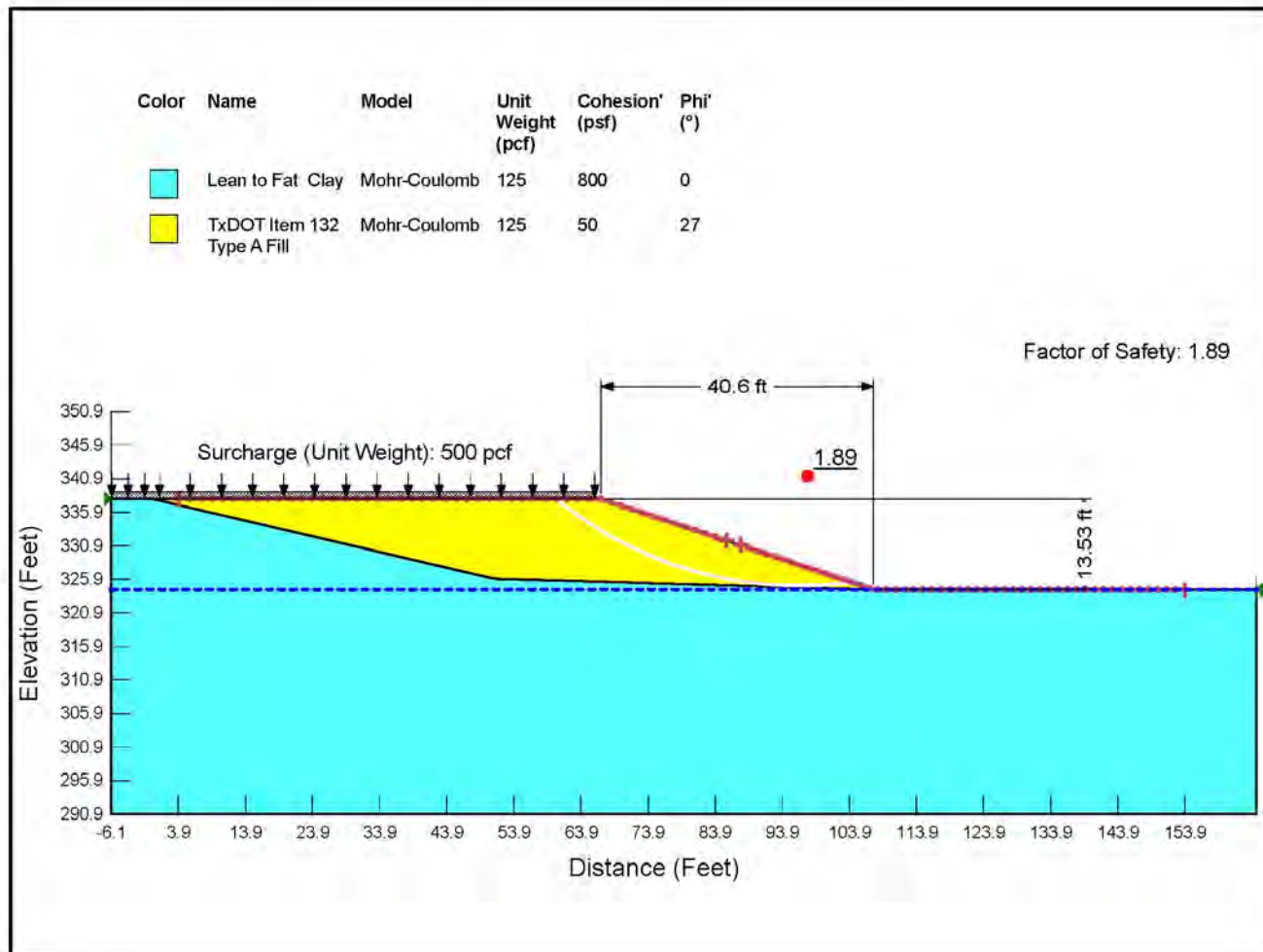
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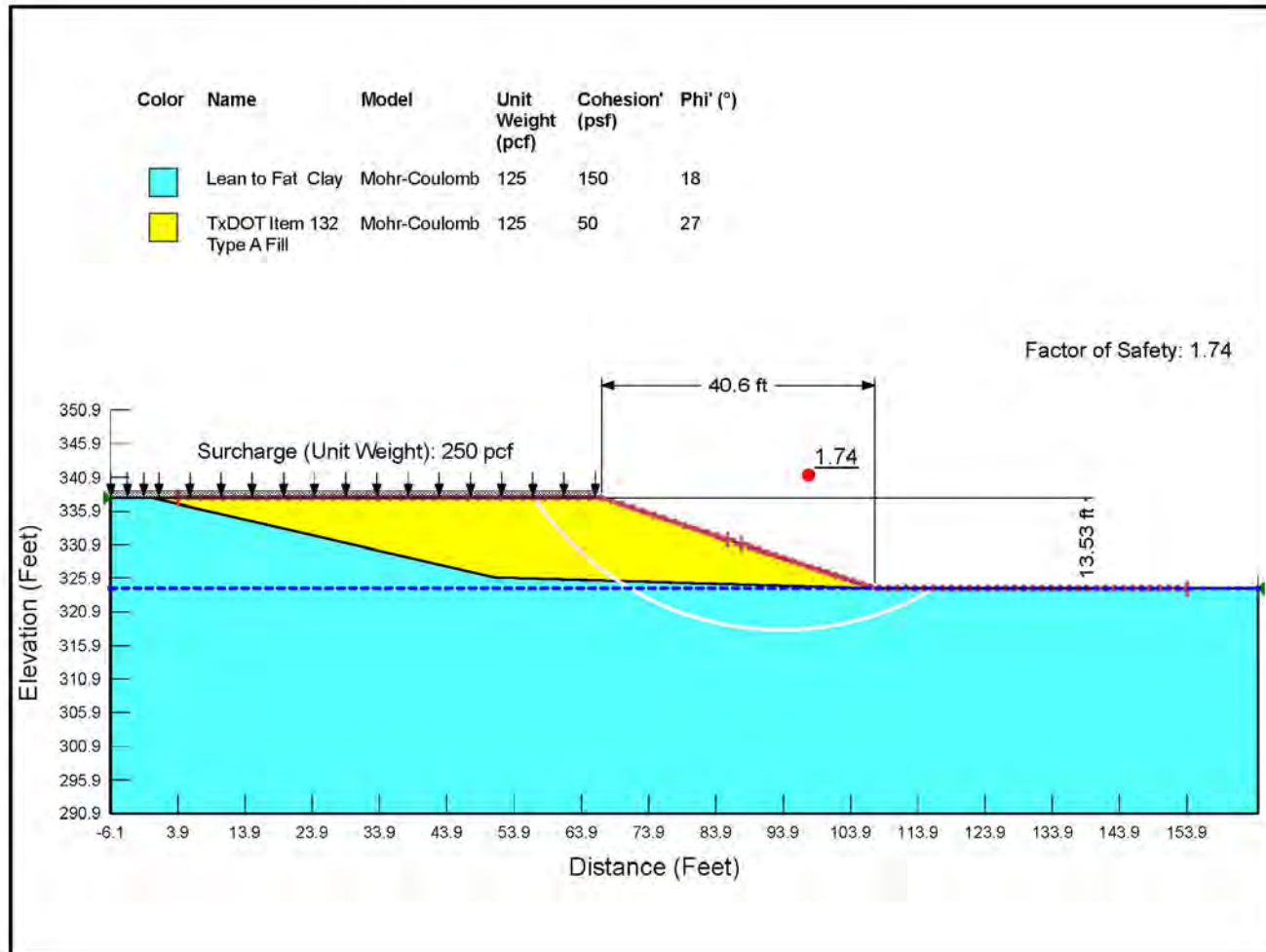
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Appendix H- Slope Stability Analysis Results

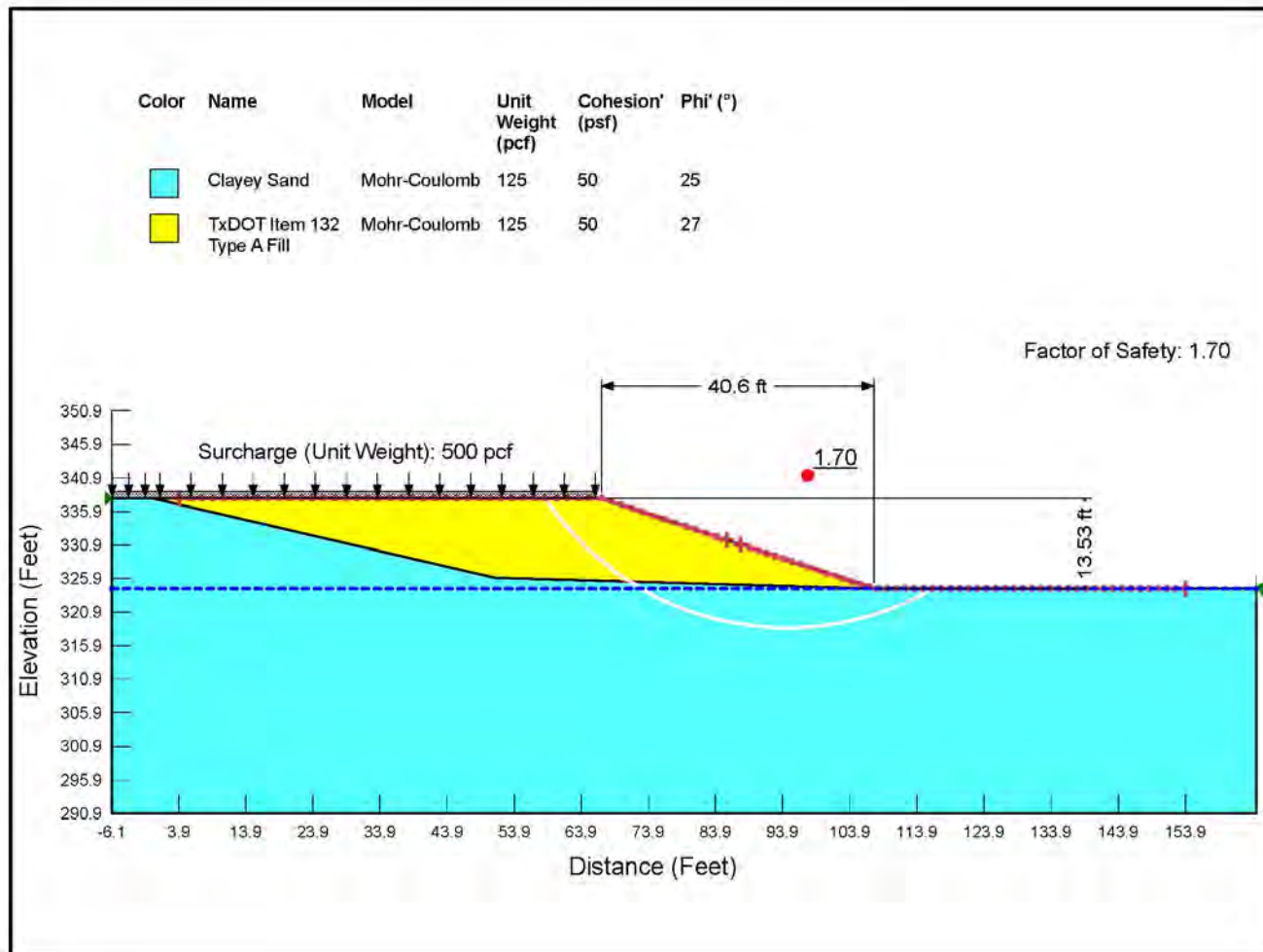
SHORT TERM GLOBAL STABILITY ANALYSIS



LONG TERM GLOBAL STABILITY ANALYSIS



SHORT TERM GLOBAL STABILITY ANALYSIS



LONG TERM GLOBAL STABILITY ANALYSIS

